Overconfidence and the Political and Financial Behavior of a Representative Sample *

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June 20, 2025

Abstract

Overprecision—excessive confidence in the accuracy of one's beliefs—affects decision-making across multiple domains. In this paper, we examine how overprecision influences financial and political behavior using a nationally representative sample from the German Socio-Economic Panel Innovation Sample (SOEP-IS). We introduce the Subjective Error Method, a simplified yet effective measure of overprecision, and find that individuals exhibiting higher overprecision make larger stock market forecasting errors, hold less diversified investment portfolios, adopt more extreme political views, and are less likely to vote. Crucially, we demonstrate that overprecise behavior in finance is systematically associated with overprecise behavior in politics, providing evidence that overprecision is a pervasive bias that permeates many aspects of people's lives rather than a domain-specific phenomenon. Finally, using two additional surveys, we validate the Subjective Error Method as a robust measure of overprecision, demonstrating its robustness across knowledge domains and its correlation with respondents' uncertainty.

Keywords Overconfidence, SOEP, Survey

JEL Classification C83 · D91 · G41

*Corresponding author: cirilbosch@gmail.com. We would like to thank Agustin Casas, Antoni-Italo de Moragas, Benjamin Enke, Tilman Fries, Thomas Gräber, Andreas Haufler, Frank Heinemann, Don Moore, Guillem Riambau, Uwe Sunde, the participants at the CESifo Area Conference in Behavioral Economics as well as the participants of seminars at CUNEF, EMLyon, and FU Berlin for their insightful comments. We acknowledge the financial support from the Deutsche Forschungsgemeinschaft (DFG) through CRC TRR 190 "Rationality and Competition" (project number 280092119). Ethical permission was provided by the Scientific Advisory Board of DIW Berlin and the Ethics Committee of the Faculty of Economics at Ludwig-Maximilians-Universität München and informed consent was obtained from all participants included in the analyses. Two online surveys were preregistered at aspredicted.org (#118284 and #123146). The authors are not presented in alphabetical order, as the first two authors contributed more to the paper.

What would I eliminate if I had a magic wand? Overconfidence.

—Daniel Kahneman, *The Guardian*, 18 July 2015

1 Introduction

Overconfidence is a pervasive and potent bias in human judgment (Mannes and Moore, 2013; Kahneman, 2011). It leads to wars (Johnson, 2004), to excessive entry into markets (Camerer and Lovallo, 1999), or to 80% of the population thinking that they are above-average drivers (Svenson, 1981). However, overconfidence is a general term that encompasses three different phenomena: overestimation, overplacement, and overprecision (Moore and Healy, 2008; Moore and Schatz, 2017). Overestimation has to do with absolute values—you think you are better than you really are. Overplacement has to do with relative values—you think your performance is better than that of others when it is not. Overprecision, the focus of this paper, has to do with the degree of certainty with the assessment of judgments and beliefs—you think your judgment is more accurate than it really is. In other words, overprecision relates to the second moment of the distribution, such that a person may hold accurate beliefs on average but underestimate the variance of the possible outcomes (Malmendier and Taylor, 2015).

Overprecision has wide-ranging consequences. From an economic perspective, overprecision has been associated with suboptimal decision-making, such as insufficient insurance purchasing (Grubb, 2015) or distortions in corporate investment decisions (Ben-David et al., 2013; Moore et al., 2015). In financial markets, overprecision contributes to portfolio underdiversification (Goetzmann and Kumar, 2008), excessive trading (Barber and Odean, 2001), and systematic forecasting errors (Deaves et al., 2019). In politics, it fosters ideological extremism, entrenches partisan identification (Ortoleva and Snowberg, 2015a,b; Stone, 2019), and increases susceptibility to misinformation or "fake news" (Thaler, 2023). However, despite these well-documented consequences, most of the existing literature has focused on the effect of overprecision within specific, isolated domains of individual behavior. As a result, it remains unclear whether the observed overprecision in each domain reflects a population-wide phenomenon—where different individuals exhibit overprecision in specific domains but not across them—or whether it is driven by a subset of individuals who consistently display overprecision across multiple contexts.

In this paper, we try to close this gap by demonstrating that overprecision is an overarching cognitive bias affecting multiple dimensions of decision-making. Specifically, we investigate its role in two key domains: financial decision-making and political behavior, introducing the "Subjective Error Method" (SEM) in the German Socio-Economic Panel Innovation Sample (SOEP-IS), a nationally representative panel of the German population. The SEM is a simplified version of the MISS approach to measuring overprecision by Soll et al. (2022) and involves two steps: first, respondents answer a numerical question, such as the year of Saddam Hussein's capture or the height of the Eiffel Tower. Second, they estimate the "distance" (in the units relevant to the question) between their initial answer and the correct one. In other words, respondents report the *absolute error* they expect to make (henceforth, subjective error) relative to their initial response. By comparing respondents' realized errors to their subjective errors, we obtain a measure of overprecision in a simple and direct way.

The richness of our data allows us to correlate respondents' overprecision with their sociodemographic characteristics. The results show that overprecision is positively correlated with narcissism and negatively correlated with age, years of education, gross income, and financial literacy, but does not differ across genders. More importantly, we document several overprecise behaviors that align with theoretical predictions regarding political and financial behavior. Specifically, overprecision is positively correlated with greater forecasting errors in respondents' stock price predictions and lower portfolio diversification, as suggested by Odean (1998) and Barber and Odean (2000). In terms of political views and behavior, overprecision is associated with a greater tendency to hold extreme political ideologies, as suggested by Ortoleva and Snowberg (2015b). However, in contrast to Ortoleva and Snowberg (2015b), overprecision is associated with voting abstention rather than an increased likelihood to vote.

The alignment of our results with the theoretical predictions in the financial and political domains provides strong support for the SEM as a reliable measure of overprecision and suggests that overprecision is a pervasive cognitive trait shaping multiple dimensions of human behavior. To confirm this overarching influence of overprecision, we conduct a two-stage analysis testing whether overprecise behavior in one domain (e.g., finance) is associated with overprecise behavior in another (e.g., politics). We find that overprecision-related underdiversification and forecasting errors are systematically correlated with overprecision-related political extremism and voter abstention, and vice versa. This finding bridges both strands of the literature and provides compelling evidence that overprecision is an underlying behavioral trait manifesting across multiple domains.

Additionally, we conduct two preregistered companion surveys administered to two representative samples of the German population to assess the robustness of our overprecision measure. The first survey elicits the SEM across five knowledge domains: contemporary history, general knowledge, economics, four-week-ahead stock price predictions, and a neutral counting task.[1] Using various statistical methods, we show that individuals consistently display overprecise behavior across different knowledge domains and that overprecision is stable within individuals across questions. This pattern suggests that overprecision is not domain-specific phenomenon, but a more general cognitive trait. The second survey compares respondents' subjective errors to their elicited probability distributions over all answers to the factual question. The results show that the SEM is not only consistent within subjects across questions, but that reported subjective errors are strongly correlated with the variance of respondents' probability distributions (i.e., their stated uncertainty), providing further validation that the SEM is a reliable measure of individual-level confidence miscalibration.

Overall, our paper contributes to the existing literature on overprecision in four ways. First, we implement the SEM in a large, nationally representative sample. While most of the existing literature on overprecision relies on university students (e.g., Alpert and Raiffa, 1982) or specialized subject pools (e.g, Glaser and Weber (2007) use finance professionals and McKenzie et al. (2008) IT professionals), our study utilizes a rich dataset comprising a representative sample of the German population. This makes our results more generalizable and allows us to examine the socioeconomic determinants of overprecision. Second, we confirm several theoretical predictions regarding respondents' financial and political behavior—two of the most consequential domains of decision-making. Third, we show that overprecision-related behavior in the financial domain is associated with overprecision-related behavior in the political domain, and vice versa. To our knowledge,

[1] These domains differ in the ability to ascertain the true answer. In contemporary history, general knowledge, and economics, a correct answer exists at the time of the survey and may be known depending on the respondents' prior knowledge. The neutral counting task also has a correct answer, but respondents can only estimate it. In contrast, for stock price predictions, the correct answer remains unknown on the survey date. Since these last two topics are independent of previous knowledge, we can test whether overprecision is robust across domains where the influence of prior knowledge is limited.

this is the first study to bridge the literature on overprecision in finance and politics, empirically demonstrating its cross-domain effects. Finally, through two additional online surveys, we show that overprecision, as measured through the SEM, is stable within individuals, robust across domains, and correlated with respondents' uncertainty levels. Taken together, our findings provide strong empirical evidence that overprecision is a pervasive cognitive trait shaping multiple aspects of individuals' lives that can be easily and effectively measured using the SEM.

The remainder of this paper is organized as follows: Section 2 discusses the notion of overprecision (section 2.1 and introduces the Subjective Error Method (section 2.2). Section 3 details the SOEP-IS data set and correlates overprecision with its various sociodemographic determinants. Section 4 connects respondents' overprecise behavior in the financial and political domains, demonstrating that the same variation in behavior explains overprecise behavior in both domains. Section 5 tests the robustness of overprecision within individuals and across domains. Section 6 concludes.

2 Overprecision Measurement and the Subjective Error Method

2.1 Overprecision Measurements

Overprecision, also known as miscalibration, is a form of overconfidence resulting from excessive confidence in one's own judgment (Moore et al., 2015). It relates to the second moment of the belief distribution and, therefore, directly affects information processing. For this reason, overprecision is widely used in finance and political science to model overconfident agents. For example, Odean (1998) finds that overconfident traders engage in excessive trading and hold underdiversified portfolios because they believe their private signals are more precise than they actually are. Scheinkman and Xiong (2003) combine a short-sale constraint with overprecise traders to explain the formation of asset market bubbles.[2] In the political science literature, Ortoleva and Snowberg (2015b) show that more overprecise individuals tend to vote more, hold more extreme political views, and exhibit stronger partisan identification. Consistent with this, Stone (2019) suggests that overprecision increases partisanship through excessively strong inferences drawn from bi-

[2] For a longer discussion on the different models of overprecision used in the finance literature, see Daniel and Hirshleifer (2015).

ased information sources. More recently, the literature has begun to study the role of overprecision in the dissemination of fake news (Pennycook et al., 2021; Thaler, 2023).

However, precisely because overprecision deals with the second moment of the belief distribution, it is difficult to measure (Moore et al., 2015). The most common way to measure it is to elicit the respondents' 90% confidence intervals (CI) for a series of numerical questions such as "How long is the Nile River?" (e.g., Alpert and Raiffa, 1982). Using this paradigm, a perfectly calibrated respondent would fail to capture the correct answer within the CI in one out of every ten questions. However, the literature has shown that this method is problematic as respondents are unfamiliar with CIs and do not fully grasp what is being asked (Moore et al., 2015). This confusion is corroborated by the finding that the elicited intervals resulting from asking 90% CIs are practically identical to those resulting from asking for 50% CIs (Teigen and Jørgensen, 2005), and that often the purported 90% CIs only contain the correct answer only 30% to 60% of the time (e.g., Russo and Schoemaker, 1992; Bazerman and Moore, 2013; Moore et al., 2015).

While there are some alternatives to CIs for measuring overprecision, these tend to be time-consuming or limited in the information they provide. For example, the two-alternative forced-choice (2AFC) method asks respondents to choose between two possible answers to a question and to indicate their level of confidence that the answer is correct. By comparing the number of correct answers to the stated confidence, one can measure whether, on average, respondents are overconfident. However, this method has several drawbacks as it cannot distinguish between overprecision and overestimation of one's own judgment (Moore et al., 2015), and it cannot capture continuous distributions (see Griffin and Brenner (2004) or Moore et al. (2015) for a further discussion of the 2AFC method and its statistical limitations). Another approach to measuring overprecision is the Subjective Probability Interval Estimates (SPIES) method by Haran et al. (2010), which elicits full probability distributions from respondents. Although the SPIES method may measure overprecision more accurately than CIs (Moore et al., 2015), it is time-consuming and requires respondents to understand the concept of probability distributions. Additionally, since distributions can only be elicited by partitioning the support into discrete bins, researchers must make *ad hoc* decisions to implement and define the desired 90% boundaries of the distribution.

More recently, Ortoleva and Snowberg (2015b) and Chapman et al. (2023) employ a parametric estimation method by regressing a six-point scale self-reported measure of confidence in the accuracy of the answer on a polynomial of the realized error. The drawback of this approach is that the measure of accuracy is quite coarse due to its limited six-point scale. Moreover, the individual measure of overprecision for each person depends on the relationship between confidence and accuracy across the entire population of respondents, which might result in the incorrect classification of subjects as over- or underprecise.[3] Soll et al. (2022) examine how expert advice of varying quality affects subjects' uncertainty and point estimates. Because they are interested in subjects' prediction confidence, Soll et al. (2022) introduce the MISS Method, a three-step procedure for measuring overprecision. The first step involves explaining and describing to subjects the concept of MISS—namely, the concept of absolute expected error.[4] In the second step, subjects make a point prediction for a numerical question (e.g., what year was John Lennon born?). Finally, in the third step, subjects are asked to estimate their MISS for the previous question.[5] By comparing the reported MISS to the realized true error, the authors develop a measure of confidence for each subject.

While the MISS measure of overconfidence is similar to that of the SEM (more thoroughly defined in Section 2.2), the introduction of the abstract concept of MISS in the first step and its use in eliciting subjects' expected absolute error make the MISS method cognitively taxing and may confuse respondents (see Footnotes 4 and 5). Indeed, as reported by the authors, subjects struggled to understand the MISS method and instructions even after being provided with numerical examples. As an alternative, Soll et al. (2022) recommend following what they call the "Within C Method". In this method, subjects are asked to report the probability that the point estimate for a numerical question is within C units

[3] See the discussion of the *Residual method* (op'''''') in Appendix C.1 for an in depth comparison of the methods including the example of Figure C.3.

[4] For example, in a task where subjects had to guess the number of calories in a dish, in the first step of the MISS method, the instructions read: "In addition to making a best guess [of calories in the picture], you will also estimate the accuracy of your guess. Accuracy is simply the distance to the truth, which we call the MISS. The MISS represents the amount by which your guess missed the truth. For example, suppose that Robin guesses the calories in a slice of cake. The slice has 600 calories, but Robin does not know this. If Robin guesses 350, then Robin's MISS is 250. If Robin guesses 1100, then the MISS is 500. The lower the MISS, the more accurate the guess. Remember, MISS is the amount by which you missed the correct answer, either too high or too low. If you guess exactly right the MISS is zero."

[5] Specifically, subjects are asked: "What is your best estimate of the MISS for the guess you just gave?". Notice that this question requires subjects to recall the definition of MISS which was given in the first step.

of the truth. This allows the authors to back out subjects' expected absolute error and degree of confidence. However, this assessment can only be done by imposing a normal distribution on subjects' errors, which may not always be appropriate (Soll and Klayman, 2004). Moreover, in the "Within C Method", subjects' levels of overconfidence are highly sensible to C, which can result in the same person being classified as overconfident or underconfident depending on the presented value of C.

To address the caveats listed above, we propose the Subjective Error Method. This method is like the MISS method but omits the first step, resulting in a procedure that does not necessarily rely on subjects' knowledge of statistical concepts, requires no parametric assumptions, is easy to understand, quick and easy to implement, and does not depend on the properties of the entire sample.

2.2 The Subjective Error Method

The Subjective Error Method consists of asking respondents two consecutive questions. The first question (a) can be on any topic but needs to have a numerical answer.[6] The second question (b) asks respondents how far away they expect their answer to question (a) to be from the true answer. In other words, the second question asks respondents to report their absolute subjective error. An example would be:

(a) *How long (in kilometers) is the Nile River?*

(b) *How far away (in kilometers) do you think your answer to (a) is from the true answer?*

By comparing the *subjective error* of respondents stated in (b) to the absolute *realized error* from question (a), we get a measure of how over- or underprecise a respondent is about their judgment.

More formally, assume that a respondent's true error is normally distributed, with mean 0 and variance σ^2 as shown by the solid curve in Figure 1. A perfectly calibrated individual would, on average, accurately assess the distribution of the true error when

[6] Some examples of numerical questions are the result of multiplying 385 by 67, the length of the Nile River, or the year of Lady Diana's death. Some examples of questions that do not work are the name of the oldest son of Lady Diana, the color of the Batmobile, or the gender of the current prime minister of the United Kingdom.

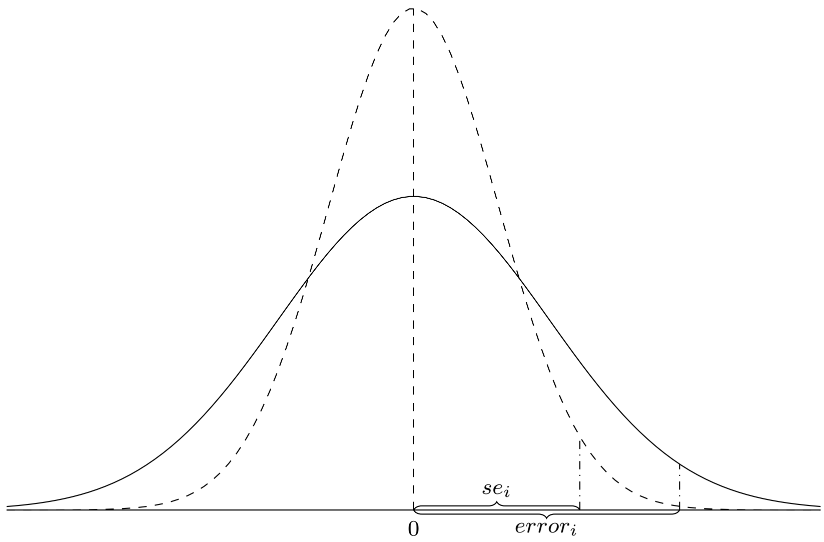


Figure 1: Two hypothetical normal distributions of the (subjective) error. The solid curve shows the true distribution of the error with a standard deviation of 2 (precision of .25). The dashed curve shows the perceived distribution by an overprecise respondent with a standard deviation of 1.25 (precision of .64). The dash-dotted vertical lines indicate the subjective error se_i and the absolute true error $error_i$ resulting with the same cumulative probability.

answering questions using the SEM. However, for most respondents, the perceived distribution may not align with the true distribution. If a respondent is overprecise, their perceived variance of the error $\hat{\sigma}^2$ is smaller than the true variance; that is, the precision $\rho = 1/\hat{\sigma}^2$ is larger, as illustrated by the dashed curve in Figure 1. In this case, the subjective error would, on average, systematically deviate from the true error in all questions.[7]

Denote the answer of respondent i to question j as $a_{i,j}$, their subjective error for question j as $se_{i,j}$, and the true answer to the question as ta_j, then our measure of overprecision for respondent i for question j is:

$$error_{i,j} = |a_{i,j} - ta_j|, \tag{1}$$

$$op_{i,j} = error_{i,j} - se_{i,j}, \tag{2}$$

where equation (1) measures the absolute realized error ($error_{i,j}$) of respondent i to question j. Importantly, we measure the *absolute error* because we care about the magnitude

[7]Note that the difference between the true error and the subjective error that would realize with the same cumulative probability is directly proportional to the difference in the precision of the underlying distributions.

of the error but not its direction. In equation (2), we calculate the difference between the subjective error ($se_{i,j}$) and the realized error ($error_{i,j}$) for respondents i to question j. In this case, the direction of the error is important—a respondent who underestimates their subjective error (i.e., $error_{i,j} > se_{i,j}$) is considered *overprecise*, while a respondent who overestimates their subjective error (i.e., $error_{i,j} < se_{i,j}$) is considered *underprecise*. Finally, respondents who accurately guess their subjective error (i.e., $error_{i,j} = se_{i,j}$) are considered perfectly calibrated for that question.

2.3 Subjective Error Method – Discussion

The Subjective Error Method elicits overprecision by asking respondents for their best numerical estimate and their perceived distance from the true answer. This raises a fundamental epistemological question: why would individuals ever report anything other than zero for the subjective error if they are offering their best guess to the first question (question (a))? Within a strictly rational Bayesian framework, the best estimate should fully reflect the individual's beliefs, leaving no anticipated deviation from the truth. Empirically, however, respondents rarely report complete certainty, even in domains where they have expertise (e.g., Yeung and Summerfield, 2012).

This discrepancy can be understood by recognizing that people are not pure Bayesian decision-makers. Instead, they are aware—through metacognitive processes and accumulated experience—that their "best guess" does not imply certainty and that there is often a gap between what they consider plausible and how confident they really are (e.g., Yeung and Summerfield, 2012). Consider the question, "How long is the Nile River?", most respondents will find it implausible that their best answer will be exactly correct.[8] In fact, for questions about future events, such as stock prices or electoral results, a correct answer does not yet exist, further heightening respondents' awareness of uncertainty.

Importantly, this metacognitive "error-awareness" does not necessarily arise from formal probability calculations or the recall of precise knowledge. Instead, it is likely shaped by heuristics, past experience, and contextual cues that guide individuals in estimating both their best guess and its probable imprecision. The SEM imposes no specific epistemological stance. While some respondents may implicitly estimate their error based

[8]One might even question whether a single "correct answer" to the length of the River Nile exists. For a single exact answer to exist, the question should be more precise, such as, "What is the length of the Nile River stated in the 1993 edition of the Encyclopedia Britannica?"

on past experiences with similar questions, others may rely on intuitive heuristics or domain-specific impressions. However, our method does not assume that respondents can explicitly recall past errors or possess formal statistical knowledge about their uncertainty; rather, it accommodates a range of inferential strategies that feel natural to respondents.[9] This diversity of inferential strategies makes the SEM highly flexible and applicable across diverse populations, providing a direct channel for capturing the extent to which individuals under- or overestimate the errors inherent in their own judgment.

Regarding the SEM elicitation, two potential concerns arise about how respondents might interpret the second question (question (b)). The first is that respondents might misinterpret the question and report their "expected error," which should be zero if they provided their best estimate in the first question. However, the data suggest otherwise; across all questions, only 9.95% of the respondents who answered the first question incorrectly state a subjective error of zero. If respondents were truly reporting their "expected error," we would expect a much higher proportion of individuals reporting zero, regardless of whether their initial response was correct. In addition, in a preregistered companion survey that elicits both subjective errors and full probability distributions over possible answers, we find that respondents' subjective errors are strongly correlated with the variance of their elicited probability distributions, confirming that the SEM effectively captures belief uncertainty (Appendix I). The second concern is that respondents may interpret the subjective error differently, depending on the confidence intervals they have in mind when answering their subjective errors. Using the data of the same companion survey, we show that respondents are largely consistent in their interpretation of the subjective error, which is symmetrically distributed around the modal confidence level of 50% (Appendix I).

Another consideration is whether the SEM might mechanically classify individuals with lower cognitive ability as more overprecise simply because they tend to make larger factual errors. However, cognitive ability and overprecision are conceptually distinct. Cognitive ability influences individuals' capacity to acquire information and solve problems. Overprecision, by contrast, reflects the misalignment between perceived and actual accu-

[9] Empirical validation of this flexibility is provided in our companion survey (Appendix H.1). While subjective and realized errors decrease with self-reported expertise, their decline is proportional, leaving overprecision unchanged. This pattern shows that the SEM robustly captures miscalibration across respondents and domains, confirming its flexibility in accommodating diverse inferential strategies.

racy. Thus, although cognitive ability improves response accuracy, overprecision captures the misalignment between the actual accuracy *and* confidence in one's judgment. To test whether education mechanically drives our overprecision measure, we regress subjective error on realized error and its interaction with education. The results show that more educated individuals adjust their subjective errors slightly more as realized errors increase, but the magnitude of this adjustment is modest, and even the most educated respondents remain substantially overprecise (Table B.1 in Appendix B). This finding is supported by the view that greater knowledge does not eliminate overprecision, because improvements in accuracy are offset by parallel reductions in reported uncertainty. This "neutral" effect has been reported in the literature (e.g., Önkal et al., 2003; McKenzie et al., 2008) and is corroborated by a second preregistered companion survey, which shows that our measure of overprecision is unaffected by topic-specific expertise (Appendix H.1). Additional robustness checks using our standardized overprecision measure, which accounts for variation in question difficulty, further confirm that the observed associations are not artifacts of realized error magnitude or education (Appendix C.1). We therefore conclude that our overprecision measure captures genuine confidence miscalibration rather than reflecting differences in cognitive ability.

Finally, in a related paper, Enke and Graeber (2023) study the "subjective uncertainty about the optimal action" of experimental subjects when confronted with choices across different economic domains. Although conceptually related to overprecision, cognitive uncertainty (as the authors call it) refers to individuals' awareness of their limitations in processing complex information or selecting the appropriate decision framework. This uncertainty fosters cautious and moderate behavior, as individuals are reluctant to draw firm conclusions when faced with ambiguity. In contrast, overprecision arises from a failure to properly account for uncertainty in one's judgment, resulting in overly narrow confidence intervals or extreme probability assessments. While distinct, overprecision and cognitive uncertainty can operate together. For example, in financial markets, an investor may acknowledge market volatility (reflecting cognitive uncertainty) but still make overly confident predictions about the performance of a specific stock (reflecting overprecision). In other words, cognitive uncertainty reflects individuals' awareness of complexity and their own informational limitations, but it does not directly explain why they form overly narrow confidence intervals or make extreme judgments under uncertainty. Overprecision fills this gap by identifying a specific miscalibration in confidence that amplifies biased decision-making behavior.

As is clear from the setup, the primary advantage of eliciting overprecision through the Subjective Error Method is its conceptual and practical simplicity. Unlike most existing approaches, the SEM does not necessarily require respondents to understand statistical constructs such as confidence intervals or probability distributions. This makes it easy to explain, quick to administer, and, crucially, easy to understand. As such, it is particularly appropriate for large-scale, demographically diverse samples. Moreover, by comparing the reported and realized errors on a question-by-question basis, our method offers an individualized measure of miscalibration that avoids the need for arbitrary reference classes or relying on population-level error distributions. Additionally, the SEM is unaffected by knowledge, as any reduction of errors in the respective first question is likely offset by a symmetric reduction of the subjective error. Furthermore, because it accommodates a range of cognitive strategies, the SEM is a versatile tool for measuring overprecision across different domains, tasks, and survey formats.

3 Individual Overprecision and its Determinants

In this section, we describe the dataset used to study the effects of overprecision on the financial and political behavior of a German sample (section 3.1), explain how we create our measure of overprecision using the Subjective Error Method (section 3.2), and report the correlation between our measure of overprecision and the sociodemographic characteristics of our sample (Section 3.3).

3.1 Data

We introduced a module in the 2018 wave of the German Socio-Economic Panel Innovation Sample (SOEP-IS). The Innovation Sample is a companion panel of the larger SOEP-Core and is designed to host and test novel survey items (see, Richter and Schupp, 2015). As the SOEP-Core, the SOEP-IS is a high-quality survey with all interviews conducted face-to-face by a professional interviewer.

Our module consists of seven contemporary history questions. In each question, we ask respondents: (a) the year in which a specific event occurred, and (b) the distance (in years) between their answer to (a) and the correct answer to the question.[10] The questions are designed to vary in difficulty, with content ranging from the year in which the Volkswagen Beetle was introduced (1938) to the year in which the US Army captured Saddam Hussein (2003) (see Table B.2 and Table B.3 in the appendix for all questions and their correct answers in English and German, respectively). The module was presented to the respondents who joined the panel in 2016 ($N = 902$). We supplement the data with additional information on personal characteristics from the survey years 2016–2018. We drop 55 respondents who did not answer any of the overprecision questions and 42 respondents with incomplete information. Our final sample consists of 805 respondents.[11]

3.2 Individual Overprecision

In Figure 2, we plot the density of answer $a_{i,j}$ for each question j. The red vertical line marks the correct answer. The spread of the densities indicates that some questions were easier for respondents than others. In Figure 3, we plot the realized error ($error_{i,j}$) on the vertical axis and subjective error ($se_{i,j}$) on the horizontal axis for each of the seven questions. Additionally, we plot a 45-degree red line; any dot above represents a respondent who is overprecise ($error_{i,j} > se_{i,j}$) in their answer, while any point below corresponds to a respondent who is underprecise ($error_{i,j} < se_{i,j}$). It is clear from the figure that respondents are, on average, overprecise in their answers across all questions, regardless of their difficulty.

We construct a measure of overprecision for each question ($op_{i,j}$) following the outline in Section 2.2. To measure the within-respondent consistency of overprecision across the seven questions, we use congeneric reliability, commonly referred to as coefficient omega (e.g., Cho, 2016). Congeneric reliability represents the proportion of variation (variance

[10]The questions are formulated in German. For example, when we ask about the year of Lady Diana's death, we ask: (a) *In welchem Jahr starb Lady Diana, die erste Frau von Prinz Charles?* and then (b) *Was schätzen Sie, wie viele Jahre Ihre Antwort von der richtigen Antwort entfernt ist?*.

[11]To test whether our estimation sample is still representative of the German population, we compare the unweighted means of personal characteristics in our sample with the weighted means according to the sampling weights in the larger SOEP-Core, which is representative of the German population. The results in Table B.5 in the appendix show that our subsample is still broadly representative of the larger SOEP-Core, with only some significant but small and nonmeaningful differences. When applying the sampling weights to our estimation sample, the differences disappear.

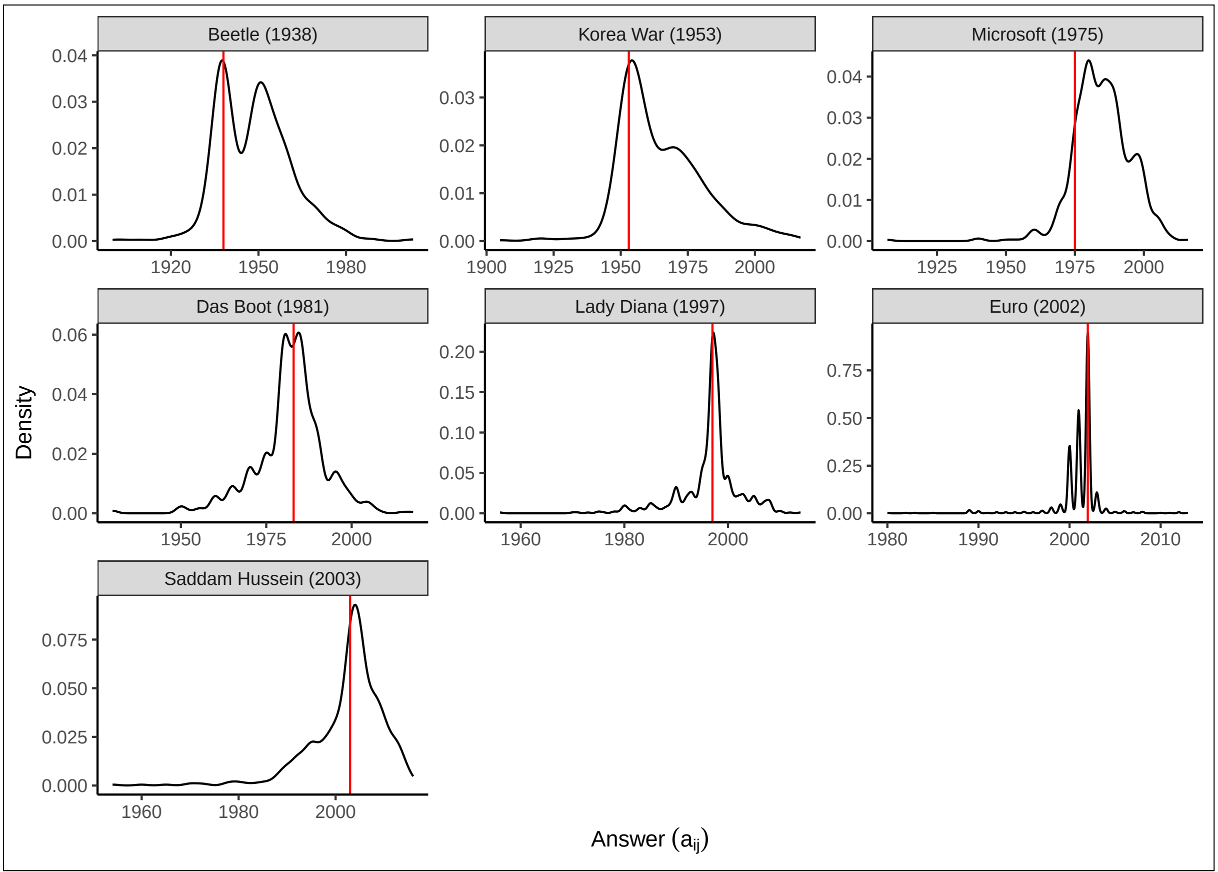


Figure 2: Density of the answers ($a_{i,j}$) for each question. The red vertical line marks the correct answer. Note that the vertical axis is different for each question.

and covariance) among a set of variables that can be explained by an unobserved factor.[12] The results indicate that 49% of the variation among the seven items can be explained by a common factor, which we interpret as overprecision.

To create a unique measure of overprecision for each respondent i (op_i), we take the average overprecision across all seven questions j.[13] We plot the density of op_i in Figure 4a,

[12]Consider a model in which each observed outcome i of item j can be expressed as $T_{i,j} = \mu_j + \lambda_j F_i + e_{i,j}$, where $T_{i,j}$ is the i^{th} outcome of item j, μ_j is a constant term, $e_{i,j}$ is the individual score error, and λ_j is the factor loading on the latent common factor F. To construct congeneric reliability, we estimate the factor loadings, $\hat{\lambda_j}$, for the overprecision measure of each question with respect to one common factor. We interpret this common factor as overprecision. Congeneric reliability is calculated using the formula $\frac{(\sum \hat{\lambda}_j)^2}{(\sum \hat{\lambda}_j)^2 + \sum \hat{\sigma}^2_{e_j}}$, where $\hat{\sigma}^2_{e_j}$ represents the estimated variance of the error. This is a generalized version of Cronbach's alpha (Cronbach, 1951) and measures the proportion of variation among the set of items j that can be explained by the latent factor. While congeneric reliability allows for different factor loadings of the latent common factor, Cronbach's alpha assumes that the latent factorloads equally on all items and, thus, serves as a lower bound for reliability. For the case of τ-equivalence, i.e., $\lambda_j = \lambda_k \; \forall k$, all factor loadings are equal and both measures coincide.

[13]Given that a principal component analysis of the seven items yields a strong first factor, an alternative

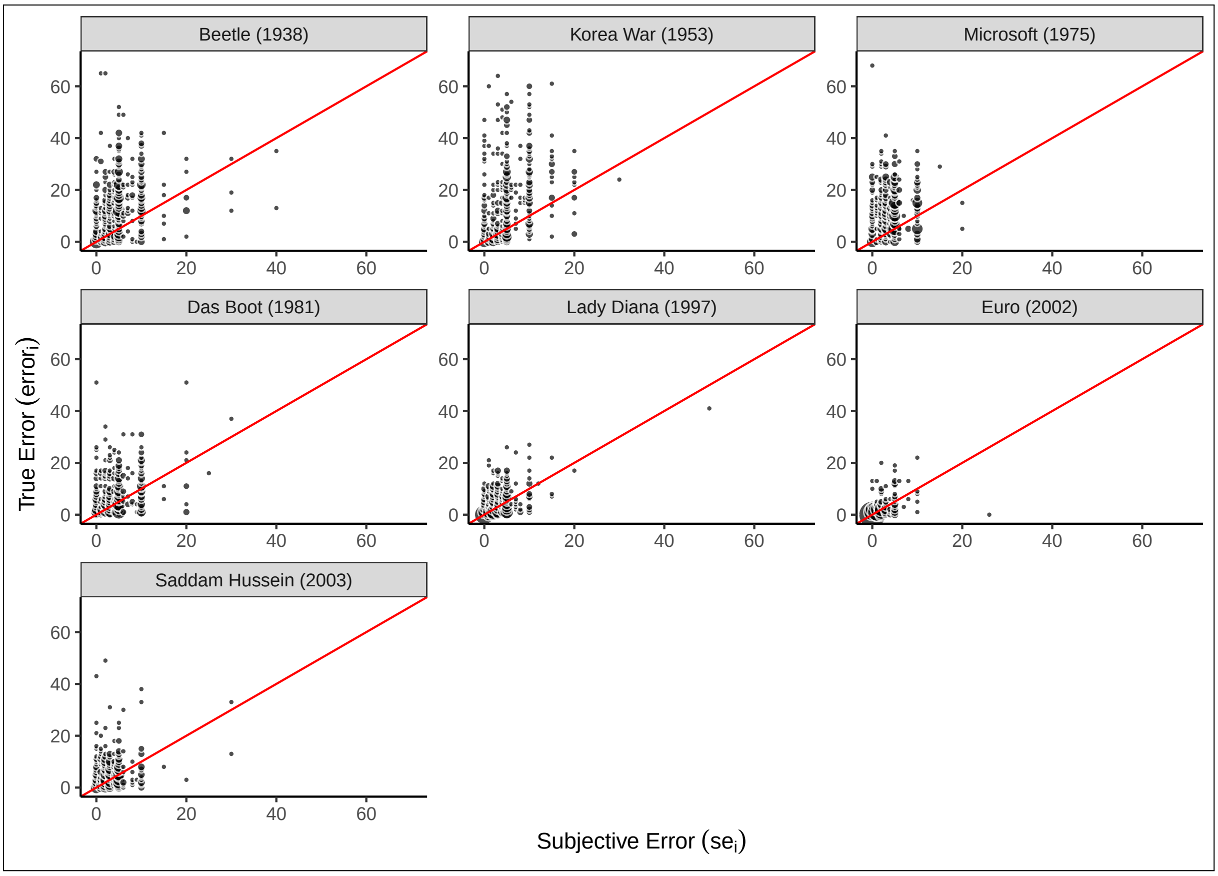
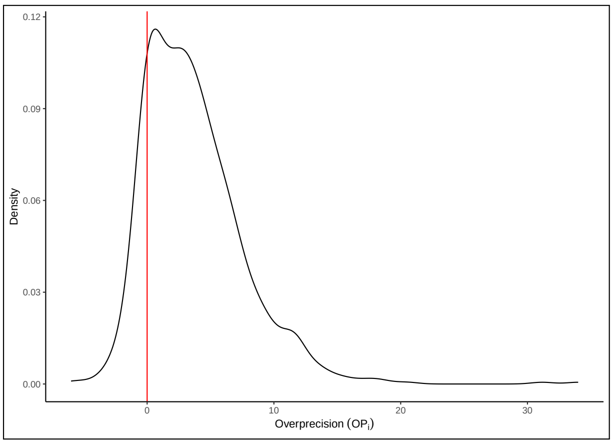


Figure 3: Relation between the realized error ($error_{i,j}$) in the vertical axis and the subjective error ($se_{i,j}$) in the horizontal axis. Any dot above (below) the 45-degree red line is an overprecise (underprecise) answer by the respondent.

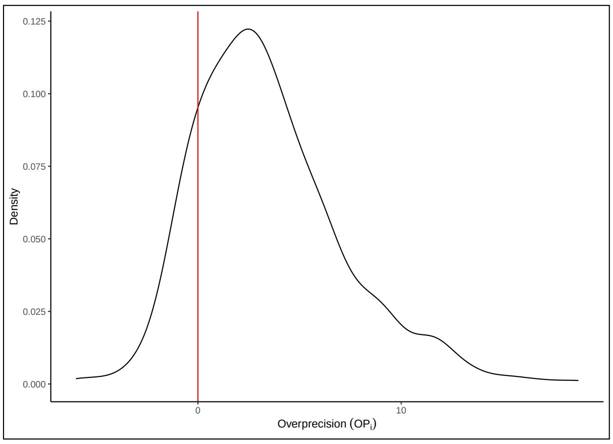
which, in line with Figure 3, shows that the great majority of respondents (82%) are overprecise. Interestingly, and in contrast to most of the literature using CIs to measure overprecision, we find a relatively large number of respondents that are underprecise (approximately 11%).

Moreover, 7% of the respondents appear to be perfectly calibrated in the aggregate measure, as indicated by the vertical red line in Figure 4a. Of these respondents, 83% are perfectly calibrated across all the questions they answer. However, note that in the SOEP-IS, respondents can choose not to answer a question. In total, 51% of the respondents answered all questions, while 5% answered only one (see Figure A.1 in the appendix for

would be to construct the composite measure op_i using the principal component as in Ortoleva and Snowberg (2015b). The composite overprecision measure resulting from using the first component is very similar to using the simple average ($\rho^{Pearson} = .88; \rho^{Spearman} = .84; N = 805$). Since the results only marginally change when using the principal component, we do not report the results. Additionally, to address concerns about different scales, we also construct a standardized measure of overprecision by standardizing each measure of overprecision ($op_{i,j}$) before aggregating, as detailed in Appendix C.



(a) Density of op_i for all respondents (N=805)



(b) Density of op_i for subset (N=410)

Figure 4: Density of Overprecision (op_i). In the left panel we plot the density of op_i, which is the average overprecision for each respondent i across all questions j. In the right panel we plot the density of op_i only for those respondents who answered all questions in the survey.

a detailed breakdown). Among the perfectly calibrated respondents, 40% answered only one question, and just 12% answered all seven. This suggests that what we observe in Figure 4a represents an "upper bound" of perfectly calibrated respondents. In Figure 4b, we plot the density function for the subset of respondents who answered all the questions. As shown, respondents are substantially less calibrated, with the mode of op_i shifting to the right, leaving only 1% of the respondents perfectly calibrated; meanwhile, there is an increase in the proportion of underprecise respondents (15%).

For ease of interpretation in the subsequent analysis, we standardize the aggregate score (op_i) to have mean zero and standard deviation one, and refer to it as sop_i. In Appendix C.1 we we discuss and compare our measure of overprecision to five alternative approaches. These are: i) a *standardized* measure, which standardizes each question before aggregating; ii) a *centered* measure, which centers the errors and subjective errors around their mean, allowing us to disentangle the second moment of the distribution (overprecision) from its first moment; iii) a *relative* approach, which takes into account the relative distance between the subjective error and the realized error; iv) an *age-robust* measure, constructed using only those questions concerning events that occurred after the respondent was born; and v) a *residual* approach following the regression methodology of Ortoleva and Snowberg (2015b).

3.3 Sociodemographic Determinants of Overprecision

In this section, we assess how overprecision is correlated with individual sociodemographic characteristics of the individual. We do so by focusing on sociodemographic characteristics that are either related to understanding the task at hand or that have been examined in the literature in the context of overconfidence in general.[14] This section is exploratory and intends to inform us about potential sociodemographic controls that need to be added in the main section of this paper.

In Table 1, we regress sop_i on a series of sociodemographic variables using five different OLS models. In all models, we control for age, gender, and years of education. In Column (2) we add the number of overprecision questions answered. In Column (3), we add the monthly gross individual income (*gross income*) measured in thousands of euros[15] as well as dummies for labor force status (e.g., employed, unemployed, maternity leave, etc.) and a dummy for those respondents who were living in East Germany in 1989. In Column (4), we add further personal characteristics: financial literacy, risk preferences, impulsivity, patience, and narcissism. Finally, in Column (5), we add federal state (*Bundesland*) and month-of-interview fixed effects.[16]

The results show a negative correlation between overprecision and age, education, or income. For example, an increase in the gross income of 2,000 euros is associated with a reduction in overprecision by almost one-tenth of a standard deviation, and every 2 years of education are associated with a reduction in overprecision by about one-tenth of a standard deviation. Furthermore, overprecision is negatively correlated with our measure of financial literacy and positively correlated with our measure of narcissism. It is also important to note that the number of questions answered by respondents, which we include in Column (2), is not random, with overprecision increasing as subjects answer more questions (see Figures A.2 and A.1 in the appendix for a graphical overview of these results).[17] In all subsequent analyses, we use the above-mentioned variables as controls.

[14]See, e.g., Deaves et al. (2009) for gender, Campbell et al. (2004) for narcissism and risk attitudes, Visser et al. (2019) for impulsivity.

[15]Since *gross income* is only available for employed individuals, we code missing variables as 0 and include a dummy that is 1 for missing observations.

[16]Note that we only report coefficients that are statistically significant in Table 1. We report all coefficients in Table B.6 in the appendix.

[17]While controlling for the number of answered questions helps mitigate biases related to systematic

Dependent Variable: *sop*	(1)	(2)	(3)	(4)	(5)
age	-0.008***	-0.008***	-0.007**	-0.005	-0.005
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
female	0.074	0.121	0.094	0.134*	0.111
	(0.070)	(0.074)	(0.074)	(0.074)	(0.076)
years education	-0.053***	-0.062***	-0.050***	-0.043***	-0.037***
	(0.013)	(0.013)	(0.014)	(0.013)	(0.014)
answered		0.064***	0.067***	0.086***	0.085***
		(0.023)	(0.022)	(0.023)	(0.023)
gross income			-0.049**	-0.037*	-0.039*
			(0.021)	(0.020)	(0.021)
fin. literacy				-0.481***	-0.398**
				(0.170)	(0.174)
narcissism				0.107**	0.100**
				(0.042)	(0.043)
N	805	805	805	805	805
adj. R^2	0.036	0.046	0.061	0.081	0.098
Constant Term	Yes	Yes	Yes	Yes	Yes
Employment & GDR 1989	No	No	Yes	Yes	Yes
Personal characteristics	No	No	No	Yes	Yes
Fixed Effects	No	No	No	No	Yes

Robust standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1: Determinants of overprecision. In Columns (1)-(5) we run an OLS with *sop* as the dependent variable. In all Columns, we include age, gender, and education. In Column (2) we add the number of answered questions. In Column (3) we include gross income and dummies for labor force status (employed, unemployed, retired, maternity leave, not working), and whether the respondent was a citizen of the GDR before 1989. In Column (4) we include further personal characteristics: financial literacy, risk preferences, impulsivity, patience, and narcissism. In Column (5) we also include fixed effects for the federal state (*Bundesland*) where the respondents live and the time at which they responded to the questionnaire.

The results in Table 1 largely align with the existing literature. For example, the positive correlation with narcissism aligns with previous results (e.g., Campbell et al., 2004; Hamurcu and Hamurcu, 2021). Similarly, we observe a negative correlation between education and overprecision, which resonates with previous results reporting a negative link

differences in response behavior, it does not fully eliminate concerns about selection effects. Non-response may still be linked to unobserved characteristics such as cognitive ability, motivation, or numerical reasoning skills. However, alternative strategies, such as excluding individuals who skip questions, would introduce selection bias by disproportionately removing those with lower confidence or greater uncertainty. By including the number of answered questions as a control, we retain the full sample and account for individual differences in confidence and response behavior.

between cognitive ability and overprecision (e.g., Duttle, 2016). The effect of gender on overprecision is far from universal in the literature. While Ortoleva and Snowberg (2015b) find that females are significantly less overprecise than males, López-Pérez et al. (2021), Deaves et al. (2009), and Wohleber and Matthews (2016) find no effect. This last result is supported by Bandiera et al. (2022), whose meta-analysis shows that both men and women are overconfident, with no significant gender differences, which aligns with our results. Concerning age, although there is some evidence suggesting that overprecision increases with age (Prims and Moore, 2017; Ortoleva and Snowberg, 2015b), we find a slight negative correlation. However, our result is likely driven by the question domain, as we also find a positive correlation between age and overprecision when controlling for age in the construction of the overprecision measure in Appendix C. Finally, the literature directly linking overprecision and financial literacy is scarce. Most existing studies do not study overprecision but other types of overconfidence. For instance, Xiao et al. (2022) find that individuals who overestimate their financial literacy are more susceptible to investment fraud, while Hamurcu and Hamurcu (2021) examine how narcissistic tendencies predict overplacement in a financial literacy quiz. Merkle (2017) analyzes investors' financial overestimation and overplacement, examining its evolution over time and shows that these biases persist rather than dissipate with experience. Kramer (2016) reports that overestimation of financial knowledge is negatively correlated with seeking financial advice, whereas objective measures of financial literacy show no such relationship. Inghelbrecht and Tedde (2024) show that overestimation of financial knowledge predicts more frequent trading, but find no evidence that it leads to systematically worse investment performance. Although none directly measure the impact of financial literacy on overprecision, they suggest a negative relationship between overconfidence, financial literacy, and incorrect financial decision-making, which aligns with our findings.

4 Overprecision and Individual Financial and Political Behavior

This section reports the main results of our paper as we examine how overprecision correlates with respondent behavior in the political and financial domains (Section 4.2) and show that the same variation in behavior that is related to overprecise behavior in the financial domain is also associated with overprecise behavior in the political domain, and *vice versa* (Section 4.3).

4.1 Methodology

To test the predictions from the theoretical literature on overprecision, which we elaborate on in more detail below, we employ three distinct procedures. First, we run a regression of each outcome (y_i) on our measure of overprecision and a vector of control variables of the form:

$$y_i = \alpha + \beta sop_i + \boldsymbol{\gamma' X_i} + \epsilon_i, \tag{3}$$

where sop_i denotes the standardized overprecision measure for respondent i, $\boldsymbol{X_i}$ is a vector of control variables, and ϵ_i is the random error term. We include all possible control variables available in the SOEP-IS that might be correlated with the outcome we are studying or with overprecision based on the previous literature. These include age, gender, years of education (used as a proxy for cognitive ability), monthly gross labor income, dummy variables for the labor force status (employed, unemployed, maternity-leave, non-working, and retired), measures of impulsivity, patience, narcissism, financial literacy, and risk preferences, a dummy variable for having lived in the German Democratic Republic in 1989, the number of overprecision questions answered by each respondent, state fixed effects, and interview date (month and year) fixed effects. The latter absorbs any variation in the outcome driven by the time the survey was run, e.g., in the development of the asset prices. Additionally, we include a measure of political interest in the political analyses.[18] We estimate (3) using OLS and present the point estimate of the coefficient on the standardized overprecision measure sop_i from the full regression and its unadjusted p-value respectively in Columns (1) and (2) of Table 2.[19] Since we test the behavior of respondents across several dimensions, we also report the Sidak-Holm adjusted p-values to correct for multiple hypothesis testing (MHT) in Column (3).

Second, we derive the "R^2 rank" of our standardized measure of overprecision sop_i using a forward stepwise regression, similar to Cobb-Clark et al. (2022). This is obtained by running a stepwise regression in which variables are added to the model sequentially. In step 1, we regress the behavior of interest on each of the K control variables in the

[18] In Table B.10 in the appendix, we also include the Big Five personality traits (Rammstedt and John, 2007). These are only available from the 2017 SOEP-IS, and because not all respondents in our sample responded to them, we lose 55 observations. Yet, the results remain robust to the inclusion of the Big Five personality traits.

[19] Adjusting the degrees of freedom by the number of questions used to construct the measure of overprecision does not significantly affect the results.

specification separately. Of these K regressions, we pick the control variable with the highest R^2, as it is the variable that can explain most of the observed variation in the data. In step 2, we regress $K-1$ times the behavior of interest on the control variable selected in the first step, plus each of the $K-1$ remaining controls. This is continued until all K variables have been added to the model. The resulting R^2 rank is determined by the step at which each control variable is added to the model. Therefore, the better the "R^2 rank" of sop_i, the more the variable can explain the variation in the outcome, i.e., rank 1 delivers the highest R^2. We report the results in Column (4) of Table 2, along with the maximum number of variables included in the model, as specified above.[20]

Third, we employ a least absolute shrinkage and selection operator (LASSO) to test whether our overprecision measure has predictive power for the outcome variable in an out-of-sample prediction. LASSO is a machine learning technique frequently applied to improve the predictive power of statistical models. The objective of the LASSO approach is to choose those variables with the highest predictive power from the set *of all possible control variables*. It does so by estimating a penalized regression by minimizing the sum of squared residuals along with a penalty term for the sum of the coefficients.[21] This is implemented via cross-validation, wherein the estimator partitions the data into different folds for training and testing, selecting the penalty term that minimizes the out-of-sample prediction error in the testing data.[22] If sop_i is included in the model, this indicates that it has predictive power for the outcome. We report the results in Column (5) of Table 2, along with the number of control variables selected by LASSO and the resulting R^2 of the model in Column (6). In Column (7), we report the number of observations, which may vary due to missing data in the outcome variables.[23]

[20]Note that each binary representation of a possible realization of the fixed effects could be included as an independent variable. This leads to 41 and 42 as the maximum number of variables to be included.

[21]Formally $\min_\beta \frac{1}{2N}\sum_{i=1}^{N}(y_i-\alpha-\sum_j \beta_j x_{ij})^2+\lambda\sum_j|\beta_j|$ for the linear case, where j are the coefficients which are included in the model and λ is a given tuning parameter. See Tibshirani (1996) for more details.

[22]The algorithm proceeds step-wise and estimates the model for each λ starting at the smallest λ that delivers zero non-zero coefficients and ending at a λ of 0.00005 in a grid of 100. Each step could add or remove a different number of variables from the model.

[23]A test of the means of personal characteristics for the estimation samples and the entire sample (N=805) shows no significant differences. The only exception is a slightly higher share of male respondents in the stock market regressions. Therefore, we consider the estimation samples to be representative of the entire sample (N=805).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) LASSO R^2	(7) N
Financial Behavior:							
DAX Error	0.083*	0.056	0.108	4/41	yes/14	0.10	548
1-year ahead	*0.529*	*0.308*		*7/41*	*yes/21*	*0.16*	*578*
2-year ahead	*3.078***	*0.016*		*4/41*	*yes/8*	*0.04*	*557*
Diversification	-0.131***	<0.001	0.002	3/41	yes/19	0.14	774
Political Behavior:							
Extremism	0.087**	0.041	0.117	6/42	yes/14	0.05	716
Left-Right	-0.008	0.854	0.854	18/42	no/14	0.08	716
Abstention	0.032**	0.011	0.041	3/42	yes/18	0.14	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Estimation results of Section 4. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the R^2 rank in the forward stepwise regression as specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as the dependent variable also include a self-reported political interest measure. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)). Variable definitions are in Table B.4 in the appendix.

4.2 Prediction Results

The results of our three analytical approaches are summarized in Table 2. We first discuss financial behavior outcomes and then outcomes regarding political behavior. For each outcome, we present the theoretical predictions from the literature. To get a sense of the relative effect sizes of the various regressors, the full regression results for column (1) can be found in Table B.7 of Appendix B. In addition, as robustness tests, Table B.8 of Appendix B presents the results of running a Seemingly Unrelated Regression (SURE) (Zellner, 1962),[24] and in Appendix C we test the robustness of our results using the five alternative approaches mentioned in Section 3.2. The results are principally replicated in all cases.

Additionally, in Appendix D, we conduct a specification curve analysis (Simonsohn

[24] We use SURE as a robustness check since some of the outcome variables are potentially correlated. Estimating them jointly via SURE increases efficiency and accounts for cross-equation error correlations. To do so, we estimate two models: One with all outcomes in Table 2 except the one- and two-year-ahead regressions (due to perfect linearity with the DAX forecast error by construction), and another that omits all DAX forecast error outcomes to minimize the loss of observations.

et al., 2020) to evaluate the robustness of our OLS estimates. This analysis systematically explores all combinations of control variables and alternative definitions of overprecision, offering a comprehensive assessment of the sensitivity of our results to different modeling choices. The results show that our key findings are robust, as the SEM consistently correlates with overprecise financial and political behavior across most specifications. This shows that the associations reported in Column (1) of Table 2 are not artifacts of specific modeling assumptions, but reflect a robust and meaningful relationship between overprecision and behavior across domains.

Financial Behavior Outcomes

The first hypothesis addresses forecast errors of asset price predictions in the stock market. Benos (1998) and Odean (1998) argue that overprecise investors hold incorrect beliefs about future asset valuations due to overweighing their private signals when forming expectations. Direct empirical support for the association between overprecision and forecast errors in financial markets is provided by Deaves et al. (2019), who correlate the predictions of German stock market forecasters with a measure of overprecision. Furthermore, Hilary and Menzly (2006) provide evidence supporting this association among North American analysts.[25] Therefore, we expect overprecise respondents to be less accurate in their predictions. To test this prediction, we use the absolute distance between the one- and two-year-ahead predictions of the German Stock Index (DAX) from the realized value.[26] Since individual forecast errors might be prone to random noise, we aggregate both errors using the principal component, which we standardize to be mean zero and standard deviation one (*DAX Error*).[27] Additionally, we report the results for each forecast error separately (*one-year-ahead* and *two-year-ahead*).[28]

[25]However, note that, unlike our method, Hilary and Menzly (2006) and Deaves et al. (2019) rely on indirect proxies to construct their measure of overprecision.

[26]We use the closing price based on the interview day to calculate exact forecast errors for each respondent.

[27]The principal component analysis shows a strong first factor with an eigenvalue of 1.65. All other factors are below the commonly used Kaiser criterion of 1.0.

[28]We include a dummy variable that indicates asset ownership as a control variable in the predictions to account for different information sets in a robustness test in Table B.11 in the appendix. The qualitative results remain unaffected by this change, although the sample size decreases.

The results in Table 2 show that our measure of overprecision is positively correlated with forecast errors in asset prices. An increase in overprecision of 1 standard deviation is associated with a 0.083 standard deviation increase in the principal component of forecast errors. The LASSO estimation results reveal that overprecision is also a strong predictor of these forecast errors, as it is selected as an explanatory variable in the models of stock market forecasts and ranks fourth in the forward stepwise regression approach.

Next, we test the theoretical prediction by Odean (1998) that overprecision is associated with underdiversified portfolios. Overprecise investors overweigh their private information, leading them to trade excessively and to concentrate on a limited number of assets. Goetzmann and Kumar (2008) provide empirical evidence supporting this prediction for traders in the US and Merkle (2017) does so for traders in the UK. While the former study relies on an asset turnover proxy, the latter directly elicits overprecision through survey questions. While both studies rely on revealed portfolio data, we test this prediction using self-reported hypothetical allocations. Specifically, we construct a standardized diversification measure (mean zero, standard deviation one) based on participants' allocation choices across five asset categories: stocks, real estate, government bonds, savings, and gold.[29,30]

Our results confirm the theoretical prediction that overprecision is associated with underdiversification. The point estimate in column (1) in Table 2 shows that a 1 standard deviation increase in overprecision is associated with a 0.131 standard deviation decrease in our diversification measure. That is, the optimal portfolio of overprecise respondents is skewed towards certain asset categories. Moreover, overprecision is included among the variables in the LASSO model and ranks third in the forward stepwise regression

[29]The measure assumes that the "optimal" portfolio diversifies evenly across the different investment options (See Table B.4 in the appendix for a details). However, one could argue that each dollar invested in stock is more diversified than its equivalent investment in gold. To test this idea, we created two different "optimal" portfolio benchmarks; in the first one, maximum diversification would be achieved by investing 35% in stocks, 35% in bonds, 10% in gold, 10% in real estate, and 10% in the savings account. The second and more extreme "optimal" portfolio invests 45% in stocks, 45% in bonds, 4% in savings, 3% in gold, and 3% in real estate. The results are robust, with both cases showing a statistically significant and negative effect of overprecision on diversification. We thank an anonymous referee for this suggestion.

[30]The measure of diversification displays an inverse-U relation with risk aversion. More risk-averse respondents skew their portfolio toward safer assets such as savings and gold. In contrast, more risk-loving respondents skew their portfolio toward riskier assets such as stocks and real estate. This lends credibility to the diversification measure. However, note that we do not make any claim on the optimal degree of portfolio diversification. Rather, we test whether, conditional on the individual degree of risk aversion, overprecision is associated with a tendency towards a certain asset category.

approach.

Political Views and Voting Behavior

According to Ortoleva and Snowberg (2015b), overprecision leads individuals to believe that their own experiences are more informative about politics than they really are. For instance, overprecise people may consult biased media outlets without fully accounting for this bias, or exchange information on social media without recognizing that much of it comes from politically like-minded peers. Against this background, the authors theoretically and empirically demonstrate that overprecision leads to ideological extremism and strengthens the identification with political parties, increasing the likelihood of voting. Yet, the literature remains inconclusive on whether these associations hold for liberals and conservatives alike. While Ortoleva and Snowberg (2015b) find that conservatives seem more susceptible to overprecision than liberals, in a different paper Ortoleva and Snowberg (2015a) show that this association holds only in election years while Moore and Swift (2011) only find a weak and non-significant correlation between overprecision and conservatism.

To test whether overprecision correlates with the political preferences of respondents, we use their self-reported ideology on a scale from 0 (extreme left) to 10 (extreme right) to construct the variable *left-right*. Using the same question, we also construct *Extremism*, which measures how far respondents perceive themselves to be from the political center on a scale from 0 to 5. We standardize both variables to be mean zero and standard deviation one. Finally, to study whether overprecise respondents are more likely to vote, we use a dummy variable that equals 1 if a respondent indicated being a nonvoter in the (ex-post) opinion poll (*Sonntagsfrage*) for the 2017 federal elections to the German *Bundestag* (*Abstention*).[31]

Confirming Ortoleva and Snowberg (2015b), the results in Table 2 suggest that overprecision is positively correlated with ideological extremism. Overprecision is among the variables chosen by the LASSO estimation and ranks high (sixth) in the forward stepwise regression approach. In line with Ortoleva and Snowberg (2015a), we do not find evidence that overprecision is more strongly associated with one side of the political spectrum, as it is not correlated with political ideology and is not selected by the LASSO estimation.

[31] The year in which the survey took place (2018) is a non-election year.

Furthermore, overprecision ranks quite low (18/42) in the R^2 rank approach. Finally, we find that overprecision is a strong predictor of voting abstention, as the LASSO estimation selects it and ranks third in the R^2 rank approach. As shown in Table 2, an increase of one standard deviation in overprecision is accompanied by a 3 percentage point increase in the likelihood of not voting.

The last result appears to contradict the result of Ortoleva and Snowberg (2015b). However, comparing the voting behavior of overprecise respondents in the United States and Europe is challenging. In Ortoleva and Snowberg (2015b), partisanship is measured within the Republican and Democratic parties. Because both of these parties have a high likelihood of winning the elections, those who strongly identify with them have greater incentives to vote for them (e.g., Miller and Conover, 2015). In contrast, in Germany, more extreme respondents gravitate towards fringe parties (e.g., Die Linke, AfD, NPD)[32] with smaller chances of winning elections, so the incentives to vote differ significantly from those in the dataset used by Ortoleva and Snowberg (2015b).[33] Therefore, the theoretical assumptions underlying the predictions made by Ortoleva and Snowberg (2015b) regarding voter turnout and overprecision accurately describe voting behavior in the two-party system of the United States but are less applicable to the more fragmented German political environment.

4.3 Linking Overprecise Behavior Across the Financial and Political Domains

The previous results show that overprecision correlates with respondents' financial and political behavior. However, an important question remains: does a single underlying cognitive bias relate to overprecise behavior across the financial and political domains, or do distinct aspects of overprecision independently shape financial and political decisions?

Identifying whether overprecision stems from a common bias or domain-specific biases has significant theoretical and practical implications. Theoretically, if overprecision reflects a stable underlying domain-general bias, this would support the view that fundamental cognitive processes, such as metacognitive monitoring and uncertainty calibration,

[32] If we pool all respondents voting for radical parties (AfD, NPD, and Die Linke) and compare them to the voters of the rest of parties, a nonparametric test confirms the tendency of radical party voters to ideological extremism (Mann-Whitney U p-value<0.001).

[33] Take as an example the explicit (self-imposed) *cordon sanitaire* that all major democratic parties have imposed around the AfD. Angela Merkel's intervention and the series of resignations that followed the Thuringian election in 2019 shows how strongly this *cordon* is enforced.

are robust and generalizable across domains (Koriat, 2007; Yeung and Summerfield, 2012). From a practical perspective, if overprecision stems from a single underlying bias, interventions aimed at recalibrating confidence, such as training programs to better estimate uncertainty or feedback mechanisms that enhance metacognitive awareness, could enhance decision-making simultaneously across financial, political, and other domains simultaneously.

To determine whether overprecise behavior across domains originates from a common cognitive bias or whether distinct facets of overprecision are independently associated with behavior in each domain, we employ a two-stage approach. This method disentangles domain-specific effects from the broader latent bias of overprecision and quantifies its cross-domain influence.

In the first stage, we estimate the coefficient of overprecision by regressing the outcome Y_{ij} of individual i in domain j (i.e., financial or political behavior), on its overprecision (sop_i) and a vector of control variables (X_i) as in eq. 3:

$$Y_{ij} = \alpha_j + \beta_j sop_i + \gamma_j' X_i + \epsilon_{ij}. \tag{4}$$

where α_1 is a constant, β_1 captures the effect of overprecision on financial behavior, and ϵ_{ij} represents the error term. From this regression, we obtain the fitted values $\hat{Y}_{1i}$, which capture the variation in outcomes attributable to overprecision in domain j.

In the second stage, we use the first-stage fitted values $\hat{Y}_{ij}$ as predictors in the models for the remaining outcomes Z_{ik} (where $j \neq k$):[34]

$$Z_{ik} = \delta_k + \phi_{j\neq k} \hat{Y}_{ij} + \tau_k' X_i + \nu_{ik}, \tag{5}$$

where $\phi_{j\neq k}$ quantifies the extent to which overprecision-related variation in one domain's outcomes (e.g., financial) is correlated with variation in another domain's outcomes (e.g., political). The key assumption is that the first-stage fitted values $\hat{Y}_{ij}$ in the second stage primarily reflect the influence of overprecision in domain j. If this is the case, then ϕ provides a meaningful estimate of cross-domain correlation attributable to overprecision.

Importantly, this approach does not assume a strict causal relationship between financial and political overprecision, nor does it impose a mediation framework where financial

[34]For example, if Y_{ij} in the first stage represents Portfolio Diversification, then in the second stage, Z_{ik} include *DAX Error*, *Extremism*, and Voting Abstention.

overprecision serves as the only channel through which overprecision influences political behavior. Instead, our method isolates the component of financial behavior attributable to overprecision and examines whether this component is systematically associated with political behavior, and *vice versa*. This allows us to assess whether overprecision is a domain-general cognitive bias that consistently manifests across decision-making contexts or instead operates independently in financial and political domains.[35]

The results are summarized in Table 3, which presents 12 estimates—two within the financial domain, two in the political domain, and eight across domains.[36] The coefficients show how a one standard deviation increase in overprecision-induced behavior in domain j impacts outcomes in domain k. For instance, consider the coefficient describing the relationship between overprecision in portfolio diversification and political extremism. In this case, the coefficient of -0.667 indicates that a one standard deviation overprecision-induced increase in political extremism is associated with a 0.667 standard deviation reduction in portfolio diversification. The significant correlations across all tested relationships suggest that a common source of overprecision is associated with biased behavior in multiple domains.

To our knowledge, this is the first study to provide evidence overprecision in one domain is systematically associated with overprecision in another. This result is important not only because it bridges the gap between the political and financial behavior literature but also because it underscores the systematic nature of overprecision in decision-making, opening the door to interdisciplinary work on its mitigation.

[35]See Appendix F for a more formal explanation of our statistical model and a discussion of alternative approaches.

[36]Given the hypothesized (and confirmed) null result for ideological inclination, we exclude this outcome from the analysis.

			Dependent variable in			
			Financial Domain		Political Domain	
			DAX Error (1)	Diversification (2)	Extremism (3)	Abstention (4)
Overprecision-induced variation in	Financial Domain	DAX Error		-1.573***	1.050**	0.380**
				(0.436)	(0.512)	(0.148)
		Diversification	-0.636*		-0.667**	-0.242**
			(0.331)		(0.325)	(0.094)
	Political Domain	Extremism	0.831*	-1.407***		0.362**
			(0.501)	(0.420)		(0.141)
		Abstention	2.295*	-3.886***	2.763**	
			(1.384)	(1.159)	(1.347)	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Coefficients from the second-stage regression (eq. 5), which regresses the dependent variable in the respective column separately on each of the predicted values in the respective row, along with a vector of controls. The predicted values in the rows are derived from regressing the variable in the respective row on overprecision and a vector of controls. The first stage corresponds to the results in Table 2. Standard errors in parentheses. Variable definitions are in Table B.4 in the appendix.

5 Overprecision Across Domains

Section 4 shows that by focusing on questions related to 20th-century history, the SEM captures a measure of overprecision that is associated with various aspects of respondents' political and financial behaviors, consistent with theoretical results from the literature. However, previous empirical literature suggests that overconfidence varies depending on the nature of the questions posed (e.g., Klayman et al., 1999). To test the effect of question domains on the results of the SEM, we conduct a preregistered online survey with a representative sample of the German population (#118284 at www.aspredicted.org). In this survey, we evaluate respondents' overprecision across five distinct domains and check whether these measures correlate within subjects and across domains using principal component analysis, partial correlations, and the leave-one-out analysis introduced in Morrison and Taubinsky (2023).[37]

[37]Formally, assume that through the SEM we elicit the prior precision estimates $\tilde{\sigma}_{i,n}$ of individual i across n different domains while the true level of prior precision of individual i is $\sigma_{i,n}$. If the stated prior precision of individual i is a linear transformation of the true prior precision, such that $\tilde{\sigma}_{i,n} = k_{i,n} \times \sigma_{i,n}$, then the parameter $k_{i,n}$ can be interpreted as the degree of overprecision of individual i in domain

5.1 Survey Design

The survey, programmed with Qualtrics, consists of blocks of five questions each within five distinct domains (see the full set of questions in Table B.12 in the appendix). All respondents go through all domains, which are:

1. ***Neutral***: In this domain, respondents are flashed five different 20×20 matrices of black triangles and gray squares for 8 seconds each. After each matrix, respondents are asked to estimate the number of black triangles in each of the five shown matrices and to report their subjective error. This task is similar to that of Bosch-Rosa et al. (2024) and has the advantage of being independent of socioeconomic traits such as wealth or education, thereby avoiding heterogeneity in experience and prior knowledge across respondents. The matrices range from 70 to 330 black triangles. See Figure A.5 in Appendix A for an example with 120 black triangles and 280 gray squares.

2. ***Contemporary history***: In this domain, we ask the five most frequently answered questions in the SOEP-IS survey. The contemporary history domain allows us to compare the results of the online survey with those of the SOEP survey in Appendix E, where we show that respondents' answers are not substantially different across both surveys. It also allows us to replicate the findings on stock price predictions from Section 4.2 in Appendix H.2.

3. ***General knowledge***: In this domain, respondents answer five general knowledge questions and report their subjective error for each. Some examples of questions in this domain include the number of teeth of an adult polar bear, the number of keys on a concert piano, or the number of African countries that are part of the UN. This domain encompasses a broad range of knowledge, covering topics commonly encountered in daily life.

4. ***Future stock prices***: In this domain, respondents are asked to predict the 28-day price forecast of five different assets (Benz, Puma, BMW, Deutsche Post, BASF)

n. However, overprecision might vary across the different domains. In this case, the elicited degree of overprecision can be expressed as $k_{i,n} = \lambda_i + e_{i,n}$, where $e_{i,n}$ represents domain-specific characteristics, such as knowledge in that respective domain. In the companion survey, we we elicit λ_i which allows us to examine the consistency of overprecision across different domains.

and report their subjective error for each. To do so, we employed a widget from *tradingview.com* that allowed us to present respondents with an interactive price chart over the past year for each stock. This domain enables us to test overprecision in the financial context and, importantly, to measure overprecision regarding future events. This distinction is important because, unlike the other domains, respondents in this domain are asked to predict something that *will* happen, rather than something that has a correct answer at the time of questioning.

5. ***Economics***: In this domain, respondents answer five questions related to the German economy and report their subjective error for each one. To maintain consistency in the range of responses, this domain is limited to percentage changes. Examples include the percentage change in the German CPI from 2011 to 2021, the percentage change in the German nominal GDP from 2006 to 2021, or the percentage change in the German DAX from 2014 to 2021. We include this domain due to the importance of overprecision in economic decision-making and its relevance to the financial behavior analyzed in the SOEP survey.

All respondents started with the neutral domain since it began with three practice rounds to familiarize them with the matrices.[38] Afterward this, we randomized the order of the remaining domains. Before each domain, respondents were asked to self-report their knowledge of the topic on a scale from 0 (not knowledgeable at all) to 100 (very knowledgeable).

To ensure data quality, we introduced attention checks throughout the survey. Respondents failing these checks were automatically removed from the survey. To account for the use of *Google* or other search engines, respondents were asked at the end of the survey whether they had used such methods to answer the questions. Additionally, in the domains of contemporary history, general knowledge, and economics, we included an additional question that served as a "*Google* control." These questions were presented in the same format as all others, but are difficult and unlikely to be known by respondents. Therefore, any respondent answering all of them correctly is likely using external help and

[38]The difference between the practice rounds and the main rounds is that in practice rounds, the correct answer is shown to the respondents after they answer the questions, which is not the case in the main rounds.

should be potentially excluded from the sample.[39] To our knowledge, this is the first time that such a method has been implemented, and it could prove valuable for online surveys that ask for publicly available information, particularly when the answers are monetarily incentivized (Haaland et al., 2023; Stantcheva, 2023).

Additionally, we collect sociodemographic variables such as age, gender, years of education, income, and nationality. We also asses mathematical literacy by asking respondents to solve three mathematical problems and have them self-report the amount of effort they put into answering the survey. Table B.14 in Appendix B lists all of the variables collected in the survey.

5.2 Data

We collected 1,000 complete responses. As preregistered, we excluded all respondents who admitted to having used a third party to answer our questions, respondents identified as 'Googlers' through our control questions, and those in the lowest five percentile on the self-reported effort measure. This leaves us with 839 respondents for the baseline analysis. In Table B.15 of Appendix B, we include the summary statistics of the full and cleaned samples. The data-cleaning process does not substantially alter the sample composition with respect to the collected variables.[40]

5.3 Individual Overprecision

In Figure 5, we plot the distributions of the answers across all five domains, similar to Figure 2. For most questions across the five domains, the answers are distributed around the true answer, indicating that respondents were attentive and capable of answering the questions presented. However, the dispersion of the answers and the distance of the mode from the true answer clearly indicate that some questions were easier than others. In

[39]Following our pre-registration, we consider a respondent to have used a search engine if two conditions are met. In any domain: i) answering the *Google* controls correctly and stating a subjective error of zero, and ii) answering correctly and stating zero subjective error for at least three other questions in this domain. If a respondent is flagged as having used search engines, then she is excluded from the analysis. The control questions we used to determine if respondents were using *Google* included: the birth year of Joachim Sauer, Angela Merkel's husband (1949), the upper weight limit in kilograms for the female Olympic boxing Bantamweight class (54kg), and the percent change of M1 in the Euro area from 2015 to 2021 (86%). Importantly, we do not use the answers to these questions to measure respondents' overprecision.

[40]In Appendix G, we use different exclusion restrictions on the sample as specified in the pre-registration and show that our results are robust to the different restrictions.

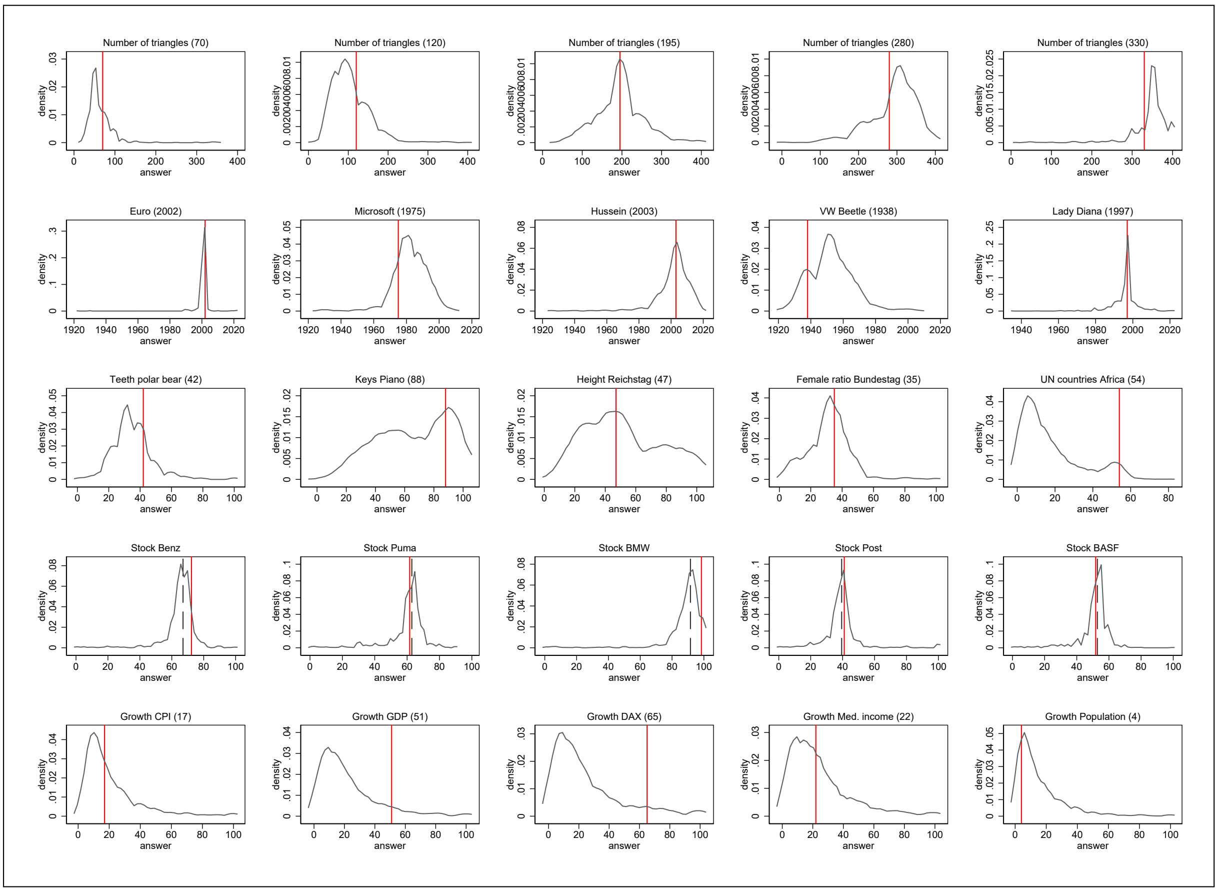


Figure 5: Density of the answers ($a_{i,j}$) for each question. The vertical red line marks the correct answer. For the stocks, the vertical red line marks the weighted average of the 28-day-ahead stock price, and the vertical black dashed line marks the weighted average of the stock price on the day of the interview. Note that the vertical axis scale differs across questions. The corresponding graphs for the additional questions used to detect the use of search engines can be found in Figure A.3 in the appendix.

Figure 6, we plot the realized error ($error_{i,j}$) on the vertical axis against the subjective error ($se_{i,j}$) on the horizontal axis for each of the five questions in each domain, similar to Figure 3. We add a 45-degree line, where any dot above is represents an overprecise observation ($error_{i,j} > se_{i,j}$) and any point below represents an underprecise observation ($error_{i,j} < se_{i,j}$). The results show that respondents are, on average, overprecise across all questions and domains.

5.4 Overprecision Across Domains

For each domain, we construct an aggregate measure of overprecision using the procedure in Section 3.2.[41] As preregistered, to have meaningful estimates and decrease noise, we

[41] Before aggregating, we divide the answers in the neutral domain by four to account for the differing scale. In our pre-registration, we specified that we would use both the simple average and the principal

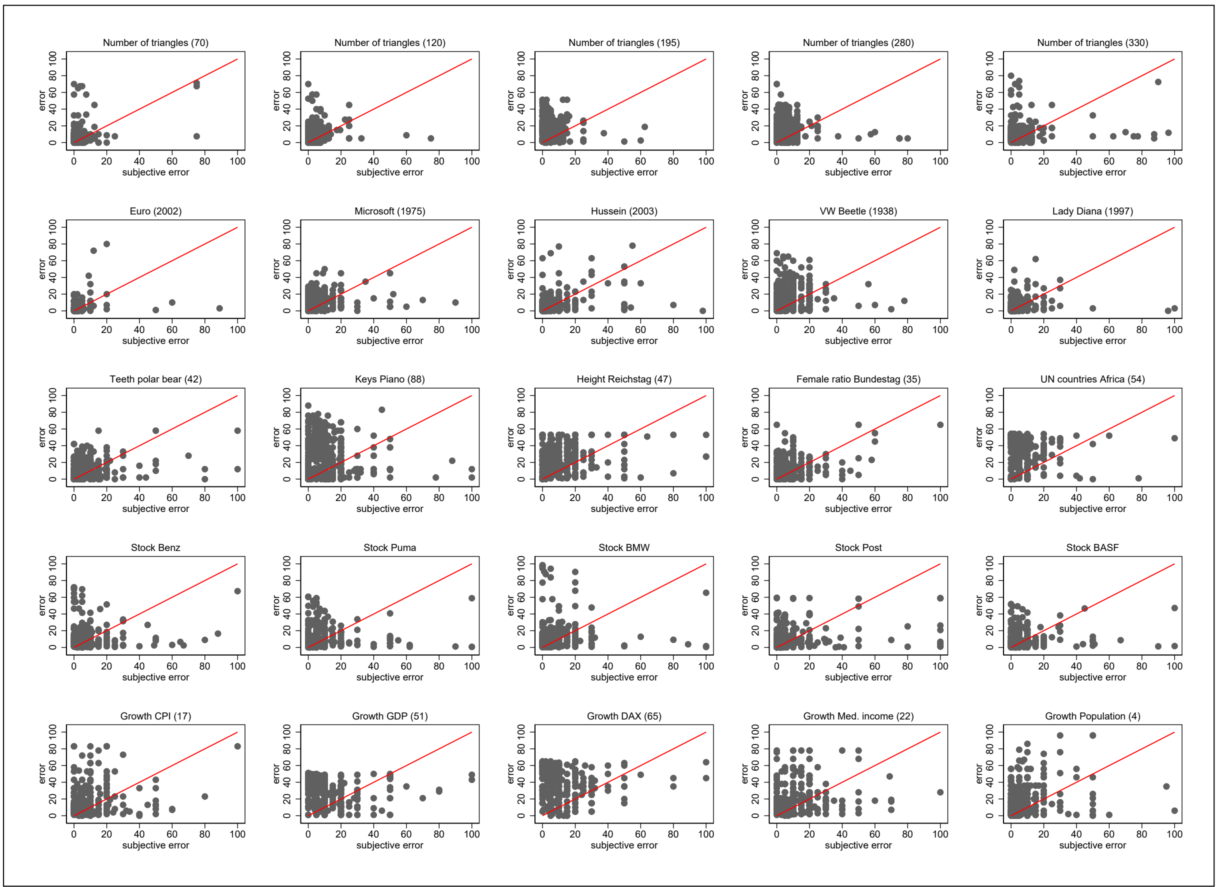


Figure 6: Relation between the realized error ($error_{i,j}$) on the vertical axis and the subjective error ($se_{i,j}$) on the horizontal axis. Any dot above the 45-degree red line is an overprecise answer by the respondent, while any dot below indicates an underprecise one. Note that the results for the neutral domain are divided by four to make the results comparable to the other domains. The corresponding graphs for the additional questions used to detect the use of search engines can be found in Figure A.4 in the appendix.

only construct the aggregate overprecision measure for each domain if the respondent answered at least four out of the five questions. For better comparability, we standardize the aggregate measure for each domain to have a mean of zero and a standard deviation of one, as we did with the SOEP-IS data.[42]

We analyze the relationship among the domain-specific overprecision measures in three ways: i) through a principal component analysis, ii) through a partial correlation analysis, and iii) through a leave-one-out analysis following Morrison and Taubinsky (2023). We confirm the robustness of the results in Appendix G using different exclusion restrictions,

component across all domains for aggregation. However, since the results only marginally change, we do not report the results using the principal component as an aggregation method; these results are available upon request.

[42]Standardizing the aggregate measure does not change the qualitative results.

as specified in the pre-registration.

Principal component analysis: Principal component analysis (PCA) reduces dimensionality by detecting common patterns in variables and generating new latent variables—principal components—that maximize the captured variance and covariance among these variables. All principal components can be ranked by their eigenvalues, with the first principal component capturing the most variation in the data. This first component identifies the most significant underlying pattern in the data and represents the most crucial latent variable in the dataset. Therefore, if it is possible to consolidate all measures of overprecision across domains into a *single* principal component, this suggests that there is one "latent trait" across domains, indicating that overprecision is a domain-free trait. The analysis is based on the 552 respondents, for whom we could calculate an aggregate score in each domain. The results show that only one factor has an eigenvalue exceeding the Kaiser criterion of 1.0. This indicates that only *one* latent variable explains the observed variance and covariance across the five domains. Moreover, the factor loadings across all five domains are positive, indicating that this trait is persistent across the five domains.[43] In other words, the PCA reveals that a single underlying factor can explain the observed variation in the overprecision measures, with all domains contributing positively to it. We interpret this factor as the individual overprecision of respondents.

Partial correlation analysis: Partial correlation analysis allows us to estimate the correlation across two domain-specific overprecision measures keeping the control variables fixed. Table 4 presents the partial correlation coefficients, controlling for age, gender, education, income, nationality, Bundesland, mathematical literacy, and self-reported domain knowledge. The results show a positive correlation of around 0.3 across the different domain-specific overprecision measures, except the Neutral domain, which exhibits only a weak positive correlation with the others.[44] While these are weak correlations, they are in line with previous literature correlating measures of overprecision (e.g., Glaser and

[43] To be more precise, the eigenvalue of this factor is 2.29, with the respective factor loadings are 0.52, 0.76, 0.74, 0.69, and 0.65 for the neutral, contemporary history, general knowledge, stock prices, and economics domains.

[44] If we winsorize the data at the 99% and 1% the average correlation drops slightly with all values getting compressed towards 0.3. This is a common result for cognitive biases since these usually follow skewed distributions. See the winsorized correlation values in Table B.9 of Appendix B.

Domain 1	Domain 2	Correlation	Standard error	N
Neutral	History	0.17	0.03	688
Neutral	Knowledge	0.21	0.04	626
Neutral	Stocks	0.30	0.04	676
Neutral	Economics	0.18	0.04	639
History	Knowledge	0.52	0.04	609
History	Stocks	0.35	0.04	638
History	Economics	0.37	0.05	620
Knowledge	Stocks	0.34	0.04	591
Knowledge	Economics	0.32	0.04	580
Stocks	Economics	0.28	0.04	618

Table 4: Partial correlations between domains. This table shows the partial correlations between the aggregate overprecision measures across the five domains. The control variables are age, gender, education, income, nationality, state, mathematical literacy, and a measure of self-reported knowledge on the topic of the domain.

Weber, 2007) and can be visually confirmed in Figure 7, which presents a binned scatter plot of aggregate overprecision measures across domain pairs.[45]

Leave-one-out analysis: The leave-one-out analysis adopts the methodology described in Morrison and Taubinsky (2023). We first divide the overprecision sample for each domain into two groups: the top 25% (*High* group) and the bottom 75% (*Low* group) based on their overprecision scores. We then pair each domain with all others and use a t-test to check if those participants in *High* show significantly higher overprecision than those in *Low* in all the other domains. The rationale behind this method is that if the SEM captures the same trait across domains, the ranking of overprecision among respondents should be consistent across domains, with those in the *High* group in one domain also being more overprecise than those in the Low group in the other domains. The results are presented in Table 5, clearly showing that the ranking of overprecision is maintained across domains, with more overprecise respondents in one domain also being significantly more overprecise in the other domains.[46]

[45] There is no consensus in the literature for "sufficiently large" correlation values. While Evans (1996) considers that a coefficient of .30 across personality traits is a "weak" correlation, Cohen (1988) considers it a "medium" correlation. In fact, he notes that "Many of the correlation coefficients encountered in behavioral science are of this order of magnitude [0.3], and, indeed, this degree of relationship would be perceptible to the naked eye of a reasonably sensitive observer" (Cohen, 1988, p. 80). In our case, Figure 7 confirms this visual perception.

[46] While Table 5 presents the results for twenty different tests, it is clear that the t-test results are not driven by any sort of multiple hypothesis testing problem, and that any correction for it would result in qualitatively identical results.

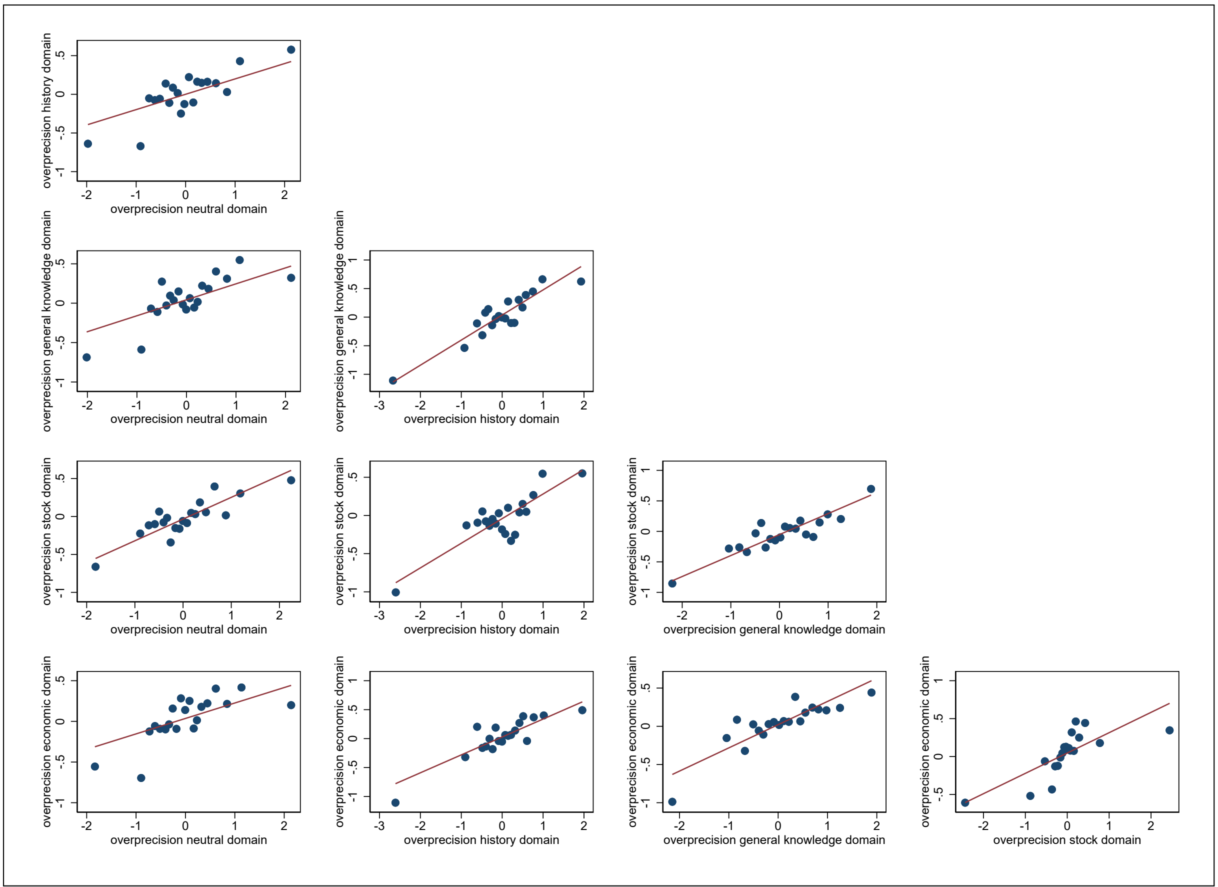


Figure 7: Correlation across domains using binned scatter plots for each pair of overprecision measures. The number of bins is 20 in each plot. The red line is a linear fit between the two dimensions.

The results from our three preregistered analyses indicate a strong relationship of overprecision across all domains. This suggests that the SEM consistently measures overprecision across domains and that overprecision is a stable trait (at a given point in time). Specifically, the results from the leave-one-out analysis suggest that it is possible to measure overprecision in a domain different from that of the outcome of interest. This supports the analysis in Section 3, where we examine the relationship between overprecision measured in the domain of contemporary history and respondents' financial and political behavior.

Sample		Lower 75%			Upper 25%			Difference	
In	Out	mean	sd	N	mean	sd	N	Difference	P-value
Neutral	History	-0.08	1.09	540	0.27	0.71	163	-0.35	< 0.01
Neutral	Knowledge	-0.06	0.88	484	0.35	0.96	149	-0.41	< 0.01
Neutral	Stocks	-0.10	0.92	522	0.24	0.99	168	-0.34	< 0.01
Neutral	Economics	-0.04	0.90	495	0.29	0.69	152	-0.32	< 0.01
History	Neutral	-0.05	0.86	621	0.16	1.23	165	-0.22	0.01
History	Knowledge	-0.09	0.90	489	0.48	0.84	144	-0.57	< 0.01
History	Stocks	-0.11	0.87	536	0.32	1.09	154	-0.43	< 0.01
History	Economics	-0.05	0.90	499	0.34	0.69	148	-0.38	< 0.01
Knowledge	Neutral	-0.05	0.92	629	0.18	1.05	157	-0.23	0.01
Knowledge	History	-0.10	1.06	552	0.35	0.79	151	-0.45	< 0.01
Knowledge	Stocks	-0.09	0.88	541	0.24	1.12	149	-0.32	< 0.01
Knowledge	Economics	-0.03	0.91	499	0.27	0.67	148	-0.30	< 0.01
Stocks	Neutral	-0.10	0.88	621	0.35	1.12	165	-0.45	< 0.01
Stocks	History	-0.09	1.05	551	0.33	0.81	152	-0.42	< 0.01
Stocks	Knowledge	-0.04	0.90	499	0.33	0.90	134	-0.37	< 0.01
Stocks	Economics	-0.05	0.87	495	0.32	0.78	152	-0.37	< 0.01
Economics	Neutral	-0.07	0.93	634	0.27	0.99	152	-0.34	< 0.01
Economics	History	-0.08	1.06	561	0.31	0.77	142	-0.38	< 0.01
Economics	Knowledge	-0.02	0.89	500	0.27	0.97	133	-0.29	< 0.01
Economics	Stocks	-0.10	0.91	542	0.28	1.01	148	-0.38	< 0.01

Table 5: Results of the leave-one-out analysis. *In sample* signifies the domain in which the quartiles based on overprecision were computed. The sample is partitioned into the lower 75% and upper 25% groups. *Out sample* signifies the domain in which overprecision is then measured and tested between both groups.

6 Conclusion

We study how overprecision relates to behavior, focusing on two particularly important domains: politics and finance. To do so, we implement the Subjective Error Method in the 2018 wave of the Innovation Sample of the German Socio-Economic Panel (SOEP-IS). The SEM is a variation of the MISS method (Soll et al., 2022) for measuring overprecision that, in contrast to most other methods, is intuitive for respondents and quick to implement.

The results show that our measure of overprecision aligns with several theoretical predictions across diverse domains. Specifically, overprecision correlates with larger forecasting errors in stock price predictions (Odean, 1998), reduced portfolio diversification (Barber and Odean, 2000), and greater political extremism (Ortoleva and Snowberg, 2015a). While prior research has studied these overprecision-driven behaviors in isolation, our results show that overprecise-driven behavior in the financial domain correlates with overprecise behavior in the political domain. In other words, individuals who exhibit overprecise behavior in finance also exhibit it in politics, and *vice versa*. By bridging the gap between these two domains, we show that overprecision is not merely a domain-specific trait, but rather an underlying aspect of decision-making that systematically shapes behavior across diverse contexts.

We also conduct two additional preregistered surveys. The first spans five different knowledge domains and we show that our measure of overprecision is valid across domains. In the second, we demonstrate that the subjective errors reported in the SEM strongly correlate with the variance in respondents' probability distributions, confirming that the SEM captures respondents' uncertainty around their beliefs. All of these findings validate the SEM as a simple yet reliable measure of overprecision, demonstrating its applicability across multiple knowledge domains and its suitability for large-scale surveys with heterogeneous respondents.

Overall, our work contributes to the literature on overprecision by demonstrating that overprecision is an underlying trait influencing various aspects of people's behavior. By linking overprecision across domains, our study advances our conceptual understanding of overprecision and highlights its role as a potential general cognitive bias rather than a context-dependent phenomenon. These findings also have practical implications: recognizing that overprecision operates as a domain-general bias suggests that interventions designed to mitigate overprecision in one domain—such as financial education programs or media literacy initiatives—may spillover on other decision-making contexts. This opens new avenues for behavioral interventions, educational efforts, and policy applications aimed at reducing the negative consequences of overprecision. However, our findings have certain limitations, as they are based on a cross-sectional snapshot, capturing only a single point in time, and do not establish causality between the Subjective Error Method and overprecise behavior. We look forward to future research addressing these questions to advance our understanding of overprecision and its impact on the behavior of the general population.

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A Extra Figures

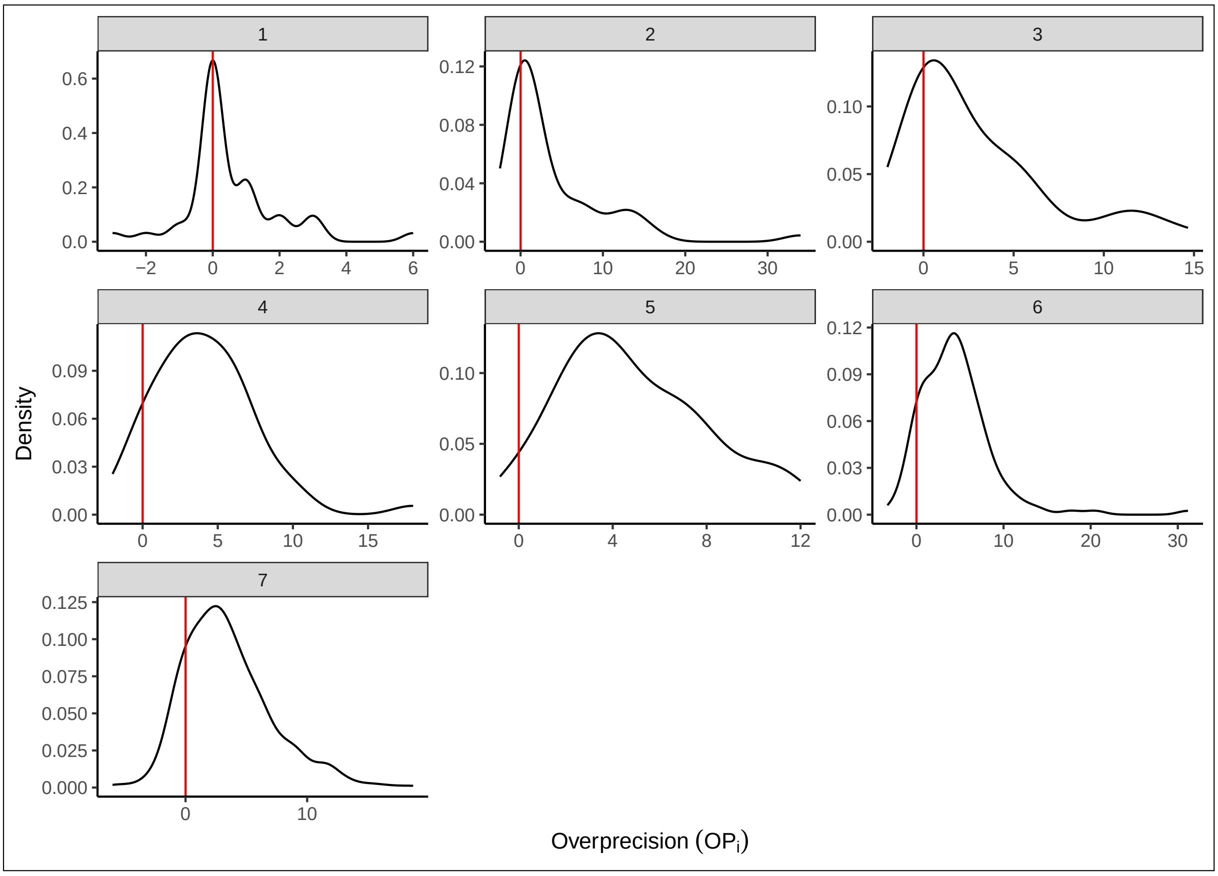


Figure A.1: Density of overprecision (op_i) for each of the subsets of questions answered by SOEP-IS respondents. We plot from left to right the densities of op_i for those respondents who answered from the minimum number of answers (1) to the maximum number of answers (7). In the title, we report the number of respondents for each density. Notice that the scale of the Y-axis changes across panels.

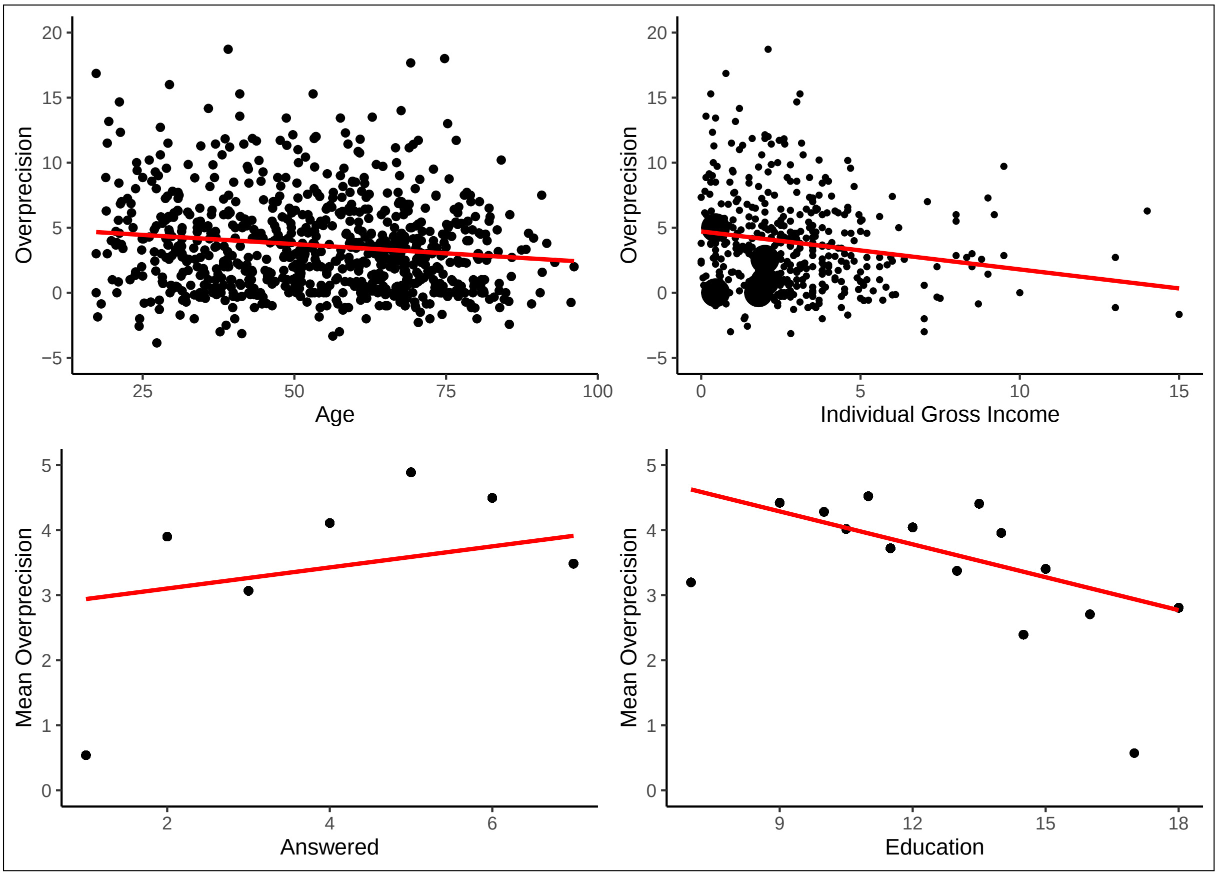


Figure A.2: Correlation of overprecision for SOEP-IS respondents. In the vertical axis of each panel, we plot the overprecision (upper row) and mean overprecision across all groups which we plot in the horizontal axis (lower row). In all four cases, the red line is the fitted linear regression. We dropped one individual outlier in all cases to make the graphs more readable.

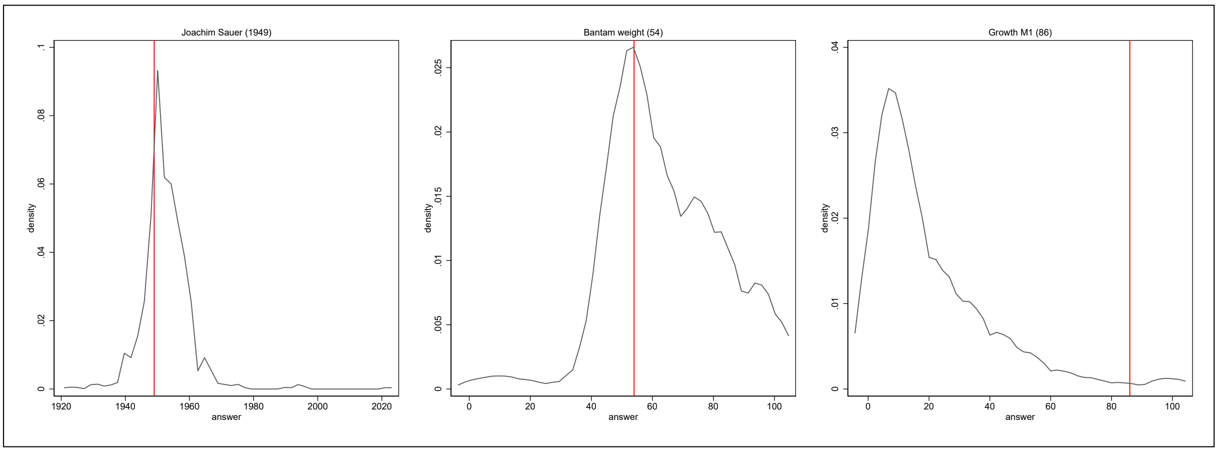


Figure A.3: Density of the answers $(a_{i,j})$ for each of the questions in the companion survey, which we use to detect respondents who we assume to have used search engines. The vertical line marks the correct answer. Note that the vertical axis is different for each question.

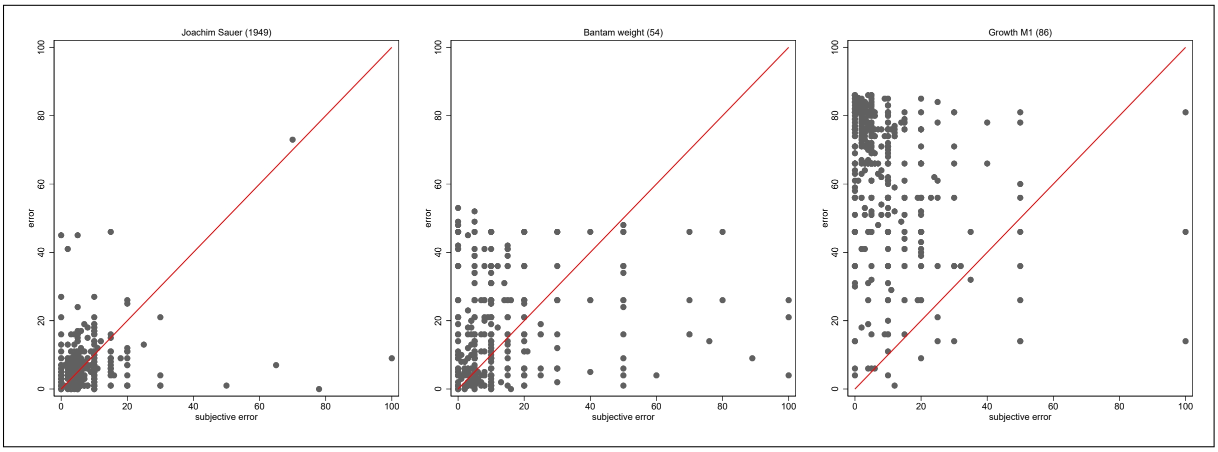


Figure A.4: Relation between the realized error ($error_{i,j}$) in the vertical axis and the subjective error ($se_{i,j}$) in the horizontal axis for each of the questions which we use to detect respondents who we assume to have used search engines to respond the companion survey. Any dot above (below) the 45-degree red line is an overprecise (underprecise) answer by the respondent.

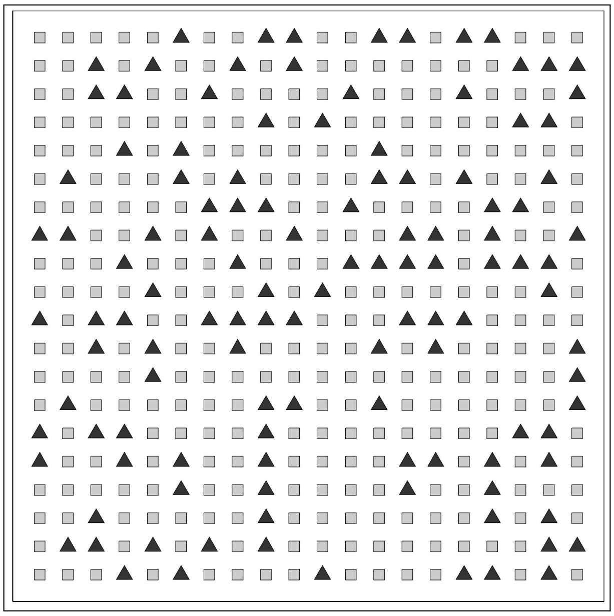


Figure A.5: Matrix example. Example of one of the matrices use in the neutral domain of the online survey. The matrix contains 120 black triangles and 280 gray squares.

B Extra Tables

	Subjective Error				
	(1) OLS	(2) Quantile	(3) Interaction	(4) Lower Third	(5) Upper Third
true error	0.154***	0.165***	0.041	0.127***	0.179***
	(0.010)	(0.005)	(0.050)	(0.017)	(0.016)
education	-0.009	-0.007	-0.073**	0.181	-0.046
	(0.030)	(0.020)	(0.031)	(0.216)	(0.059)
education×*true error*			0.009**		
			(0.004)		
N	4562	4562	4562	1223	1950
Controls and Fixed Effects	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.1: Effect of education and the interaction of education and true error on subjective error. The dependent variable in each column is the true error for each question. All columns include the full set of socio-demographic controls and fixed effects used in Table 1. We only present the coefficients for the variables of interest: education, true error, and their interaction. Column (1) is an OLS, Column (2) a median quantile regression, Column (3) an OLS with an interaction between true error and education, Column (4) uses only respondents the lower tertile of education, and Column (5) uses respondents in the upper tertile of education. Standard errors are clustered at the subject level.

SOEP-IS Code	Question (a)	Answer
Q467 - IGEN02a	In which year were euro notes and coins introduced?	2002
Q470 - IGEN03a	In which year was Microsoft (Publisher of the software package Windows) founded?	1975
Q473 - IGEN04a	In which year was the movie “Das Boot” (directed by Wolfgang Peterson) first shown in German cinemas?	1981
Q476 - IGEN05a	In which year was Saddam Hussein captured by the US army?	2003
Q479 - IGEN06a	In which year was the first Volkswagen Type 1 (also known as “Volkswagen Beetle”) produced?	1938
Q482 - IGEN07a	In which year did the Korean War end with a truce?	1953
Q485 - IGEN08a	In which year did Lady Diana, Prince Charles’ first wife, die?	1997
	Question (b)	
	What do you think: How far is your answer from the correct answer?	

Table B.2: Original history questions in English language from the 2018 SOEP-IS

SOEP-IS Code	Questions (a)	Answer
Q467 - IGEN02a	In welchem Jahr wurden Euro-Geldscheine und -Münzen eingeführt?	2002
Q470 - IGEN03a	In welchem Jahr wurde das Unternehmen Microsoft (Herausgeber des Betriebssystems Windows) gegründet?	1975
Q473 - IGEN04a	In welchem Jahr kam der Film "Das Boot" (Regie: Wolfgang Petersen) in die deutschen Kinos?	1981
Q476 - IGEN05a	In welchem Jahr wurde Saddam Hussein von der US-Armee gefangen genommen?	2003
Q479 - IGEN06a	In welchem Jahr wurde der erste Volkswagen Typ 1(auch bekannt als "Käfer") produziert?	1938
Q482 - IGEN07a	In welchem Jahr endete der Korea-Krieg mit einem Waffenstillstand?	1953
Q485 - IGEN08a	In welchem Jahr starb Lady Diana, die erste Frau von Prinz Charles?	1997
	Question (b)	
	Was schätzen Sie: wie viele Jahre liegt Ihre Antwort von der richtigen Antwort entfernt?	

Table B.3: Original history questions in German language from the 2018 SOEP-IS

Variable	Definition
Financial Behavior:	
DAX forecast error	Principal component of the absolute distance between one- and two-year-ahead prediction of the DAX realization and the actual realization over the horizon. The closing price of the date of the respective interview was used. Standardized to have mean 0 and standard deviation 1.
portfolio diversification	Aggregate diversification measure over five asset classes. For each asset class, a penalty score is calculated expressing the distance to an equally diversified portfolio. Diversification equals the maximum attainable penalty score less the actual penalty. The diversification measure is standardized to have mean 0 and standard deviation 1.

Variable	Definition
Political Behavior:	
extremeness	Absolute distance to the center of an ideology scale from 0 (left) to 10 (right). Standardized to have mean 0 and standard deviation 1.
left-right	Location on an ideology scale from 0 (left) to 10 (right). Standardized to have mean 0 and standard deviation 1.
non-voter	=1 if respondent indicated not to vote in the Sonntagsfrage (ex-post) for the Bundestagswahl 2017.
Controls:	
age	Difference between interview month/year and birth month/year in years.
female	=1 if female.
GDR 1989	=1 if living in East Germany in 1989.
years education	Years of education (including any further education after primary and secondary education)=.
gross income	Monthly gross labor income in thousands. Missings are coded with a zero.
missing income	=1 if missing gross income.
fin. literacy	Share of correct answers to 6 questions related to financial knowledge.
risk aversion	Location on a risk scale from 0 (risk avers) to 10 (risk loving). Standardized to have mean 0 and standard deviation 1.
narcissism	Average narcissism measure over 6 items on a scale from 1 to 6. Standardized to have mean 0 and standard deviation 1.
impulsivity	Location on impulsivity scale from 0 (not impulsive) to 10 (fully impulsive). Standardized to have mean 0 and standard deviation 1.
patience	Location on the patience scale from 0 (not patient) to 10 (fully patient). Standardized to have mean 0 and standard deviation 1.
employed	=1 if employed.
unemployed	=1 if unemployed.
nonwork	=1 if non-working.

Variable	Definition
matedu	=1 if on maternity, educational, or military leave.
retired	=1 if retired.
answered	Number of questions answered for overprecision.
Additional Controls:	
assets	=1 if owning financial assets.
pol. interest	Political interest on a scale from 1 (high) to 4 (low). Reversed and standardized to have mean 0 and standard deviation 1.

Table B.4: Overview and definition of the variables from the SOEP used in the analysis.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	SOEP-IS		SOEP Core		Difference		
	mean	sd	mean	sd	difference	p-value	N[Core]
age	53.914	(0.627)	50.535	(0.180)	-3.379	0.000	30,997
female	0.508	(0.018)	0.508	(0.005)	0.000	0.989	30,997
german	0.933	(0.009)	0.877	(0.003)	-0.056	0.000	30,997
east (current)	0.174	(0.013)	0.172	(0.003)	-0.001	0.916	30,997
GDR 1989	0.186	(0.014)	0.198	(0.004)	0.012	0.404	24,591
years education	12.704	(0.098)	12.276	(0.027)	-0.428	0.000	28,482
employed	0.534	(0.018)	0.593	(0.005)	0.058	0.001	30,967
retired	0.229	(0.015)	0.221	(0.004)	-0.007	0.627	30,967
gross income	2.943	(0.112)	2.837	(0.029)	-0.106	0.359	17,829
married	0.568	(0.017)	0.521	(0.005)	-0.047	0.009	30,896
N[SOEP-IS]	805						

Table B.5: Representativeness of the SOEP-IS subsample. This table shows the descriptives of selected personal characteristics of the respondents for the subsample of the SOEP-IS and the SOEP-Core. The results for the SOEP-IS in Columns (1) and (2) are unweighted, whereas the results for the SOEP-Core in Columns (3) and (4) are weighted using the sampling weights provided. Columns (5) and (6) show a simple t-test on the difference between the means. Column (7) shows the sample size of the SOEP-Core. The sample size varies due to missing observations.

Dependent Variable: *sop*	(1)	(2)	(3)	(4)	(5)
age	-0.008***	-0.008***	-0.007**	-0.005	-0.005
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
female	0.074	0.121	0.094	0.134*	0.111
	(0.070)	(0.074)	(0.074)	(0.074)	(0.076)
years education	-0.053***	-0.062***	-0.050***	-0.043***	-0.037***
	(0.013)	(0.013)	(0.014)	(0.013)	(0.014)
answered		0.064***	0.067***	0.086***	0.085***
		(0.023)	(0.022)	(0.023)	(0.023)
gross income			-0.049**	-0.037*	-0.039*
			(0.021)	(0.020)	(0.021)
employed			0.012	0.079	0.139
			(0.163)	(0.156)	(0.164)
unemployed			0.696**	0.618*	0.622*
			(0.347)	(0.334)	(0.339)
retired			-0.017	-0.008	-0.005
			(0.109)	(0.109)	(0.112)
maternity leave			0.024	0.076	0.090
			(0.217)	(0.207)	(0.209)
east 1989			0.067	0.045	-0.169
			(0.100)	(0.098)	(0.116)
fin. literacy				-0.481***	-0.398**
				(0.170)	(0.174)
risk aversion				0.056	0.048
				(0.038)	(0.038)
impulsivity				-0.019	-0.014
				(0.038)	(0.040)
patience				0.035	0.038
				(0.035)	(0.035)
narcissism				0.107**	0.100**
				(0.042)	(0.043)
N	805	805	805	805	805
adj. R^2	0.036	0.046	0.061	0.081	0.098
Constant Term	Yes	Yes	Yes	Yes	Yes
Fixed Effects	No	No	No	No	Yes

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.6: Determinants of overprecision. Full version of Table 1 including all coefficients. In Columns (1)-(5) we run an OLS with *sop* as the dependent variable. In all Columns, we include age, gender, and education. In Column (2) we add the number of answered questions. In Column (3) we include gross income and dummies for labor force status (employed, unemployed, retired, maternity leave), and whether the respondent was a citizen of the GDR before 1989. In Column (4) we include further personal characteristics: financial literacy, risk preferences, impulsivity, patience, and narcissism. In Column (5) we also include fixed effects for the federal state (*Bundesland*) where the respondents live and the time at which they responded to the questionnaire.

	(1) DAX Error	(2) DAX 1yr	(3) DAX 2yr	(4) Diversification	(5) Extremism	(6) Left-Right	(7) Abstention
sop	0.083*	0.529	3.078**	-0.131***	0.087**	-0.008	0.032**
	(0.043)	(0.518)	(1.269)	(0.036)	(0.043)	(0.042)	(0.012)
age	-0.001	-0.036	0.020	-0.007**	-0.007**	0.007**	-0.001
	(0.004)	(0.042)	(0.104)	(0.003)	(0.003)	(0.003)	(0.001)
female	-0.008	-0.663	0.755	0.021	-0.000	-0.224***	-0.026
	(0.094)	(1.107)	(2.721)	(0.076)	(0.084)	(0.082)	(0.025)
years educ.	-0.002	-0.045	0.106	0.056***	0.029*	-0.031**	-0.013***
	(0.017)	(0.206)	(0.503)	(0.014)	(0.016)	(0.016)	(0.005)
gross income	-0.013	-0.215	-0.265	-0.009	-0.018	-0.003	0.002
	(0.026)	(0.315)	(0.756)	(0.023)	(0.027)	(0.026)	(0.008)
miss income	0.002	-0.665	0.741	-0.239	0.023	0.540**	0.058
	(0.289)	(3.408)	(8.179)	(0.207)	(0.226)	(0.222)	(0.069)
answered	-0.099***	-0.720**	-3.221***	0.064***	-0.033	-0.013	0.001
	(0.031)	(0.359)	(0.897)	(0.022)	(0.024)	(0.024)	(0.007)
employed	-0.078	-2.018	-0.212	-0.285	0.071	0.429*	0.065
	(0.295)	(3.466)	(8.338)	(0.208)	(0.226)	(0.222)	(0.069)
unemployed	0.141	3.764	-0.227	0.020	0.277	-0.087	0.149**
	(0.242)	(2.935)	(7.066)	(0.209)	(0.230)	(0.226)	(0.068)
retired	0.186	1.618	5.572	-0.003	0.104	0.068	0.040
	(0.154)	(1.819)	(4.436)	(0.121)	(0.128)	(0.126)	(0.037)
maternity	-0.043	-1.523	0.865	-0.050	-0.044	-0.344	0.028
	(0.295)	(3.504)	(8.616)	(0.231)	(0.263)	(0.258)	(0.077)
east 1989	-0.289*	-3.603*	-6.514	-0.057	0.046	-0.160	0.030
	(0.166)	(1.987)	(4.851)	(0.141)	(0.150)	(0.147)	(0.045)
fin. literacy	0.059	-3.086	9.537	0.385**	0.007	0.142	-0.053
	(0.235)	(2.698)	(6.858)	(0.162)	(0.175)	(0.171)	(0.051)
risk aversion	0.030	-0.078	2.102	0.039	0.006	-0.032	-0.011
	(0.047)	(0.557)	(1.369)	(0.037)	(0.042)	(0.041)	(0.012)
impulsivity	-0.009	0.133	-0.754	-0.081**	0.007	0.031	0.021*
	(0.047)	(0.561)	(1.362)	(0.038)	(0.042)	(0.041)	(0.012)
patience	0.083*	0.887*	2.242*	-0.027	0.005	-0.004	0.022*
	(0.045)	(0.532)	(1.297)	(0.036)	(0.040)	(0.039)	(0.011)
narcissism	0.000	-0.395	0.593	0.058	0.077*	0.050	0.010
	(0.045)	(0.539)	(1.319)	(0.038)	(0.041)	(0.041)	(0.012)
pol. interest					0.153***	-0.106**	-0.051***
					(0.044)	(0.043)	(0.013)
constant	-0.078	12.253*	12.364	-0.909**	0.231	-0.300	0.239*
	(0.528)	(6.293)	(15.214)	(0.382)	(0.431)	(0.423)	(0.125)
N	548	578	557	774	716	716	706
adj. R^2	0.076	0.125	0.035	0.112	0.024	0.062	0.111

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.7: Estimation results of Section 4. The number of observations varies due to missing observations in the outcome variable. The maximum number of observations is 805. The results come from the full regression as specified in Section 4.1. Variable definitions are in Table B.4 in the appendix.

	DAX error	diversification	extremism	left-right	non-voter
overprecision	0.098*	-0.112**	0.076	-0.059	0.025*
	(0.051)	(0.048)	(0.051)	(0.050)	(0.014)
Observations	465	465	465	465	465
R^2	0.140	0.197	0.110	0.150	0.197
overprecision		-0.099**	0.098**	-0.042	0.030**
		(0.043)	(0.044)	(0.043)	(0.012)
Observations		653	653	653	653
R^2		0.158	0.080	0.123	0.165

Notes: Standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

Table B.8: Seemingly Unrelated Regression Equations (SURE). The upper table presents the results for the SURE regression with DAX forecast error, portfolio diversification, political extremism, left-right preferences, and non-voter as dependent variable. The lower table drops the DAX forecast regression to maximize the number of observations. All regressions include Bundesland and month-of-interview fixed effects

Domain 1	Domain 2	Correlation	Standard error	N
Neutral	History	0.19	0.03	688
Neutral	Knowledge	0.23	0.04	626
Neutral	Stocks	0.24	0.04	676
Neutral	Economics	0.20	0.04	639
History	Knowledge	0.42	0.04	609
History	Stocks	0.27	0.04	638
History	Economics	0.32	0.05	620
Knowledge	Stocks	0.29	0.04	591
Knowledge	Economics	0.30	0.04	580
Stocks	Economics	0.25	0.04	618

Table B.9: Partial correlations between the aggregate overprecision measures across the five domains using winsorized data at the 99% and 1%. The control variables are age, gender, education, income, nationality, state, mathematical literacy, and a measure of self-reported knowledge on the topic of the domain.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.104**	0.024	0.071	4/46	yes/13	0.09	510
1-year ahead	*0.631*	*0.251*		10/46	yes/18	0.14	*537*
2-year ahead	*3.944****	*0.004*		4/46	no/0	0.00	*519*
Diversification	-0.12***	0.002	0.009	4/46	yes/15	0.13	719
Political Behavior:							
Extremism	0.077*	0.059	0.115	8/47	yes/18	0.07	716
Left-Right	-0.019	0.643	0.643	24/47	no/17	0.10	716
Abstention	0.029**	0.015	0.060	3/47	yes/10	0.12	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.10: This table shows the estimation results of Section 4 including the Big Five personality traits. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 750. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure as specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) LASSO R^2	(7) N
Financial Behavior:							
DAX Error	0.081*	0.065	0.125	6/42	yes/16	0.10	545
1-year ahead	*0.477*	*0.360*		8/42	yes/24	0.17	*574*
2-year ahead	*3.044***	*0.018*		5/42	no/0	0.00	*553*
Diversification	-0.131***	0.000	0.002	4/42	yes/18	0.13	763
Political Behavior:							
Extremism	0.082*	0.057	0.161	6/43	yes/14	0.06	706
Left-Right	-0.002	0.966	0.966	18/43	no/14	0.08	706
Abstention	0.031**	0.014	0.054	3/43	yes/10	0.10	694

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.11: This table shows the estimation results of Section 4 including asset ownership as control. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 791. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure as specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the R^2 of the estimated model (Column (6)).

Qualtrics Code	Question (a)	
PX110 a	How many black triangles were in the matrix?	70
PX110 b	How many black triangles were in the matrix?	120
PX110 c	How many black triangles were in the matrix?	195
PX110 d	How many black triangles were in the matrix?	280
PX110 e	How many black triangles were in the matrix?	330
PX120 a	In which year were euro notes and coins introduced?	2002
PX120 b	In which year was Microsoft (Publisher of the software package Windows) founded?	1975
PX120 c	In which year was Saddam Hussein captured by the US army?	2003
PX120 d	In which year was the first Volkswagen Type 1 (also known as "Volkswagen Beetle") produced?	1938
PX120 e	In which year did Lady Diana, King Charles' first wife, die?	1997
PX120 f	In which year was Joachim Sauer (husband of Angela Merkel) born?	1949
PX120 g	Which year is between 1993 and 1995?	1994
PX130 a	How many teeth does an adult polar bear have?	42
PX130 b	How many keys (black AND white) does a grand piano have?	88
PX130 c	How high (in meters) is the Reichstag building in Berlin?	47
PX130 d	What percentage of seats in the 20th German Bundestag (elected on September 26, 2021) are occupied by female members of the Bundestag	35
PX130 e	How many countries on the African continent are members of the United Nations?	54
PX130 f	What is the upper bantamweight limit in women's Olympic boxing weight classes?	54
PX130 g	What whole number is between 83 and 85?	84
PX140 a-e	What do you estimate: Where (in euros) will the share price be in exactly four weeks?	+ 28 days
PX140 f	In this question, the graph is empty. Please just enter the number 42 as your answer. What do you estimate: where (in euros) will the share price be in exactly four weeks?	42

PX150 a	The Consumer Price Index (CPI) is a measure of the average percentage change in the price level of certain goods and services purchased by households for consumption. The change in the consumer price index compared to the same month of the previous year or the previous year is also referred to as the rate of inflation. By how much (in percent) did the German CPI increase from the beginning of 2011 to the end of 2021?	17
PX150 b	The gross domestic product (GDP) indicates the total value of all goods and services that were produced as end products within the national borders of an economy during a year, after deduction of all intermediate consumption. By how much (in percent) is the German GDP at market prices (nominal) increased from 2006 to 2021?	51
PX150 c	The DAX (Deutscher Aktienindex) is a stock index that measures the performance of the 40 largest companies in the German stock market. By how much (in percent) did the DAX rise from the beginning of 2014 to the end of 2021?	65
PX150 d	The middle income or median income in a society or group describes the income level at which the number of households (or persons) with lower incomes is equal to the number of households with higher incomes. By how much (in percent) did median income in Germany increase from 1991 to 2018?	22
PX150 e	A census is a legally ordered survey of statistical population data. The last census in Germany took place in 2022. By how much (in percent) did the German population grow from 1991 to 2021?	4
PX150 f	M1 money supply describes the amount of cash in circulation and the amount of sight deposits (e.g. savings accounts). By how much (in percent) did the M1 money supply in the euro zone increase from the beginning of 2015 to the end of 2021?	86
	Question (b)	
	What do you think: How many [unit] is your answer away from the correct answer?	

Table B.12: Original questions in English language from the online companion survey

Qualtrics Code	Questions (a)	Answer
PX110 a	Wie viele schwarze Dreiecke waren in der Matrix?	70
PX110 b	Wie viele schwarze Dreiecke waren in der Matrix?	120
PX110 c	Wie viele schwarze Dreiecke waren in der Matrix?	195
PX110 d	Wie viele schwarze Dreiecke waren in der Matrix?	280
PX110 e	Wie viele schwarze Dreiecke waren in der Matrix?	330
PX120 a	In welchem Jahr wurden Euro-Geldscheine und -Münzen eingeführt?	2002
PX120 b	In welchem Jahr wurde das Unternehmen Microsoft (Herausgeber des Betriebssystems Windows) gegründet?	1975
PX120 c	In welchem Jahr wurde Saddam Hussein von der US-Armee gefangen genommen?	2003
PX120 d	In welchem Jahr wurde der erste Volkswagen Typ 1 (auch bekannt als "Käfer") produziert?	1938
PX120 e	In welchem Jahr starb Lady Diana, die erste Frau von König Charles III.?	1997
PX120 f	In welchem Jahr wurde Joachim Sauer (Ehemann von Angela Merkel) geboren?	1949
PX120 g	Welches Jahr liegt zwischen 1993 und 1995?	1994
PX130 a	Wie viele Zähne hat ein ausgewachsener Eisbär?	42
PX130 b	Wie viele Tasten (schwarz UND weiß) hat ein Konzertflüegel?	88
PX130 c	Wie hoch (in Metern) ist das Berliner Reichstagsgebäude?	47
PX130 d	Wie viel Prozent der Sitze im 20. Deutschen Bundestag (gewählt am 26. September 2021) sind durch weibliche Bundestagsabgeordnete besetzt?	35
PX130 e	Wie viele Staaten auf dem Afrikanischen Kontinent sind Mitglied der Vereinten Nationen?	54

PX130 f	Bei wie viel Kilogramm liegt die Obergrenze des Bantamgewichts in den Gewichtsklassen der Frauen beim Olympischen Boxen?	54
PX130 g	Welche ganze Zahl liegt zwischen 83 und 85?	84
PX140 a-e	Was schätzen Sie: wo (in Euro) liegt der Aktienkurs in genau vier Wochen?	+28 Tage
PX140 f	In dieser Frage ist die Grafik leer. Bitte geben Sie einfach die Zahl 42 als Antwort ein. Was schätzen Sie: wo (in Euro) liegt der Aktienkurs in genau vier Wochen?	42
PX150 a	Der Verbraucherpreisindex (VPI) ist ein ist ein Maß der durchschnittlichen prozentualen Veränderung des Preisniveaus bestimmter Waren und Dienstleistungen, die von privaten Haushalten für Konsumzwecke gekauft werden. Die Veränderung des Verbraucherpreisindex zum Vorjahresmonat bzw. zum Vorjahr wird auch als Teuerungsrate oder als Inflationsrate bezeichnet. Um wie viel (in Prozent) ist der deutsche VPI von Anfang 2011 bis Ende 2021 gestiegen?	17
PX150 b	Das Bruttoinlandsprodukt (BIP) gibt den Gesamtwert aller Waren und Dienstleistungen an, die während eines Jahres innerhalb der Landesgrenzen einer Volkswirtschaft als Endprodukte hergestellt wurden, nach Abzug aller Vorleistungen. Um wie viel (in Prozent) ist das deutsche BIP zu Marktpreisen (nominal) von 2006 bis 2021 gestiegen?	51
PX150 c	Der DAX (Deutscher Aktienindex) ist ein Aktienindex, der die Wertentwicklung der 40 größ ten Unternehmen des deutschen Aktienmarkts misst. Um wie viel (in Prozent) ist der DAX von Anfang 2014 bis Ende 2021 gestiegen?	65
PX150 d	Das mittlere Einkommen oder Medianeinkommen in einer Gesellschaft oder Gruppe bezeichnet die Einkommenshöhe, von der aus die Anzahl der Haushalte (bzw. Personen) mit niedrigeren Einkommen gleich groß ist wie die der Haushalte mit höheren Einkommen. Um wie viel (in Prozent) ist das Medianeinkommen in Deutschland von 1991 bis 2018 gestiegen?	22

PX150 e	Eine Volkszählung oder auch Zensus ist eine gesetzlich angeordnete Erhebung statistischer Bevölkerungsdaten. Der letzte Zensus in Deutschland fand 2022 statt. Um wie viel (in Prozent) ist die deutsche Bevölkerung von 1991 bis 2021 gewachsen?	4
PX150 f	Die Geldmenge M1 bezeichnet die Menge an Bargeld im Umlauf sowie die Höhe an Sichteinlagen (bspw. Sparkonten). Um wie viel (in Prozent) ist die Geldmenge M1 in der Eurozone von Anfang 2015 bis Ende 2021 gestiegen?	86
	Question (b)	
	Was schätzen Sie: wie viele [Einheit] liegt Ihre Antwort von der richtigen Antwort entfernt?	

Table B.13: Original questions in German language from the online companion survey

Variable	Definition
Controls:	
age	Reported age.
gender	=1 if female.
education	Years of primary and secondary education.
income	Monthly gross labor income in thousands, reported in bins.
nationality	Reported primary nationality.
mathematical literacy	Share of correct answers to three statistical problems.
expertise	Self-reported expertise on specific domain on a scale from 0 to 100.

Table B.14: Overview and definition of the variables from the online companion survey used in the analysis.

	count	mean	sd	p25	p50	p75
Full sample						
age	1000	44.950	14.315	33.000	46.000	58.000
gender	998	0.489	0.500	0.000	0.000	1.000
german	1000	0.962	0.191	1.000	1.000	1.000
east (current)	1000	0.200	0.400	0.000	0.000	0.000
education	988	10.685	1.871	10.000	10.000	12.000
gross income	924	2.456	1.856	1.500	2.500	4.000
N	1000					
Subsample						
age	839	44.897	14.213	33.000	46.000	57.000
gender	837	0.483	0.500	0.000	0.000	1.000
german	839	0.969	0.173	1.000	1.000	1.000
east (current)	839	0.203	0.402	0.000	0.000	0.000
education	831	10.693	1.863	10.000	10.000	12.000
gross income	778	2.485	1.898	1.500	2.500	4.000
N	839					

Table B.15: Summary statistics of the online companion survey. This table shows summary statistics for the main variables in the survey.

C Alternative Measures of Overprecision

To test the robustness of our overprecision measure, in Section C.1 we discuss five alternative measures of overprecision, which are variations of our measure. In Section C.2 we use these alternative measures to test the robustness of our results from Section 3.3 regarding the sociodemographic characteristics and Section C.3 the robustness of the predictions in Section 4.2.

C.1 Alternative Measures

Standardized measure (op_i'): Since the overprecision measure of Ortoleva and Snowberg (2015b) standardizes the measure with respect to the entire population, we further construct a *standardized* measure op_i' of overprecision where we standardize the absolute measure op_i of the respective question to be mean zero and standard deviation one before aggregation to avoid the aggregated measure to be biased by a specific question and to relate the level to the entire population. The mean is used again to aggregate across the seven questions.

Centered measure (op_i''): Respondents might not only differ with respect to the perceived variance of the distribution of the error to their answer, but also with respect to the mean of the distribution. Hence, the baseline overprecision measure might capture both overprecision and a miscalibration of the mean. To separate both of them, we construct a *centered* measure of overprecision. To correct for the difference in the means of the distributions and center the distributions around zero, for each question, we subtract the sample mean from the true and subjective error. Any remaining systematic deviation of the subjective error from the realized error should be exclusively due to over- or underprecision.

Relative measure (op_i'''): we compute a *relative* measure op_i''' by dividing the absolute measure op_i in a specific question with the respective subjective error. Taking the relative distance into account makes the measure more comparable across respondents while still keeping the relative distance between the subjective error and the realized error (see Figure C.1).

Assume that, similar to the example in Figure 1, the true error is normally distributed with mean 0 and variance σ^2 (solid curve). Moreover, the perceived distribution by the

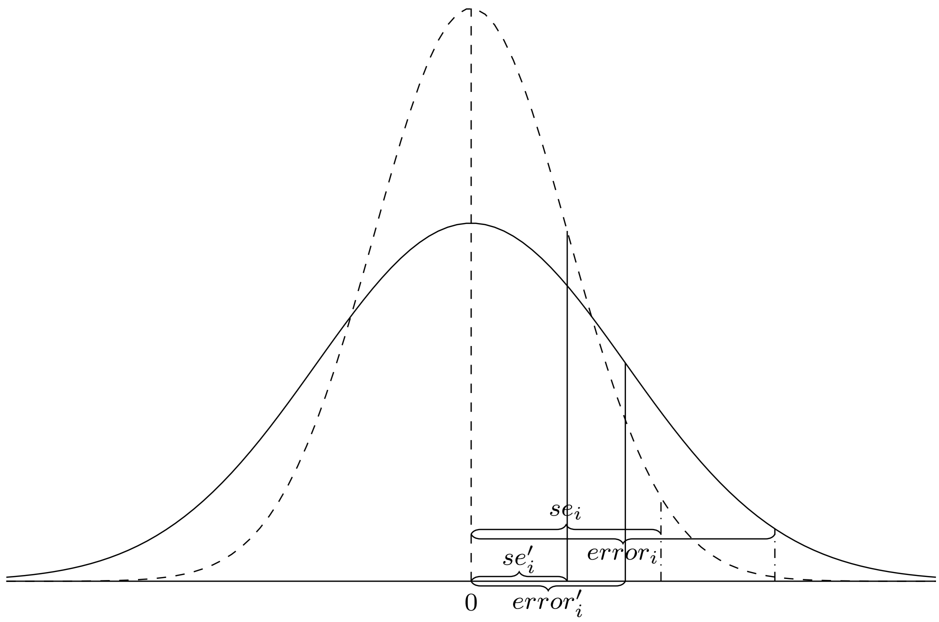


Figure C.1: Two distributions of the (subjective) error. The solid curve shows the true distribution of the error with a standard deviation of 2 (precision of .25). The dashed red curve shows the perceived distribution by an overprecise respondent with a standard deviation of 1.25 (precision of .64). The solid and dashed vertical lines indicate the subjective errors (se) and the true errors ($error$) resulting from respondents with two different ideas about the nature of the subjective error asked in the second question.

respondents might not necessarily coincide with the true distribution. If the perceived variance $\hat{\sigma}^2$ is smaller, i.e., the precision $\rho = 1/\hat{\sigma}^2$ is larger, then we call this respondent overprecise (dashed curve). As long as respondents have the same idea in mind when asking for the error they expect to make, the absolute overprecision measure is comparable across subjects. However, when respondents substantially differ, e.g., by having different confidence intervals in mind, the ranking as computed with the absolute measure might not be consistent anymore whilst the sign of the deviation still being correct. Taking the example in Figure C.1, where the respondents have the same degree of overprecision since the perceived precision of .64 deviates from the true precision of .25, for a respondent with having 95% confidence in mind (se and $error$) the absolute overprecision measure would yield 1.47 whereas for the respondent with having 68% confidence in mind (se' and $error'$) it would yield .75. Thus, the second respondent would incorrectly be classified as less overprecise.

The relative measure corrects this inconsistency by scaling the absolute overprecision measure with the subjective absolute error, making the measure comparable across subjects. In the above example, the relative measure yields .6 in both cases, which is precisely the relative difference between the standard deviations of the respective distributions and, thus, directly proportional to the relative difference between the degree of precision.

Turning to the SOEP data, the correlation between the absolute and relative measure across the seven questions ranges from $\rho^{Spearman} = .91$ to $\rho^{Spearman} = .96$ which is consistent with the respondents interpreting the subjective error question in the same way.[47] Given the high correlation between both approaches, using the absolute measure is preferable as it avoids having to drop the observations of respondents whose subjective error is zero.

Age-robust measure (op_i''''): The negative correlation between age and overprecision in our sample is likely to be driven by the type of questions that were asked in the survey. Since we asked about specific historical events within the last 100 years, respondents who lived during these events might be better calibrated. This becomes obvious in Figure C.2 where, for every question, we split the density of our overprecision measure $op_{i,j}$ between those respondents born before and after the event. As expected, those subjects born before the event are better calibrated than those born after. As a robustness test, we construct, for every respondent, an *age-robust* measure of overconfidence (op_i''''). We construct this measure following the formulation described in Section 2.1, but using only those questions about events that happened *after* the respondent was born. The drawback of this approach is that we lose a substantial amount of information and give more weight to events that occurred later in time. Taking fewer questions into account also comes at the risk that the aggregate measure is biased by one specific question.

Residual measure (op_i'''''): The *residual* measure is a measure of overprecision obtained by the estimation method of Ortoleva and Snowberg (2015b). Ortoleva and Snowberg (2015b) construct their measure of overconfidence by asking respondents about their assessment of the current and one-year-ahead inflation rate and the unemployment rate as well as their confidence about the respective answers using a six-point scale. They then regress participants' confidence on a fourth-order polynomial of accuracy to isolate the effect of knowledge. The principal component of the four residuals is then used as their measure of overconfidence. To replicate their approach as closely as possible, we construct a measure of respondent confidence by inverting the reported subjective error and computing quintiles. We then regress the respondents' implied "confidence" about the answer

[47] Note that the relationship between the absolute and the relative measure is non-linear. Therefore, we report the Spearman correlation coefficient only.

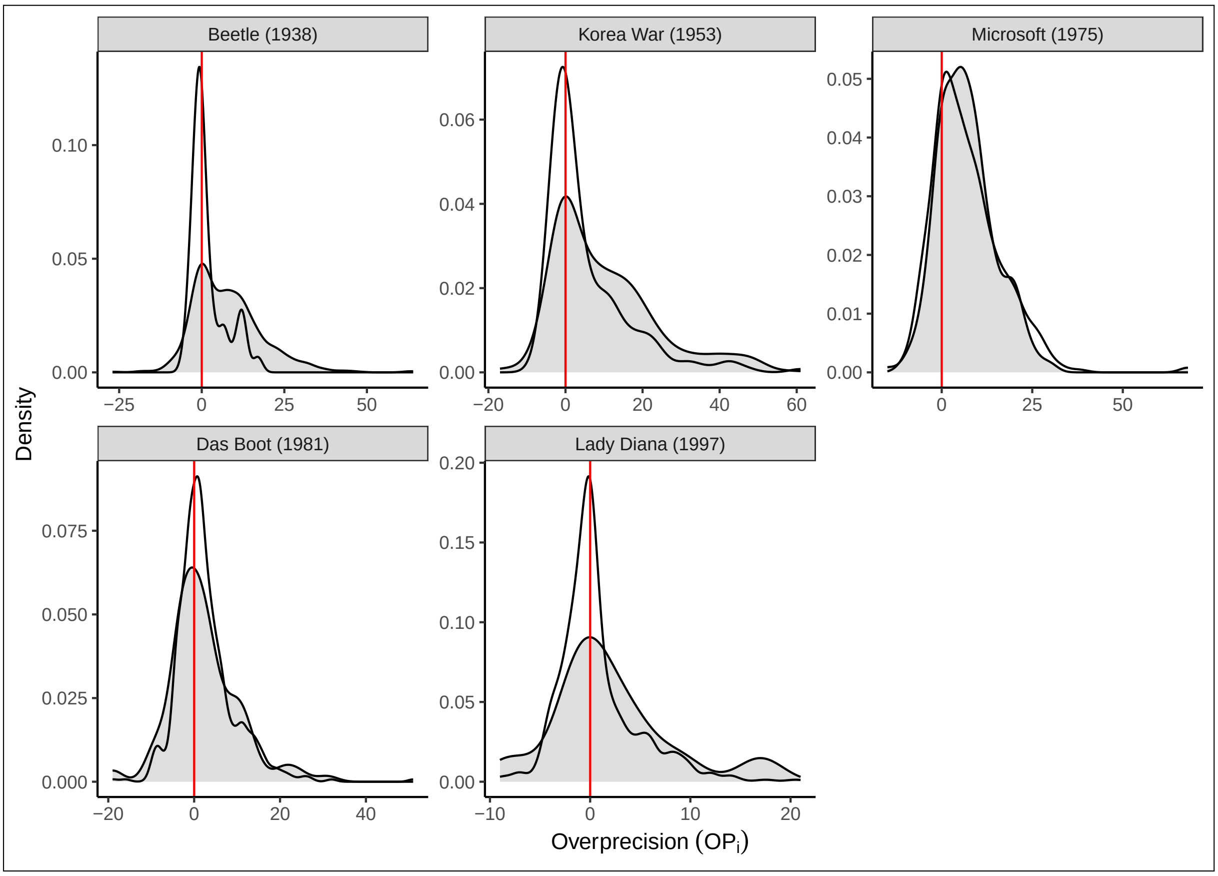
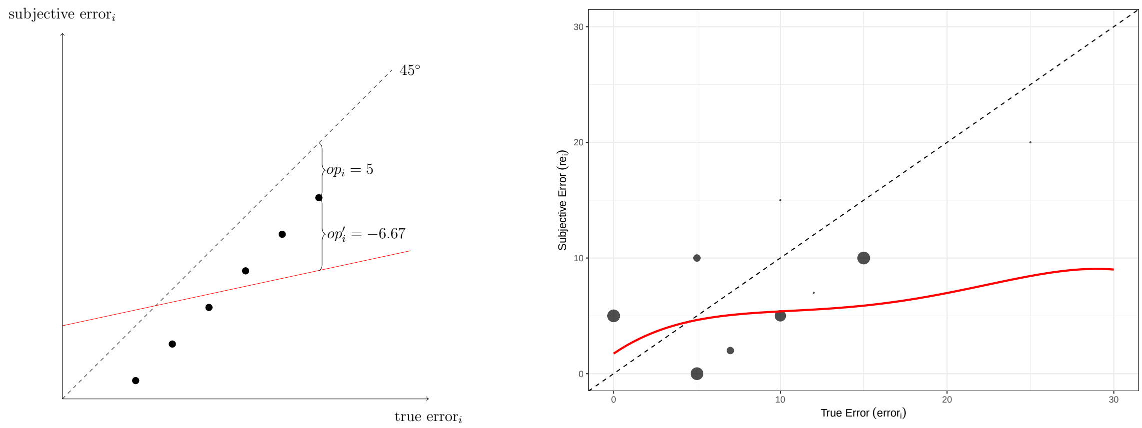


Figure C.2: Density of overprecision (op_{ij}) and age. From left (less recent) to right (more recent) We plot the density of the measured overprecision ($op_{i,j}$) for each question j. In gray, we plot the density of the measured overprecision for the question ($op_{i,j}$) of those subjects that were born after the event took place. With no color, we plot the density of all respondents born at the year of the event or before. Note that the scale of the vertical axis is different across the five plots. Questions with (correct) answers after 2000 are omitted as there were no underage respondents.

on a fourth-order polynomial of the realized error and take the principal component of the residuals across all seven questions to create our new individual measure of overprecision op_i''''''.

The residual measure of overprecision (op_i'''''') mechanically differs from our baseline measure (op_i) because it effectively calculates the distance between the subjective error and the fitted fourth-order polynomial instead of the distance between the subjective error and the realized error. This approach comes with the caveat that, if a respondents deviation is small relative to that of the population, then, when computing the residuals for the seven questions, the measure classifies the respondent as underconfident even if their realized error is larger than their subjective error (for an illustrative example see Figure C.3). Thus, for every measure of op_i'''''', the residual measure takes into account the relationship between the subjective error and the realized error for the entire *population*



(a) Illustration

(b) Data

Figure C.3: Misspecification of participants. This figure shows the difference between the Subjective Error Method and the *residual* approach for a theoretical illustration in (a) and for the answers to one of the overprecision questions in (b). Any observation in both panels above the 45° line represents underprecise individuals and any observation below represents overprecise individuals. Note that the axes are changed as compared to Figure 3. In panel (a), the dots represent observations for respondents for whom, in the example, the Subjective Error Method yields $op_i = error_i - se_i = 5$ in a specific question in the set of questions. The red line illustrates the fitted line of a simplified version of the *residual* approach using only a first order polynomial ($se_i = \alpha + \beta error_i + \epsilon_i$). In panel (b), the dots represent respondents for whom the Subjective Error Method yields an overprecision of 5 and -5 respectively. The red line indicates the fitted line of the *residual* approach using a fourth-order polynomial.

of respondents. In contrast, our approach focuses on the respondent's signal processing only by comparing the realized error with the subjective error.

C.2 Robustness of Descriptive Results

In Table C.1 we replicate Table 1 using each of the measures described in Section C.1 (Columns (2) to (6)) and our baseline measure sop_i in Column (1).

Column (2) of Table C.1 shows the results for the *standardized* measure (op'_i). The results show no qualitative changes with respect to the baseline except for the coefficient of the number of questions that were answered. However, the results are less significant. Column (3) shows the results using the *centered* measure (op''_i). The results remain largely robust with the coefficient for *gender* becoming larger and the coefficient for *answered* turning negative and insignificant.

Column (4) replicates the baseline estimations using the *relative* approach (op_i'''). The qualitative results remain similar except for less precisely estimated coefficients which can be explained by the decreased sample size. Column (5) uses the *age-robust* measure (op_i''''). The results show that, if we exclude the mechanical effect of age, then overprecision and age are positively correlated which is consistent with the earlier results from the literature (e.g., Ortoleva and Snowberg, 2015a,b; Prims and Moore, 2017). Otherwise, all of our results remain robust.

Column (6) of Table C.1 shows the results for the *residual* approach (op_i'''''). For the most part, the outcome replicates the results of Ortoleva and Snowberg (2015b), with females being less overprecise and income and education not showing up as statically relevant. Moreover, age is positively correlated with the estimated overprecision. Surprisingly, the number of answered questions has a negative effect on overprecision. In other words, contrary to the observed measure of overprecision, if we estimate overprecision using the methodology of Ortoleva and Snowberg (2015b), then the more questions a respondent answers, the less overprecise she is.

Given the results in Table C.1, we believe that our baseline measure is the best alternative. It is a simple and straightforward approach that can easily be implemented and which does not require the specification of an econometric model such as the approach of Ortoleva and Snowberg (2015b). It does not miss-classify respondents and uses all of the available information into account. Moreover, it is highly correlated to both the standardized measure ($\rho^{Pearson} = .85$; $\rho^{Spearman} = .86; N = 805$), the centered measure ($\rho^{Pearson} = .96$; $\rho^{Spearman} = .93; N = 801$), as well as the relative measure ($\rho^{Pearson} = .68$; $\rho^{Spearman} = .82; N = 801$) and therefore robust to transformations. All of these results are confirmed in Appendix C.3 where we test the predictive power of all robustness measures.

	Baseline (1)	Standardized (2)	Centered (3)	Relative (4)	Age robust (5)	Residual (6)
age	-0.005	-0.000	-0.005*	0.003	0.018***	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
female	0.111	0.087	0.177**	0.079	-0.013	-0.199***
	(0.075)	(0.076)	(0.075)	(0.083)	(0.074)	(0.074)
years education	-0.037***	-0.014	-0.037***	-0.003	-0.008	0.002
	(0.014)	(0.014)	(0.014)	(0.016)	(0.014)	(0.014)
answered	0.085***	-0.031	-0.010	0.033	0.075***	-0.104***
	(0.021)	(0.022)	(0.021)	(0.027)	(0.021)	(0.021)
gross income	-0.039*	-0.040**	-0.033	-0.019	-0.032	0.010
	(0.023)	(0.024)	(0.023)	(0.025)	(0.023)	(0.023)
fin. literacy	-0.398**	-0.306*	-0.364**	-0.469*	-0.347**	-0.149
	(0.155)	(0.158)	(0.155)	(0.175)	(0.154)	(0.153)
risk aversion	0.048	0.037	0.036	-0.016	0.017	0.007
	(0.037)	(0.038)	(0.037)	(0.043)	(0.037)	(0.037)
impulsivity	-0.014	-0.005	-0.018	-0.006	0.020	0.034
	(0.037)	(0.038)	(0.037)	(0.041)	(0.037)	(0.037)
patience	0.038	0.042	0.038	0.058*	0.033	0.044
	(0.035)	(0.036)	(0.035)	(0.039)	(0.035)	(0.035)
narcissism	0.100**	0.079*	0.098**	0.118**	-0.003	0.042
	(0.037)	(0.038)	(0.037)	(0.041)	(0.037)	(0.037)
N	805	805	805	702	800	805
adj. R^2	0.098	0.066	0.100	0.045	0.121	0.117
Constant Term	Yes	Yes	Yes	Yes	Yes	Yes
Employment & GDR 1989	Yes	Yes	Yes	Yes	Yes	Yes
Personal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.1: Determinants of overprecision using alternative measures of overprecision. For comparison, in Column (1) we run an OLS with the baseline measure. In Column (2) - (6), we run an OLS using the *standardized* measure, the *centered* measure, the *relative* measure, the *age-robust* measure, and the *residual* measure respectively. All include dummies for the labor force status (employed, unemployed, retired, maternity leave, non-working), whether the respondent was a GDR citizen before 1989, and further personal characteristics. We also control for the federal state (*Bundesland*) where the respondent lives and the time at which he/she responded to the questionnaire.

C.3 Predictions Using Alternative Overprecision Measures

In the following, we will show the results for the *standardized*, the *centered*, the *residual*, *relative*, and the *age-robust* measure of overprecison as well as using the approach following Ortoleva and Snowberg (2015b).

Table C.2 shows the results from the predictions using the *standardized* measure of overprecision. The results only slightly change with respect to the baseline, with the coefficients for the prediction errors becoming insignificant. However, the sign of the coefficient remains unchanged. The predictive power with respect to the LASSO estimations remains strong despite a slight decrease in the ranking as calculated by the R^2 method.

Table C.3 shows the results from the predictions using the *centered* measure of overprecision. Since the Pearson correlation between the centered and the baseline measure is .96, the results remain mostly unchanged.

Table C.4 shows the results from the predictions using the *relative* measure of overprecision instead. The advantage is that it makes the measure more comparable across subjects. However, we lose those observations with a reported zero subjective error due to mathematical reasons. The results, as compared to those in the baseline in Table 2, remain qualitatively similar.

Table C.5 shows the results from the predictions using the *age-robust* measure of overprecision instead. The results are at large in line with the results of the baseline estimations. The *age-robust* overprecision measures still predicts the outcomes according to the LASSO estimations. The point estimates slightly decrease in size and significance. However, as pointed out above, this measure considers fewer answers of the respondents and puts more weight on the more recent events since it only considers the questions on events after the respondent was born. Thus, the aggregate measure is calculated across fewer answers which might bias the measure. Therefore, these results have to be taken with a grain of salt.

Table C.6 shows the results from the predictions using the residual approach. Compared to the baseline measure, the alternative measure does not significantly predict any of the predictions derived from the theory. This is most likely because, applied to our data, this approach misclassifies certain respondents in the data as discussed in Appendix C.1.

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.11***	0.010	0.039	3/41	yes/15	0.10	548
1-year ahead	*0.382*	*0.450*		*19/41*	*no/15*	*0.14*	*578*
2-year ahead	*4.776****	*0.000*		*1/41*	*no/0*	*0.00*	*557*
Diversification	-0.104***	0.004	0.019	3/41	yes/18	0.13	774
Political Behavior:							
Extremism	0.09**	0.025	0.072	3/42	yes/13	0.06	716
Left-Right	-0.017	0.662	0.662	18/42	no/14	0.08	716
Abstention	0.026**	0.025	0.050	4/42	yes/19	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.2: This table shows the estimation results of Section 4 using the *standardized* overprecision measure. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.08*	0.066	0.127	4/41	yes/14	0.10	548
1-year ahead	*0.387*	*0.456*		*12/41*	*yes/21*	*0.16*	*578*
2-year ahead	*3.202***	*0.012*		*4/41*	*no/1*	*0.00*	*557*
Diversification	-0.125***	0.001	0.003	3/41	yes/19	0.13	774
Political Behavior:							
Extremism	0.081*	0.059	0.166	6/42	yes/14	0.05	716
Left-Right	0.003	0.944	0.944	18/42	no/14	0.08	716
Abstention	0.026**	0.034	0.128	4/42	yes/18	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.3: This table shows the estimation results of Section 4 using the *centered* overprecision measure. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.062	0.166	0.421	3/41	yes/16	0.11	501
1-year ahead	*0.19*	*0.721*		*9/41*	*no/20*	*0.16*	*530*
2-year ahead	*2.822***	*0.033*		*2/41*	*no/1*	*0.01*	*510*
Diversification	-0.068*	0.080	0.284	18/41	yes/14	0.11	681
Political Behavior:							
Extremism	0.113***	0.007	0.033	2/42	yes/15	0.06	624
Left-Right	-0.03	0.465	0.713	18/42	no/15	0.08	624
Abstention	0.003	0.800	0.800	3/42	no/4	0.09	616

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.4: This table shows the estimation results of Section 4 using the *relative* overprecision measure. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)) which is slightly less conservative than the Bonferroni adjustment. Column (4) displays the result from the R^2 procedure specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	0.036	0.399	0.399	8/41	no/8	0.07	546
1-year ahead	*-0.442*	*0.379*		*4/41*	*no/16*	*0.15*	*576*
2-year ahead	*2.868***	*0.021*		*5/41*	*no/0*	*0.00*	*555*
Diversification	-0.048	0.200	0.359	6/41	yes/16	0.12	769
Political Behavior:							
Extremism	0.065	0.102	0.350	5/42	yes/13	0.05	712
Left-Right	-0.061	0.119	0.315	6/42	no/11	0.07	712
Abstention	0.026**	0.022	0.106	3/42	yes/19	0.14	701

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.5: This table shows the estimation results of Section 4 using the *age-robust* overprecision measure. The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

	(1) Point estimate	(2) Unadj. p-value	(3) MHT p-value	(4) R^2 rank	(5) LASSO included	(6) R^2	(7) N
Financial Behavior:							
DAX error	-0.021	0.626	0.948	14/41	no/12	0.09	548
1-year ahead	*-0.269*	*0.606*		*9/41*	*no/20*	*0.16*	*578*
2-year ahead	*-0.267*	*0.835*		*12/41*	*no/7*	*0.04*	*557*
Diversification	-0.029	0.440	0.902	13/41	no/19	0.12	774
Political Behavior:							
Extremism	-0.046	0.256	0.773	8/42	yes/13	0.05	716
Left-Right	-0.008	0.831	0.831	17/42	no/14	0.08	716
Abstention	-0.003	0.798	0.959	18/42	no/17	0.13	706

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.6: This table shows the estimation results of Section 4 using the *residual* aggregation method of Ortoleva and Snowberg (2015a). The number of observations (Column (7)) varies due to missing observations in the outcome variable. The maximum number of observations is 805. Column (1) lists the point estimate of the coefficient on the standardized overprecision measure *sop* from the full regression as specified in Section 4.1 along with the unadjusted p-value (Column (2)) and the Sidak-Holm adjusted p-value for multiple hypothesis testing (MHT) (Column (3)). Column (4) displays the result from the R^2 procedure specified in Section 4.1 along with the maximum possible variables to be included in the model. The regressions with political outcomes as dependent variable additionally include a self-reported measure of political interest. Column (5) specifies the result of the LASSO procedure as specified in Section 4.1 along with the number of control variables chosen by LASSO and the R^2 of the estimated model (Column (6)).

D Specification Curve Analysis

To assess the robustness of our results concerning overprecision, we conduct a comprehensive specification curve analysis, following the methodology of Simonsohn et al. (2020). This method systematically evaluates the stability of estimated coefficients across a large set of plausible model specifications, thereby addressing concerns related to researcher degrees of freedom and the sensitivity of empirical findings to modeling choices.

The analysis is conducted separately for each of our four primary outcomes: *DAX Error*, *Diversification*, *Extremism*, and *Abstention*. Across all specifications, the key independent variable is the overprecision measure used in the specification. Each specification estimates a model of the form:

$$y_i = \alpha + \beta \cdot \mathrm{op}_i + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i, \tag{6}$$

where y_i denotes the outcome of interest, op_i represents the overprecision measure, $\mathbf{X}_i$ is a vector of control variables, and ε_i is the error term. For each estimate, we iterate over a grid of empirical specifications by varying control variables (e.g., age, education, income), fixed effects (e.g., month of interview), definitions of the measure of overprecision (e.g., standardized PCA, standardized mean), or outcome values (e.g., by winsorizing or standardizing). Additionally, for voting abstention, we change the estimation method from linear to Logit and for the portfolio diversification outcome we use different diversification benchmarks. An exhaustive list of all variables used for the specification curves can be found in Table D.1.

Each specification is estimated and the resulting coefficient $\hat{\beta}$ and its associated p-value are stored. These results are visualized using a specification curve, which plots the distribution of $\hat{\beta}$ estimates across specifications and their respective statistical significance. The resulting outcomes are plotted in Figures D.1 to D.4. Each dot corresponds to one model specification, with the vertical axis plotting the estimated coefficient of the variable of interest (overprecision) and the horizontal axis representing variations in model design, including alternative sets of controls, and alternative operationalizations of the overprecision measure or the outcome. By examining the distribution of estimates, one can assess whether the estimated effect remains directionally consistent and statistically significant across model specifications. The plotted coefficients are sorted by magnitude

Category	Variable	DAX	Divers.	Extrem.	Abst.
Demographics	Age	✓	✓	✓	✓
	Female	✓	✓	✓	✓
	Education (years)	✓	✓	✓	✓
	Gross income	✓	✓	✓	✓
	Missing income	✓	✓	✓	✓
	Employed	✓	✓	✓	✓
	Unemployed	✓	✓	✓	✓
	Retired	✓	✓	✓	✓
	Maternity leave	✓	✓	✓	✓
	GDR 1989 dummy	✓	✓	✓	✓
	Financial literacy	✓	✓	✓	✓
Psych. Traits	Risk preferences	✓	✓	✓	✓
	Narcissism	✓	✓	✓	✓
	Self control	✓	✓	✓	✓
	Risk preferences and financial lit	✓	✓	✓	✓
	Big Five	✓	✓	✓	✓
Other Controls	# Questions answered	✓	✓	✓	✓
	Political interest	–	–	✓	✓
	Asset ownership	✓	✓	–	–
Fixed Effects	State fixed effects	✓	✓	✓	✓
	Interview month FE	✓	✓	✓	✓
Spec. Variants	SE type (normal/robust)	✓	✓	✓	✓
	Logit model	–	–	–	✓
	Standardized outcome	✓	✓	✓	✓
	Winsorized outcome	✓	✓	✓	✓
	OP measure variants	✓	✓	✓	✓
Diversification Options	Equal weights	–	✓	–	–
	35/35/10/10/10	–	✓	–	–
	45/45/4/3/3	–	✓	–	–

Table D.1: Variables used in specification curves. The columns indicate the variables that are included in the specification curve permutations, along with the type of outcome analyzed in each. The diversification benchmarks indicate the percentage invested in each type of asset to be considered optimally diversified.

(from smallest to largest) to enhance the visibility of the estimated distribution. Additionally, because the number of specifications for each outcome reaches into the hundreds of thousands (e.g., for portfolio diversification, we ran 737,280 independent regressions), to avoid overplotting and to facilitate interpretation, each figure presents 500 estimates composed of:

- the smallest/largest estimated coefficients,
- a random sample of 9 of the smallest/largest 1,000 observations, excluding the smallest/largest estimate,
- a random sample of 479 estimates from the remaining interior values,
- and our reported estimate presented in the main body of the paper.

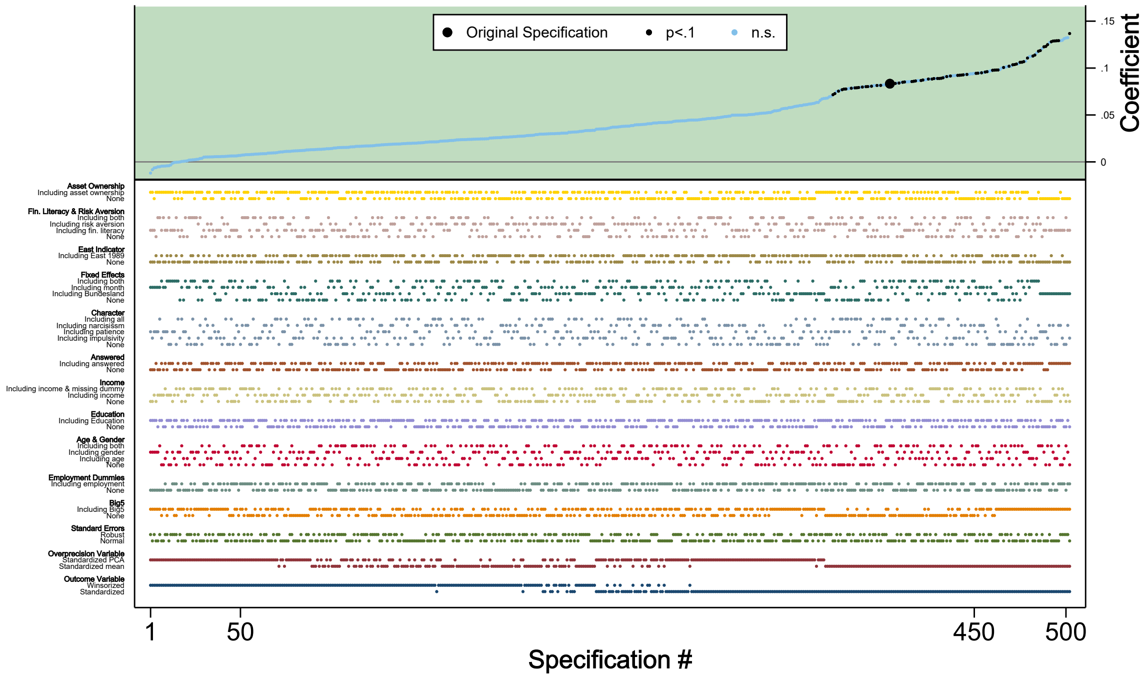


Figure D.1: Specification curve for *DAX Error*. Each point represents a different specification estimate. Black dots indicate statistically significant coefficients at the 10% level. The x-axis lists the individual specifications, while the y-axis shows the corresponding point estimates. Control variables are indicated by presence or absence bars beneath the x-axis. The big dark dot represents the specification reported in Table 2.

The four specification curves offer robust visual evidence supporting the robustness of the main findings. In all cases, the sign of the estimated effect remains the same:

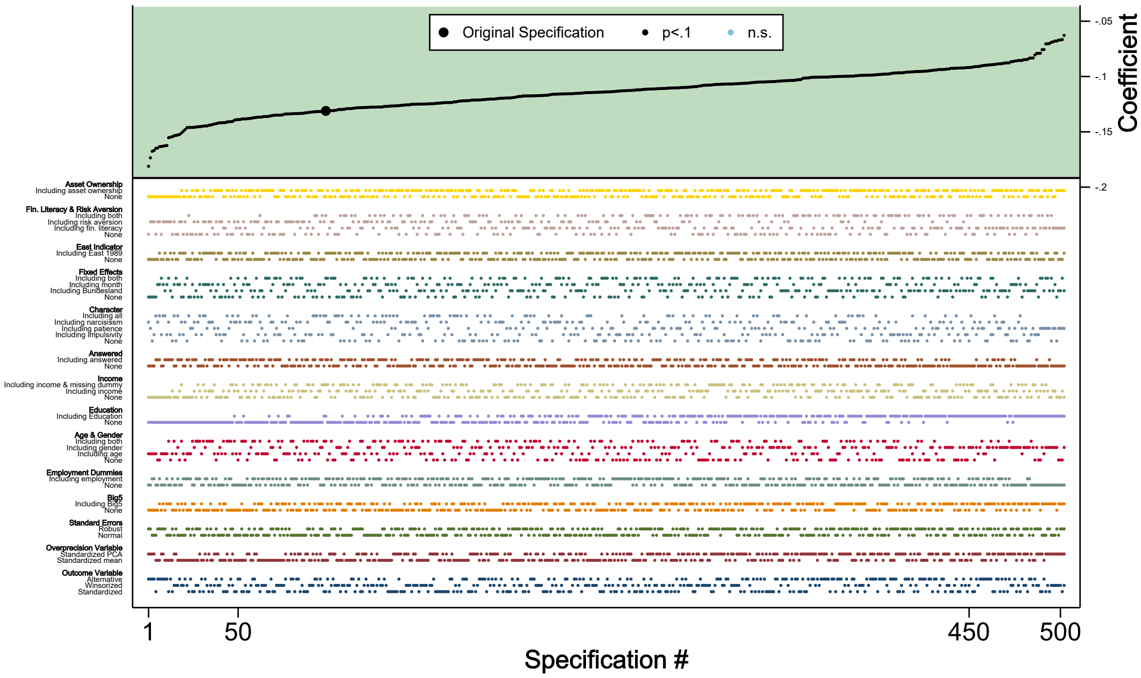


Figure D.2: Specification curve for *Diversification*. Each point represents a different specification estimate. Black dots indicate statistically significant coefficients at the 10% level. The x-axis lists the individual specifications, while the y-axis shows the corresponding point estimates. Control variables are indicated by presence or absence bars beneath the x-axis. The big dark dot represents the specification reported in Table 2.

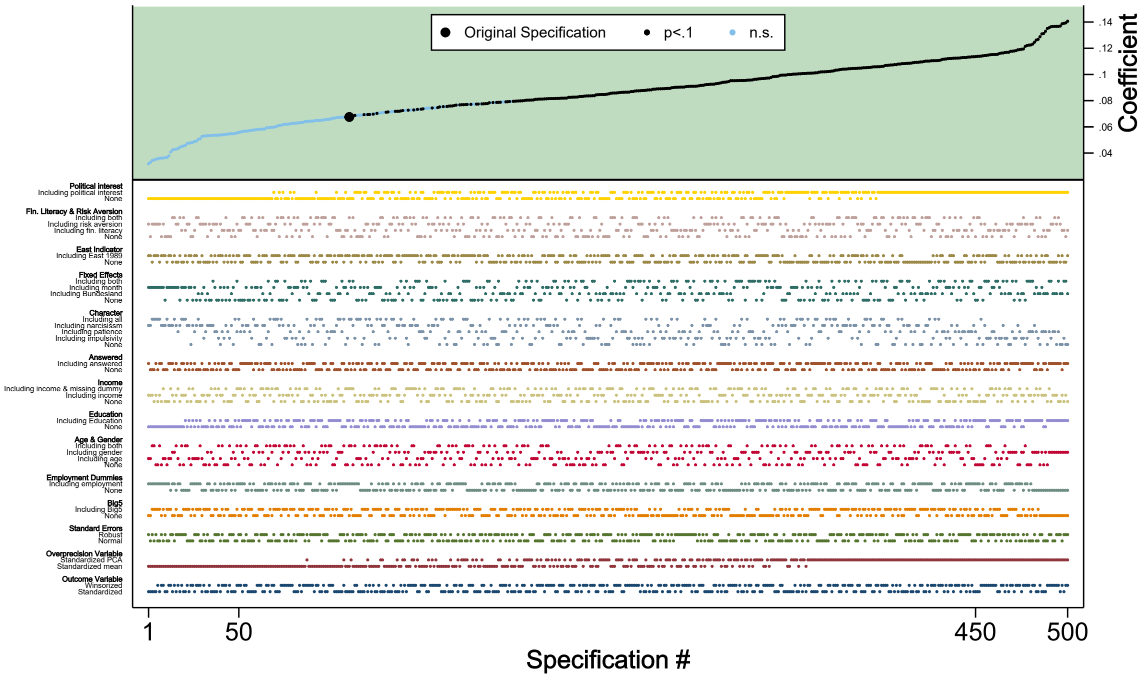


Figure D.3: Specification curve for *Extremism*. Each point represents a different specification estimate. Black dots indicate statistically significant coefficients at the 10% level. The x-axis lists the individual specifications, while the y-axis shows the corresponding point estimates. Control variables are indicated by presence or absence bars beneath the x-axis. The big dark dot represents the specification reported in Table 2.

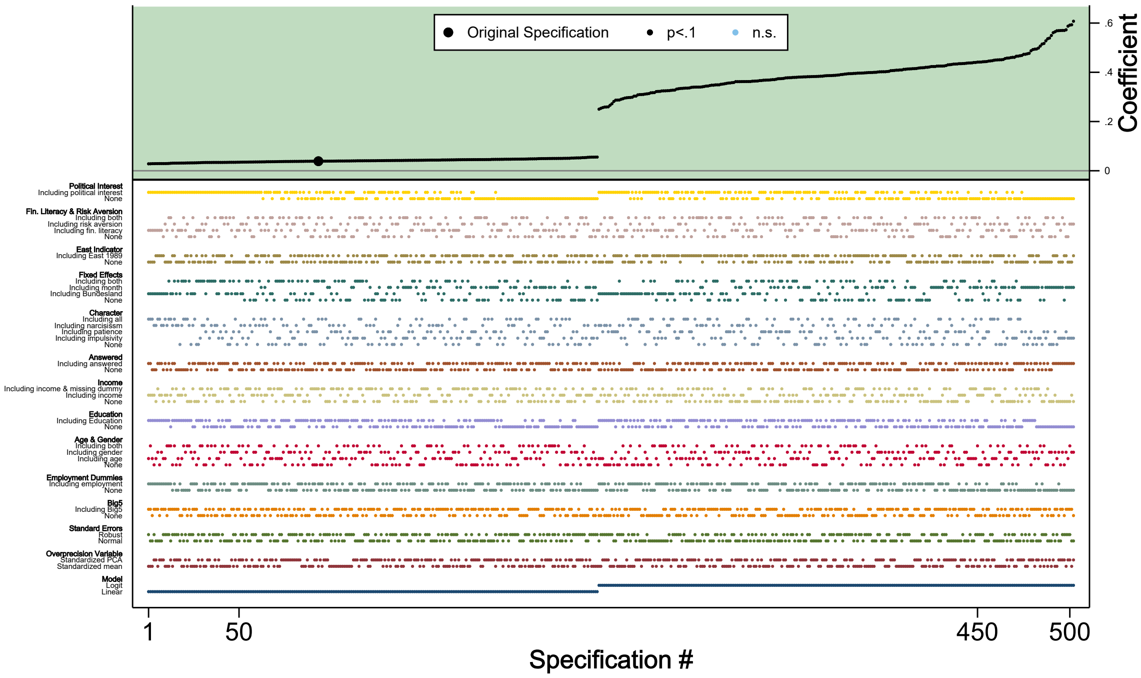


Figure D.4: Specification curve for *Abstention*. Each point represents a different specification estimate. Black dots indicate statistically significant coefficients at the 10% level. The x-axis lists the individual specifications, while the y-axis shows the corresponding point estimates. Control variables are indicated by presence or absence bars beneath the x-axis. The big dark dot represents the specification reported in Table 2.

positive for *DAX Error*, *Extremism*, and *Abstention*, and negative for *Diversification*. Such consistency in sign across specifications is notable and suggests that the link between overprecision and each primary outcome is not sensitive to estimation choices. Moreover, the presence of a substantial number of statistically significant estimates (black dots) across the distributions indicates that the observed associations are not the product of a few outlier specifications or narrow subsets of covariates. The only partial exceptions occur for the *DAX Error* and *Extremism* outcomes: for *DAX Error*, winsorizing the dependent variable reduces the magnitude of the estimated effects, occasionally rendering them insignificant or negative; for *Extremism*, specifications that omit political interest and rely on the standardized mean overprecision measure tend to yield smaller and mostly insignificant estimates. Despite these deviations, the estimated effects remain positive in the vast majority of specifications, reinforcing the overall directional consistency of the findings.

Another key feature of the plotted specification curves is the narrow range of coefficient estimates. For diversification and political extremism, estimates are tightly bunched and relatively stable, while for *DAX Error* the range is somewhat broader, though the direction of the effect remains unidirectional. For voting abstention, a mechanical jump in coefficient values results from the shift of linear estimation method to Logit; otherwise, the coefficient range is quite narrow. Perhaps most importantly, inspecting the variable combinations along the x-axis reveals no clear pattern whereby a specific control or group of controls consistently appears at either end of the distribution. Variables such as income, education, gender, political interest, or financial literacy appear throughout the curves, sometimes in specifications with smaller coefficients, and sometimes in those with larger ones. If any variable systematically weakens or strengthens the observed relationship, it would tend to cluster at one end of the coefficient range. Instead, the scattered inclusion pattern provides compelling evidence that the estimated effects of overprecision are robust to modeling choices and are not driven by any particular covariate or set of controls.

E Comparing the Survey Data to the SOEP Data

In the following, we compare the Subjective Error Method answers to the contemporary history questions for the online survey with those in the SOEP survey. To avoid any distortion from outliers, all variables are trimmed at the 1 and 99 percentile in both datasets.

Figure E.1 plots the distributions of answers to the history questions in the SOEP and the online survey. With the exception of the Beetle question, the distributions are relatively similar across samples. Table E.1 shows the results of a standard t-test on the means and of a test on the equality of the standard deviations of both samples. Except for the mean of the question regarding the VW Beetle (1938), the means of the answers to the questions differ by little, even if this difference is statistically significant in most cases. Interestingly, in most cases, the SOEP sample is closer to the correct answer. This could be explained by more effort of the respondents in the face-to-face interviews than in the online survey and speaks in favor of the 'Google' filters we introduced to filter our online sample. The dispersion of the answers around the mean is relatively similar for both samples except for the question regarding Hussein (2003) where SOEP respondents are more closely around the mean. However, the analysis suggests that knowledge about contemporary history is relatively similarly distributed among both samples.

Figure E.2 plots the distributions of the overprecision measure for the respective history questions contained in both the SOEP survey and the companion online survey. The distributions are graphically relatively similar in both samples. The statistical tests in Table E.2, however, show that the respondents in the SOEP sample, on average, tend to be slightly more overprecise for most of the questions. One explanation for this result could be that respondents try to show off in front of the interviewers in the face-to-face interviews by stating lower subjective errors. Taken together, the distributions of the overprecision measures only marginally differ among both samples, pointing to the robustness of the Subjective Error Method.

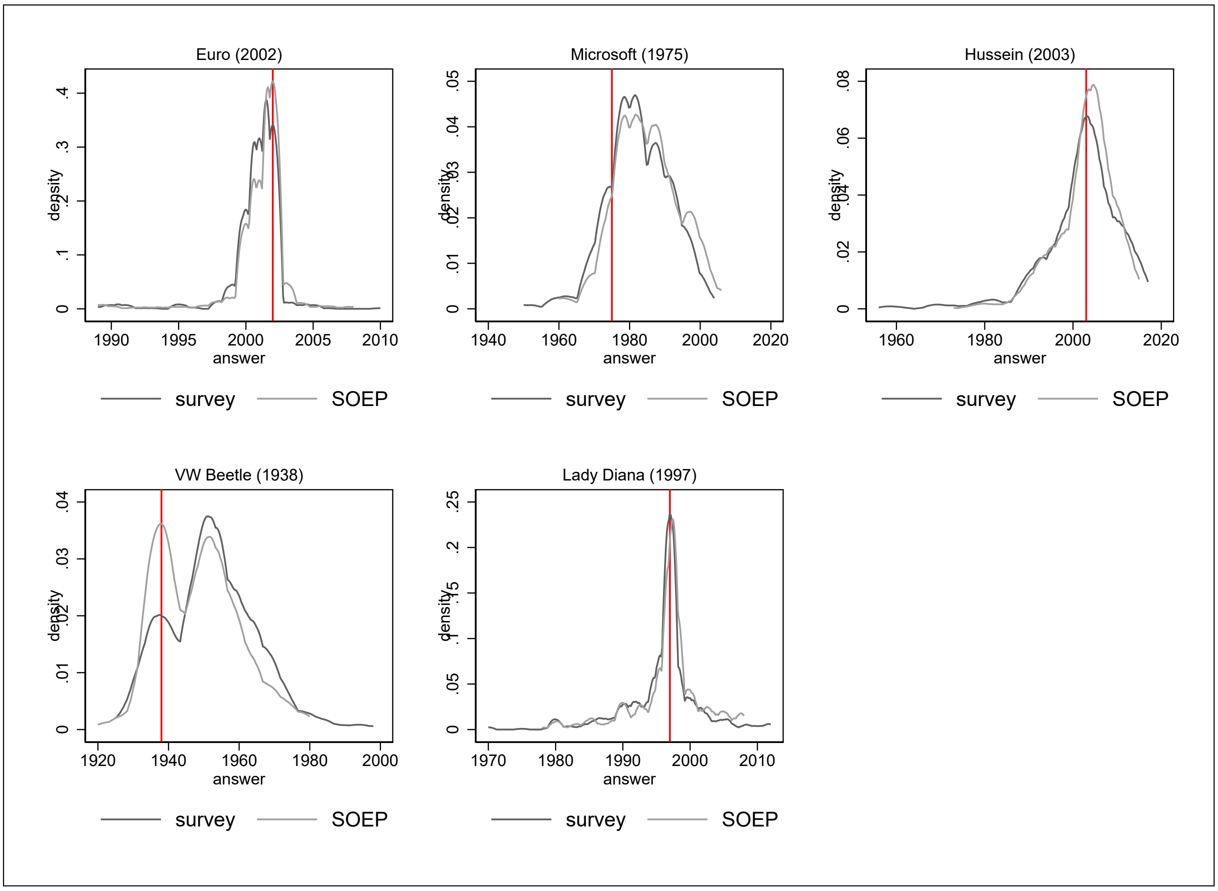


Figure E.1: Density of the answers ($a_{i,j}$) for each of the history questions contained in both the SOEP survey and the companion online survey. The vertical line marks the correct answer. Note that the vertical axis is different for each question.

question	mean (survey)	mean (SOEP)	Δ	p-val	sd (survey)	sd (SOEP)	Δ	p-val
Euro (2002)	2000,835	2001,124	-0,289	0,004	1,980	2,055	-0,075	0,294
Microsoft (1975)	1982,691	1984,929	-2,237	0,000	8,974	9,108	-0,134	0,697
Hussein (2003)	2002,212	2002,955	-0,742	0,077	8,718	6,568	2,150	0,000
VW Beetle (1938)	1952,268	1948,535	3,734	0,000	12,561	11,502	1,059	0,016
Lady Diana (1997)	1995,975	1996,680	-0,704	0,010	5,367	5,141	0,226	0,240

Table E.1: Statistical comparison of the answers between the companion online and SOEP survey. This table shows the statistical comparison of the answers to the history questions between the survey and the SOEP sample.

question	mean (survey)	mean (SOEP)	Δ	p-val	sd (survey)	sd (SOEP)	Δ	p-val
Euro (2002)	0,143	0,106	0,037	0,728	2,474	1,562	0,913	0,000
Microsoft (1975)	2,953	6,662	-3,708	0,000	8,687	7,979	0,708	0,031
Hussein (2003)	0,293	1,562	-1,269	0,000	6,707	4,554	2,153	0,000
VW Beetle (1938)	8,222	7,238	0,984	0,076	11,352	9,290	2,062	0,000
Lady Diana (1997)	0,096	0,663	-0,567	0,008	4,412	3,622	0,790	0,000

Table E.2: Statistical comparison of the overprecision measures between the companion online and SOEP survey. This table shows the statistical comparison of the overprecision measures by question between the survey and the SOEP sample.

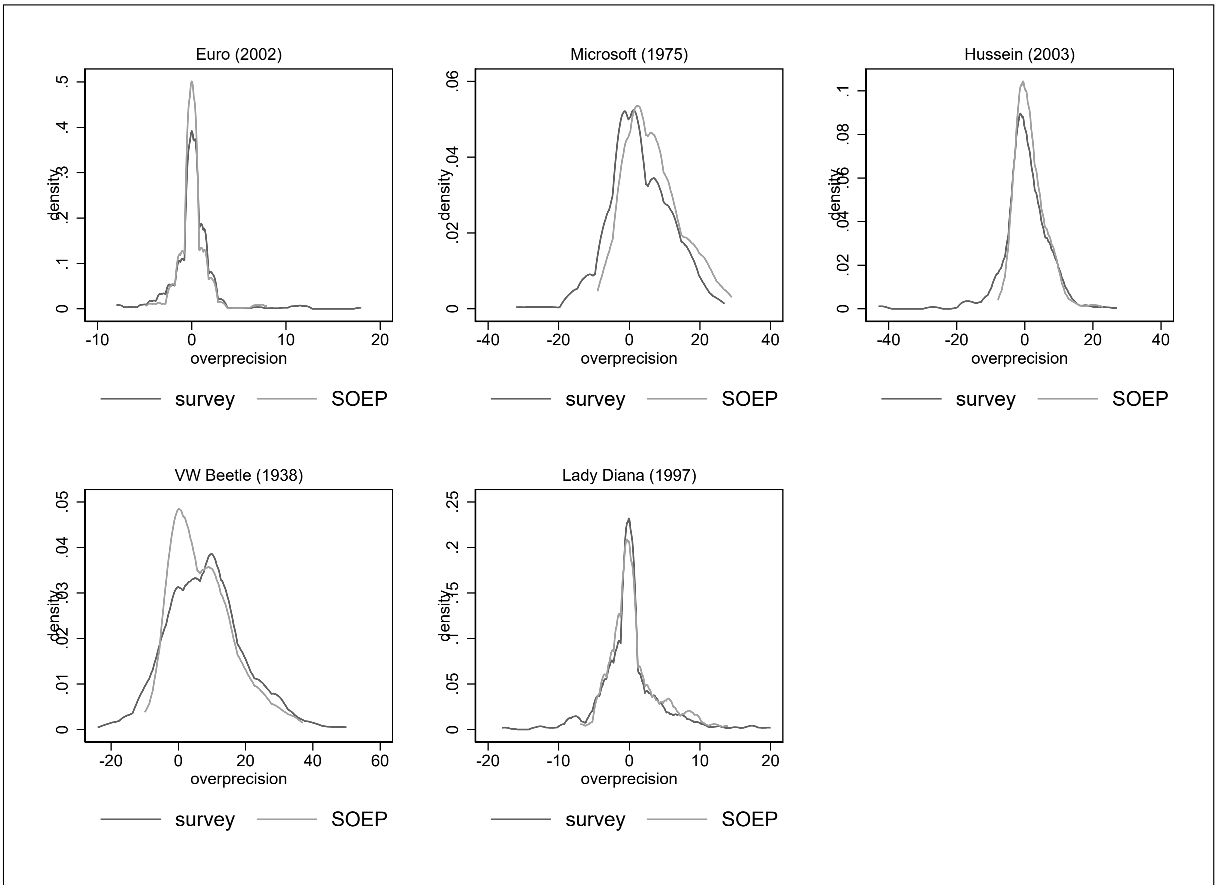


Figure E.2: Density of the overprecision measure ($op_{i,j}$) for each of the history questions contained in both the SOEP survey and the companion online survey. Note that the vertical axis is different for each question.

F Linking Overprecise Behavior Across the Financial and Political Domains

As shown in section 4, overprecision correlates with respondents' financial and political behaviors. However, it remains possible that *distinct* aspects of overprecision are related to behavior in each domain separately. To formalize this possibility, assume that our measured overprecision (op) comprises two orthogonal factors,[48] F_1 and F_2 plus an idiosyncratic error term such as:

$$op = F_1 + F_2 + \epsilon_{op}. \tag{7}$$

Consider two outcome variables, Y_1 (say, a financial behavior) and Y_2 (a political behavior) and suppose that domain 1 depends only on F_1 while domain 2 depends only on F_2. Mathematically, this would imply:

$$Y_1 = \beta_1 op + \eta_1 = \beta_1(1 \cdot F_1 + 0 \cdot F_2 + \epsilon_{op}) + \eta_1 = \beta_1 F_1 + \epsilon_1, \tag{8}$$

$$Y_2 = \beta_2 op + \eta_2 = \beta_2(0 \cdot F_1 + 1 \cdot F_2 + \epsilon_{op}) + \eta_2 = \beta_2 F_2 + \epsilon_2. \tag{9}$$

When estimating the two relationships separately, the estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ would capture the effect of distinct sources of variation, or factors, in the overprecision measure. Instead, we propose a two-stage procedure to determine whether a common source of variation in the overprecision measure correlates with both outcomes.

Let us take domain 1 as an example. In the first step, we regress the outcome Y_1 on the measure of overprecision:

$$Y_1 = \alpha_1 + \beta_1 op + \eta_1 = \alpha_1 + \beta_1 F_1 + \epsilon_1. \tag{10}$$

From it, we obtain the fitted values,

$$\hat{Y}_1 = \hat{\alpha}_1 + \hat{\beta}_1 F_1, \tag{11}$$

[48] The assumption of orthogonality is for ease of exposition; non-orthogonal factors introduce additional covariances but do not alter the fundamental logic of our two-step procedure.

where the fitted values $\hat{Y}_1$ capture the component of *op* variation that predicts Y_1.

In the second step, we regress the outcome in the other domain (Y_2) on the fitted values from the first step such that:

$$\begin{aligned} Y_2 &= \alpha_2 + \delta_2 \hat{Y}_1 + \mu_2 = \alpha_2 + \delta_2(\hat{\alpha}_1 + \hat{\beta}_1 F_1) + \mu_2, \\ &= (\alpha_2 + \delta_2 \hat{\alpha}_1) + \delta_2(\hat{\beta}_1 F_1) + \mu_2. \end{aligned} \tag{12}$$

If the estimated coefficient $\hat{\delta}_2$ is statistically significant, this suggests that the same variation in overprecision correlates with outcomes in both domains.

In practice, true behavior is likely more complex than the simplified factorization in Equations (8) and (9). We consider this approach a diagnostic tool for detecting a *common* source of overprecision rather than as a test of causality. Additionally, this two-stage approach does not fully account for latent variables that may jointly influence overprecision across domains. A structural equation model would provide a complementary approach to address this concern by explicitly modeling both direct and indirect pathways between financial and political behavior. Unlike our two-stage regression framework, structural models allow for the simultaneous estimation of multiple relationships and incorporate latent constructs that account for unobserved factors affecting both domains.

However, structural equation models also require strong distributional assumptions. Additionally, they necessitate predefined theoretical pathways, specifying whether financial overprecision directly influences political overprecision or if the relationship is mediated by another factor. By contrast, our two-stage approach is assumption-light, providing an empirical test of whether overprecision-related behavior in one domain predicts overprecision-related behavior in another while allowing the data to determine the nature of these relationships. This approach is ideal for exploratory analysis and opens the door to future research that can establish *a priori* links, to further disentangle causal mechanisms and uncover unbiased correlations between overprecision in financial, political, and other domains.

G Robustness Tests Companion Online Survey

In the following, we test the robustness of the analyses of the online experiment in Section 5 as outlined in the pre-registration. All robustness tests are concerned with the exclusion restrictions specified in the pre-analysis plan. In the baseline analysis, we exclude all respondents who admit to using a third party to answer our questions and respondents who we identify as 'Googlers' by using our control questions. We do so by excluding from the entire analysis those respondents who answer the Google controls correctly and state a subjective error of zero while displaying the same behavior for at least three other questions within the same domain. Additionally, we exclude the lowest fifth percentile on the self-reported effort measure to ensure data quality.

In total we have seven different robustness tests. Robustness tests (1) and (2) change the exclusion restriction to detect 'Googlers'. Robustness test (1) drops all respondents that have one of the Google control questions correct while stating a subjective error of zero, as opposed to the original restriction where we required respondents to answer all Google control questions correctly. Robustness test (2) drops all respondents who get at least four of the five answers in any domain correct. Robustness test (3) also excludes all respondents who are in the lowest decile of time spent on each survey question at least 20% of the time across the entire survey, while robustness test (4) repeats the same for the highest decile. Robustness test (5) increases the threshold of self-reported quality to 10% and robustness test (6) to 20%. Robustness test (7) additionally excludes those respondents who gave the same answer to more than two of the respective first questions within a domain to control for response patterns.

Table G.1 shows the robustness test results for the factor analysis, Table G.2 for the partial correlation analysis, and Table G.3 for the leave-one-out analysis. The results show that by adding more stringent exclusion restrictions concerning 'Googlers' does not substantially change the results. Adding further restrictions, which improve the quality of answers but reduce the number of observations, only marginally changes the results, predominantly improving the results.

		Robustness						
	Baseline	1	2	3	4	5	6	7
Eigenvalue	2.29	2.30	2.28	2.34	2.23	2.27	2.32	2.25
Loading neutral	0.52	0.52	0.52	0.53	0.60	0.51	0.49	0.48
Loading history	0.76	0.76	0.76	0.76	0.72	0.76	0.78	0.77
Loading knowledge	0.74	0.75	0.74	0.75	0.72	0.73	0.75	0.74
Loading stocks	0.69	0.69	0.68	0.71	0.68	0.68	0.67	0.68
Loading economics	0.65	0.65	0.65	0.65	0.61	0.66	0.67	0.65
N	552	544	547	487	440	535	476	500

Table G.1: Robustness of factor analysis. Column (Baseline) shows the baseline results with the exclusion restrictions specified in Section 5.2. Each of the robustness tests changes one of the exclusion restrictions. Robustness tests (1) and (2) change the exclusion restriction to detect 'Googlers'. Robustness test (1) drops all respondents that have one of the hard questions, which we use to detect respondents who we assume to have used search engines, correct while stating a subjective error of zero. Robustness test (2) drops all respondents who get at least four of the five answers in any domain correct. Robustness test (3) additionally excludes all respondents who are in the lowest decile of time spent on each survey question at least 20% of the time across the entire survey, while robustness test (4) repeats the same for the highest decile. Robustness test (5) increases the threshold of self-reported quality to 10% and robustness test (6) to 20%. Robustness test (7) additionally excludes those respondents who gave the same answer to more than two of the respective first questions within a domain to control for response patterns.

				Robustness													
		Baseline		1		2		3		4		5		6		7	
Domain 1	Domain 1	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N	ρ	N
Neutral	History	0.17	688	0.17	680	0.17	679	0.18	588	0.19	560	0.16	658	0.18	578	0.16	632
Neutral	Knowledge	0.21	626	0.22	618	0.20	617	0.23	539	0.22	503	0.19	603	0.18	538	0.18	570
Neutral	Stocks	0.30	676	0.30	668	0.29	670	0.32	577	0.33	546	0.28	648	0.24	575	0.25	619
Neutral	Economics	0.18	639	0.18	632	0.18	633	0.17	552	0.20	518	0.17	613	0.19	542	0.17	580
History	Knowledge	0.52	609	0.52	600	0.52	600	0.52	531	0.49	488	0.52	586	0.55	522	0.53	554
History	Stocks	0.35	638	0.35	629	0.35	632	0.37	557	0.29	514	0.35	614	0.37	544	0.35	584
History	Economics	0.37	620	0.37	612	0.37	614	0.37	542	0.31	499	0.37	596	0.38	526	0.37	565
Knowledge	Stocks	0.34	591	0.34	582	0.33	585	0.37	516	0.32	471	0.33	573	0.34	509	0.32	537
Knowledge	Economics	0.32	580	0.33	572	0.32	574	0.33	507	0.28	461	0.32	562	0.35	502	0.30	525
Stocks	Economics	0.28	618	0.28	610	0.27	613	0.30	540	0.24	500	0.29	594	0.29	529	0.28	561

Table G.2: Robustness of partial correlation analysis. Column (Baseline) shows the baseline results with the exclusion restrictions specified in Section 5.2. Each of the robustness tests changes one of the exclusion restrictions. Robustness tests (1) and (2) change the exclusion restriction to detect 'Googlers'. Robustness test (1) drops all respondents that have one of the hard questions, which we use to detect respondents who we assume to have used search engines, correct while stating a subjective error of zero. Robustness test (2) drops all respondents who get at least four of the five answers in any domain correct. Robustness test (3) additionally excludes all respondents who are in the lowest decile of time spent on each survey question at least 20% of the time across the entire survey, while robustness test (4) repeats the same for the highest decile. Robustness test (5) increases the threshold of self-reported quality to 10% and robustness test (6) to 20%. Robustness test (7) additionally excludes those respondents who gave the same answer to more than two of the respective first questions within a domain to control for response patterns.

				Robustness													
		Baseline		1		2		3		4		5		6		7	
In	Out	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p	Δ	p
Neutral	History	-0.348	< 0.01	-0.335	< 0.01	-0.363	< 0.01	-0.394	< 0.01	-0.335	< 0.01	-0.346	< 0.01	-0.352	< 0.01	-0.335	< 0.01
Neutral	Knowledge	-0.406	< 0.01	-0.422	< 0.01	-0.373	< 0.01	-0.519	< 0.01	-0.368	< 0.01	-0.397	< 0.01	-0.409	< 0.01	-0.396	< 0.01
Neutral	Stocks	-0.339	< 0.01	-0.343	< 0.01	-0.341	< 0.01	-0.354	< 0.01	-0.338	< 0.01	-0.301	< 0.01	-0.273	< 0.01	-0.328	< 0.01
Neutral	Economics	-0.323	< 0.01	-0.322	< 0.01	-0.340	< 0.01	-0.326	< 0.01	-0.338	< 0.01	-0.311	< 0.01	-0.364	< 0.01	-0.311	< 0.01
History	Neutral	-0.218	0.01	-0.184	0.03	-0.221	0.01	-0.243	0.01	-0.288	< 0.01	-0.207	0.02	-0.273	< 0.01	-0.163	0.07
History	Knowledge	-0.566	< 0.01	-0.572	< 0.01	-0.547	< 0.01	-0.605	< 0.01	-0.547	< 0.01	-0.549	< 0.01	-0.534	< 0.01	-0.556	< 0.01
History	Stocks	-0.432	< 0.01	-0.436	< 0.01	-0.453	< 0.01	-0.439	< 0.01	-0.477	< 0.01	-0.421	< 0.01	-0.368	< 0.01	-0.387	< 0.01
History	Economics	-0.385	< 0.01	-0.393	< 0.01	-0.383	< 0.01	-0.366	< 0.01	-0.439	< 0.01	-0.370	< 0.01	-0.405	< 0.01	-0.390	< 0.01
Knowledge	Neutral	-0.230	0.01	-0.244	< 0.01	-0.215	0.01	-0.247	0.01	-0.186	0.05	-0.206	0.02	-0.150	0.10	-0.213	0.02
Knowledge	History	-0.451	< 0.01	-0.480	< 0.01	-0.463	< 0.01	-0.381	< 0.01	-0.454	< 0.01	-0.468	< 0.01	-0.419	< 0.01	-0.488	< 0.01
Knowledge	Stocks	-0.323	< 0.01	-0.336	< 0.01	-0.314	< 0.01	-0.349	< 0.01	-0.341	< 0.01	-0.315	< 0.01	-0.271	< 0.01	-0.325	< 0.01
Knowledge	Economics	-0.301	< 0.01	-0.307	< 0.01	-0.291	< 0.01	-0.301	< 0.01	-0.308	< 0.01	-0.281	< 0.01	-0.320	< 0.01	-0.279	< 0.01
Stocks	Neutral	-0.449	< 0.01	-0.418	< 0.01	-0.436	< 0.01	-0.451	< 0.01	-0.467	< 0.01	-0.421	< 0.01	-0.406	< 0.01	-0.407	< 0.01
Stocks	History	-0.421	< 0.01	-0.408	< 0.01	-0.421	< 0.01	-0.416	< 0.01	-0.467	< 0.01	-0.416	< 0.01	-0.411	< 0.01	-0.361	< 0.01
Stocks	Knowledge	-0.368	< 0.01	-0.408	< 0.01	-0.360	< 0.01	-0.432	< 0.01	-0.381	< 0.01	-0.364	< 0.01	-0.352	< 0.01	-0.325	< 0.01
Stocks	Economics	-0.371	< 0.01	-0.395	< 0.01	-0.369	< 0.01	-0.437	< 0.01	-0.394	< 0.01	-0.397	< 0.01	-0.459	< 0.01	-0.348	< 0.01
Economics	Neutral	-0.338	< 0.01	-0.313	< 0.01	-0.358	< 0.01	-0.344	< 0.01	-0.346	< 0.01	-0.323	< 0.01	-0.302	< 0.01	-0.319	< 0.01
Economics	History	-0.384	< 0.01	-0.367	< 0.01	-0.386	< 0.01	-0.336	< 0.01	-0.445	< 0.01	-0.376	< 0.01	-0.359	< 0.01	-0.341	< 0.01
Economics	Knowledge	-0.292	< 0.01	-0.306	< 0.01	-0.277	< 0.01	-0.273	0.01	-0.320	< 0.01	-0.281	< 0.01	-0.301	< 0.01	-0.262	< 0.01
Economics	Stocks	-0.381	< 0.01	-0.386	< 0.01	-0.330	< 0.01	-0.364	< 0.01	-0.360	< 0.01	-0.363	< 0.01	-0.359	< 0.01	-0.340	< 0.01

Table G.3: Robustness of leave-one-out-analysis. Column (Baseline) shows the baseline results with the exclusion restrictions specified in Section 5.2. Each of the robustness tests changes one of the exclusion restrictions. Robustness tests (1) and (2) change the exclusion restriction to detect 'Googlers'. Robustness test (1) drops all respondents that have one of the hard questions, which we use to detect respondents who we assume to have used search engines, correct while stating a subjective error of zero. Robustness test (2) drops all respondents who get at least four of the five answers in any domain correct. Robustness test (3) additionally excludes all respondents who are in the lowest decile of time spent on each survey question at least 20% of the time across the entire survey, while robustness test (4) repeats the same for the highest decile. Robustness test (5) increases the threshold of self-reported quality to 10% and robustness test (6) to 20%. Robustness test (7) additionally excludes those respondents who gave the same answer to more than two of the respective first questions within a domain to control for response patterns.

H Further Exploratory Analyses in the Companion Survey

Given the rich data we collect in the companion online survey in Section 5, we conduct further exploratory analyses which were not included in the pre-analysis plan.

H.1 Overprecision and Expertise

The literature on overconfidence and overprecision has shown that knowledge and expertise increase the accuracy of answers, but also the confidence of respondents. These two effects cancel out making experts and non-experts equally overconfident. An example of this effect is Önkal et al. (2003), who show that newcomers and experts have similar levels of overconfidence when predicting foreign exchange rates. While experts' predictions are more accurate, they also show more confidence in their answers, resulting in similar degrees of overconfidence. Similarly, McKenzie et al. (2008) show that the confidence intervals of experts are closer to the truth but also narrower than those of newcomers.

In the following, we test this relationship in our data using a self-reported measure of expertise. To do so, we regress the mean error, the mean subjective error, and the aggregate measure of overprecision in each domain on self-reported expertise in the same domain. Because self-reported expertise is likely to be influenced by overprecision, we control for the aggregate measure of overprecision. That is, for each domain, we predict the first component of a principal component analysis of the standardized aggregate overprecision measures across all domains, excluding the one that is analyzed.

The results in Table H.1 show that self-reported expertise in a domain results in both a decrease in the mean realized error but also in a decrease of the mean subjective error. Importantly, the results in Columns (3) and (4) show that overprecision also affects self-reported expertise. Controlling for overprecision, the coefficient sizes on the mean error and the mean subjective error are relatively similar. Therefore, consistent with the literature, we do not find an effect of expertise on the overprecision measure within the same domain.

	mean error		mean subjective error		mean overprecision	
	(1)	(2)	(3)	(4)	(5)	(6)
History:						
expertise	-0.036***	-0.039***	-0.044***	-0.041***	0.001	0.000
	(0.008)	(0.008)	(0.013)	(0.014)	(0.002)	(0.002)
overprecision		0.389*		-3.437***		0.581***
		(0.208)		(0.856)		(0.129)
N	703	552	703	552	703	552
adj. R^2	0.032	0.042	0.025	0.312	-0.001	0.291
Knowledge:						
expertise	-0.030*	-0.030*	-0.054***	-0.056***	0.002	0.002
	(0.015)	(0.017)	(0.017)	(0.017)	(0.002)	(0.002)
overprecision		1.287***		-4.277***		0.484***
		(0.414)		(0.596)		(0.055)
N	633	552	633	552	633	552
adj. R^2	0.005	0.028	0.023	0.370	0.001	0.272
Stocks:						
expertise	-0.026**	-0.028**	-0.058***	-0.046***	0.003**	0.002
	(0.011)	(0.012)	(0.012)	(0.011)	(0.001)	(0.001)
overprecision		0.983*		-3.655***		0.443***
		(0.555)		(0.735)		(0.075)
N	690	552	690	552	690	552
adj. R^2	0.005	0.020	0.024	0.214	0.004	0.219
Economics:						
expertise	-0.039***	-0.043***	-0.066***	-0.050***	0.002	0.001
	(0.011)	(0.011)	(0.015)	(0.014)	(0.002)	(0.001)
overprecision		0.429*		-3.839***		0.376***
		(0.250)		(0.747)		(0.067)
N	647	552	647	552	647	552
adj. R^2	0.019	0.025	0.030	0.230	0.002	0.188

Robust standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table H.1: Self-reported expertise and the overprecision measure. This table reports the results of regressing the mean error, the mean subjective error, and the overprecision measure of one domain on the self-reported expertise in the same domain. In Columns (2), (4), and (6) we additionally control for overprecision, which is measured as the first component of a principal component analysis of the overprecision measures across all domains but the one that is analyzed.

H.2 Overprecision and Forecast Errors

This section replicates the results for the stock price forecast errors in Section 4.2 on a different time horizon. We calculate the absolute one-month-ahead forecast error for each stock and regress the resulting forecast error on a measure of overprecision with and without control variables (Columns (1) to (10) in Table H.2). We also compute the principal component across all five stocks and regress this aggregate forecast error on a measure of overprecision and control variables in Column (11).

The first panel replicates the analysis in Section 4.2 using the standardized aggregate overprecision measure in the contemporary history domain. The results show that overprecision measured in the history domain is weakly positively correlated with the forecast error in two of the stocks but not significantly correlated with the forecast error in the other three stocks. The correlation with the principal component across all forecast errors is positive but insignificant. The results, hence, do not contradict the findings for the one- and two-year-ahead forecasts in Section 4.2. In the second panel, we use the standardized aggregate overprecision measure from the stock forecast domain instead of the history domain. The results for all stocks and the principal component show a positive and significant correlation between overprecision measured in the stock forecast domain and the forecast errors. However, it is likely that the measured relationship is mechanical.

Because of the potential mechanical relationship, in the third and fourth panel, we use the principal component of the standardized aggregate overprecision measures across all five domains.[49] This reduces the number of observations since it requires an overprecision measure for each domain per respondent, however, since the measure is based on more questions it becomes more reliable.[50] The results once more show a positive and significant correlation between overprecision and the absolute forecast error for most of the stocks and the principal component. In the fourth panel, we winsorize the forecast errors at 5th and 95th percentile to avoid outliers influencing the results. As expected, the coefficients are smaller in terms of size, but the qualitative result is unaffected.

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[49] The principal component analysis only yields one factor with an eigenvalue of 2.29 and relatively similar factor loadings across domains (0.52,0.76,0.74,0.69,0.65).

[50] The principal component calculated across all questions, instead of across all domain-specific aggregate overprecision measures, exhibits a Spearman correlation coefficient of 0.96 with the principal component calculated across all domain-specific aggregate overprecision measures.

Forecast error in:	Benz		Puma		BMW		Post		BASF		PC
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Overprecision in history domain:											
sop	0.924**	0.779*	0.303	0.167	0.066	-0.074	-0.431	-0.577	0.581*	0.441	0.018
	(0.466)	(0.446)	(0.432)	(0.428)	(0.498)	(0.501)	(0.557)	(0.568)	(0.297)	(0.297)	(0.041)
N	665	663	669	667	665	663	657	655	664	662	641
adj. R^2	0.008	0.036	-0.000	0.005	-0.001	0.031	0.000	0.028	0.005	0.001	0.028
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Overprecision in stock forecast domain:											
sop	5.400***	5.141***	4.981***	4.831***	6.460***	6.025***	2.718***	2.467***	3.631***	3.463***	0.566***
	(1.000)	(0.963)	(0.831)	(0.845)	(1.503)	(1.462)	(0.794)	(0.819)	(0.660)	(0.666)	(0.106)
N	687	683	687	683	686	682	683	679	687	683	666
adj. R^2	0.272	0.276	0.231	0.225	0.224	0.225	0.063	0.071	0.198	0.175	0.314
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Principal component of overprecision:											
sop	2.769***	2.638***	1.970**	1.862**	2.990**	2.740**	0.845	0.555	1.864***	1.736***	0.243**
	(0.903)	(0.885)	(0.806)	(0.806)	(1.241)	(1.217)	(0.781)	(0.788)	(0.577)	(0.570)	(0.098)
N	550	548	551	549	549	547	548	546	551	549	539
adj. R^2	0.083	0.094	0.043	0.043	0.052	0.070	0.005	0.021	0.057	0.041	0.075
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Principal component of overprecision & winsorized outcomes:											
sop	1.011***	0.880***	1.328***	1.215**	0.521	0.421	0.969*	0.767	1.134***	1.016***	0.094**
	(0.323)	(0.313)	(0.510)	(0.505)	(0.332)	(0.331)	(0.516)	(0.513)	(0.321)	(0.313)	(0.043)
N	550	548	551	549	549	547	548	546	551	549	539
adj. R^2	0.029	0.045	0.032	0.042	0.005	0.015	0.016	0.040	0.040	0.025	0.036
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
Principal component of overprecision without finance domain:											
sop	1.515**	1.339**	0.705	0.508	1.405*	1.029	0.143	-0.188	0.971**	0.788*	0.086
	(0.636)	(0.618)	(0.648)	(0.649)	(0.842)	(0.835)	(0.693)	(0.709)	(0.441)	(0.438)	(0.071)
N	556	554	557	555	556	554	552	550	557	555	542
adj. R^2	0.024	0.039	0.004	0.008	0.010	0.033	-0.002	0.018	0.014	0.001	0.023
controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes

Robust standard errors in parentheses
$^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table H.2: Forecast errors and the overprecision measure. This table reports the results of regressing the forecast errors for each of the stocks (Columns (2) to (10)) and the principal component across all stocks on overprecision. The respective first columns show the results without including control variables, the respective second columns the results including control variables.

In the fifth panel, we exclude the stock forecast domain from the aggregate overprecision measure. The relationship between the forecast errors and overprecision measured in all other domains but the stock forecast domain is weaker, however, the qualitative results remain similar.

I Calibration Analysis

Individuals may interpret the second question of the Subjective Error Method differently when answering the question. Consider the example in Figure I.1, which illustrates a hypothetical distribution of the subjective error for two individuals. Both individuals possess the same knowledge about the question, correctly assess the distribution of the subjective error (dotted normal distribution), and make the same error (vertical red line). Since the Subjective Error Method asks for the absolute error, we transform the dotted subjective error distribution to a half-normal distribution and its cumulative distribution (both represented by continuous lines).[51] As shown in the graph, individual i, who interprets the question using a 90% confidence interval, reports a larger subjective error than individual j, who has a 68% confidence interval in mind. Consequently, two persons with the same expected error distribution and realized error could be classified as over or underprecise, depending on their interpretation of the Subjective Error Method's second question.

To test what percentile respondents have in mind when answering the second question of the Subjective Error Method, we run a second online companion survey using a representative sample of the German population. The pre-registration and its analysis plan are at aspredicted.com the number is #123146, and can found in https://aspredicted.org/2QD_7R1.

I.1 Survey Design

As in the first companion survey, this second survey asks respondents to estimate the year in which the five most frequently answered questions in the SOEP survey took place, as well as to estimate their subjective error. The difference is that, in this survey, subjects not only provided a point prediction for the year they believe the event occurred, but also reported a full probability distribution of their beliefs using the click-and-drag interface of Crosetto and De Haan (2022).[52] By comparing the point estimate to the probabilistic distribution reported by the subject, we obtain a full distribution of the subjective errors,

[51] Note that this assumes that the distribution is symmetric around zero.

[52] Each respondent is shown a graph with 20 years above and below their answer to the first question on the x-axis. By clicking and dragging points for each bin in the graph, the respondents can build a complete probability distribution, which is normalized to 100% after submitting the response.

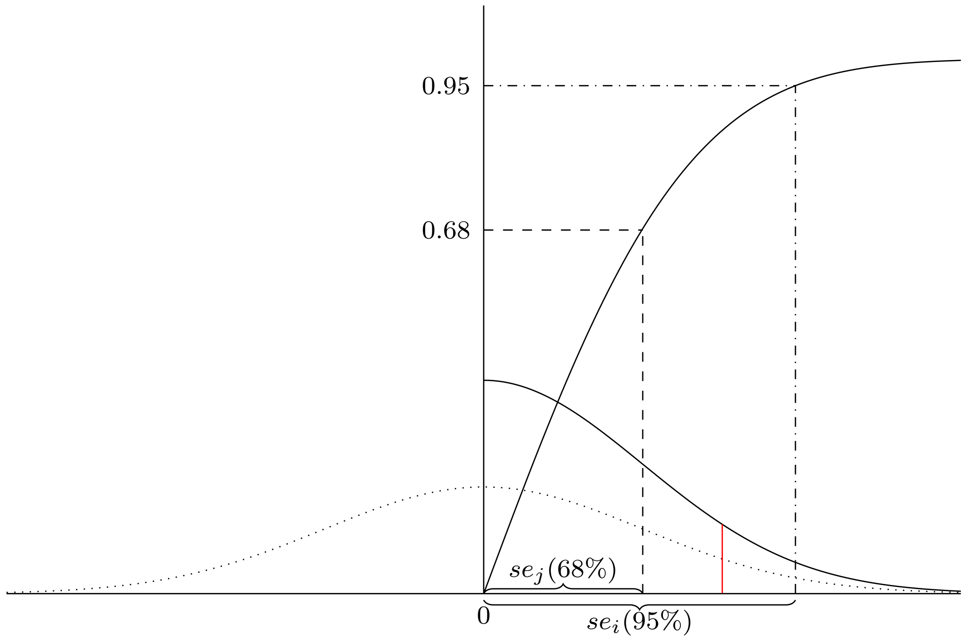


Figure I.1: Hypothetical distribution of the subjective error. The dotted normal distribution with a mean of zero and a standard deviation of two represents the subjective error for subjects i and j. The vertical red line marks the realized error for respondents i and j. The dashed line marks the 68% confidence interval of respondent i while the dot-dashed line marks the 95% confidence interval of respondent j.

with errors of +/-20 years or more summarized in the outer bins.[53] Half of the respondents received the distribution question before reporting their estimated subjective error, while the other half received them in reverse order.

To ensure the quality of the data, we incorporated attention checks during the survey, automatically excluding any respondent who failed them. To account for the use of Google or other search engines, respondents were asked at the end of the survey whether they had used such methods to answer the questions. Additionally, we included an extra question as a "Google control." This question followed the same format as all other questions but was intentionally more difficult and unlikely to be known by respondents. Specifically, we asked the year in which the first budgetary treaty of the EU was signed (1970).[54]

[53] For example, say a subject answers 1995 to the question about the year of the death of Lady Diana with a subjective error of 1 year. If she then builds a distribution that has 60% mass at 1995, 10% at 1994 and 1996 each, 5% at 1993 and 1997, and 2.5% at 1992 and 1998 then we have a full distribution of the subjective errors. By comparing the reported subjective error to the distribution, we can infer that when answering the subjective error question, she was reporting a 80% confidence interval. Notice that subjects were not required to build symmetric distributions.

[54] Following our pre-registration, we consider a respondent to have used a search engine if two conditions are met: i) answering the Google controls correctly and stating a subjective error of zero, and ii) answering correctly and stating zero subjective error for at least three other questions. If a respondent is flagged as

Importantly, in this survey we preregistered that we would use the answers to the Google control question in the analysis of the data, as we feared being underpowered.

At the end of the survey, all respondents answered a demographic survey consisting of age, gender, years of education, income, nationality, and mathematical literacy determined by solving three mathematical problems. We also asked respondents to self-report their knowledge of contemporary history on a scale of 0 (not knowledgeable at all) to 100 (very knowledgeable). Finally, we asked respondents to self-report the amount of effort they put into answering the survey.

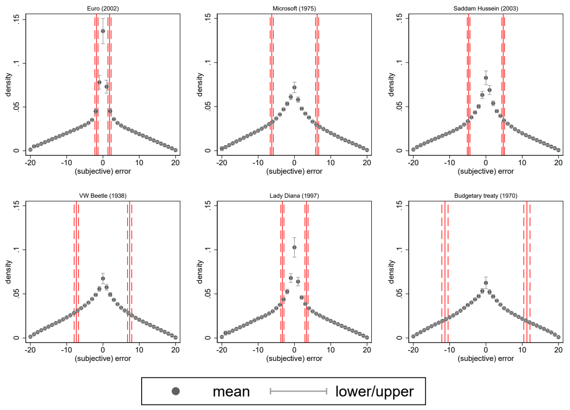
I.2 Data

We collected 1.000 complete responses. As preregistered, and to ensure that we only keep informative responses, we exclude from the analysis all respondents who admit to using a third- party to answer our questions. Additionally, we exclude respondents identified as 'Googlers' based on the control questions described in section 5.1 and drop the lowest five percentiles on the self-reported effort measure. This leaves us with 818 respondents for the baseline analysis. Of these, we keep respondents who provided at least four probability distributions, reducing the sample to 650. Finally, we exclude 6 respondents who submitted inconsistent probability distributions.

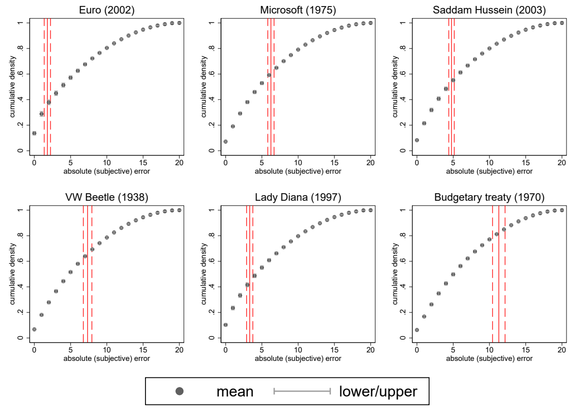
I.3 Results

Figure I.2a presents the probability density functions of the subjective error, averaged across the sample for each question. The gray dots represent the mean probability of each potential subjective error, with the 95% confidence interval of the mean shown as gray bars. The solid red vertical lines indicate the average subjective error, while the dashed vertical red lines mark the corresponding 95% confidence interval of the mean. It is clear that knowledge affects both the distributions provided by respondents and their subjective errors. For the presumably easier questions (e.g., the introduction of the Euro in 2002), the distributions are narrower, and the subjective errors are smaller. Figure I.2b plots the corresponding cumulative distribution functions of the *absolute* subjective error, averaged across the sample, along with the average subjective error. The raw data indicates that the location of the answer to the subjective error question varies within the subjective

having used search engines, then she is excluded from the analysis.



(a) PDF



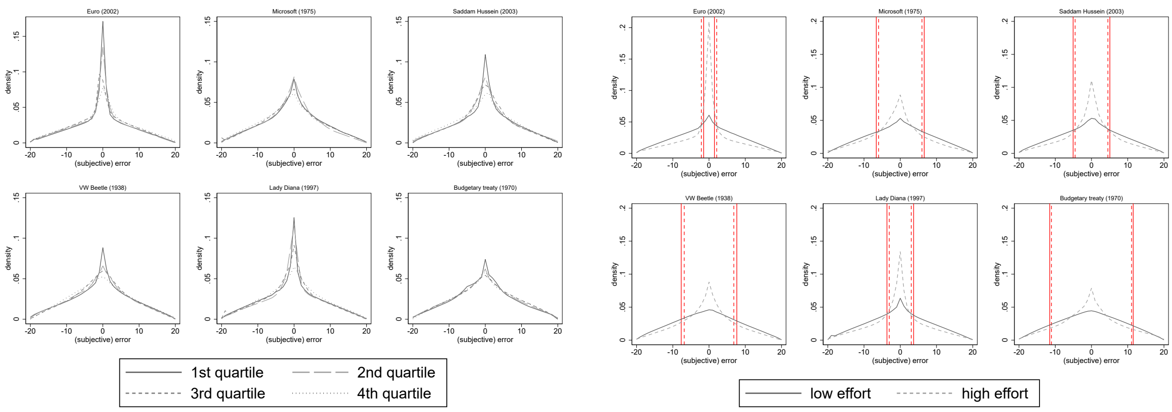
(b) CDF

Figure I.2: Subjective error and the probability distributions of the subjective error in the raw data. Lower/upper denotes the 95% confidence interval of the mean, shown as gray bars. The solid vertical red line denotes the average subjective error, while the dashed vertical red line plots the 95% confidence interval of the mean.

error distribution depending on the question's difficulty. For example, while the average subjective error corresponds to approximately the 40th percentile of the subjective error distribution for the Euro question, it reaches the 80th percentile for the 'hard' question designed to detect search engine use. This suggests that for more difficult questions, where respondents on average are more uncertain about the correct answer and provide wider distributions, they also give relatively larger subjective errors.

However, the raw data does not account for respondents' knowledge or effort in answering the distribution questions. To address the first issue, Figure I.3a plots the average provided distributions for four quartiles of the realized error for each question, which is used as a proxy for knowledge. The results of this exploratory analysis show that the lower the realized error, i.e., the better the knowledge of the respondent, the narrower the provided distributions.

The click-and-drag interface allows eliciting probability distributions in a user-friendly manner by allowing respondents to click and drag points along a distribution. However, a drawback of this interface is that the first click automatically produces a triangular distribution. Many respondents clicked only a few times on the interface, suggesting low effort levels. To address this, we run a second exploratory analysis. Figure I.3b



(a) Knowledge

(b) Effort

Figure I.3: Subjective error distributions and knowledge/effort. The solid vertical red line denotes the average subjective error, while the dashed vertical red line plots the 95% confidence interval of the mean. Knowledge is defined as the four quartiles of the absolute error in the respective question. Effort is defined as below/above the median time spent on the respective question.

proxies effort by splitting the sample according to the median time spent on building the respective distribution. For the low-effort group, i.e., those who spent less time answering, the distributions tend to be close to triangular, consistent with respondents only clicking once. In contrast, for the high-effort group, the distributions are narrower. This is mechanical since respondents who are more certain about their answers typically require more clicks to produce narrower distributions. Interestingly, however, the subjective error does not differ substantially between the two groups.

To further support this finding, we conducted an additional exploratory analysis by splitting respondents into two groups based on the shape of their provided distributions, as shown in Figure I.4. The 'Low effort' group includes all respondents who provide a perfectly triangular distribution, which requires only one click, while the 'high effort' group includes all other respondents. For the low-effort group (dashed lines), the percentile corresponding to the subjective error varies between 15% and 85%. In contrast, for the high-effort group (solid lines), it remains relatively stable around 60%. Hence, on average, after controlling for effort, the subjective error corresponds to the 60th percentile of the subjective error distribution. Given the two previous findings, it is important to control for both knowledge and effort in the subsequent analysis. As this has not been registered,

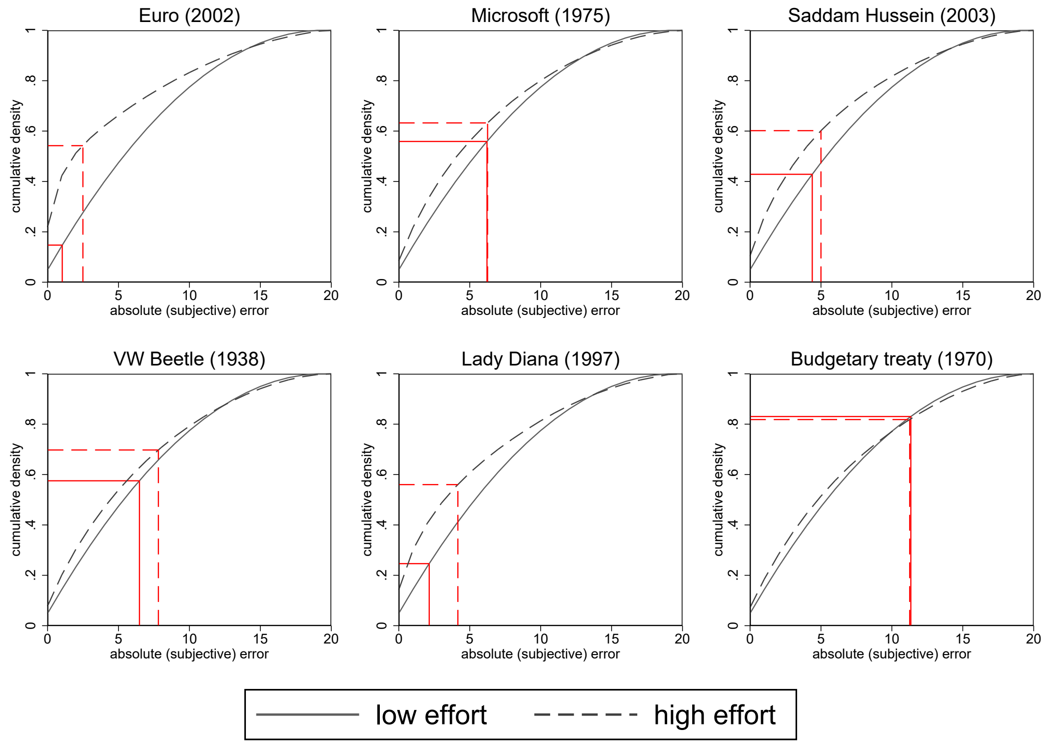
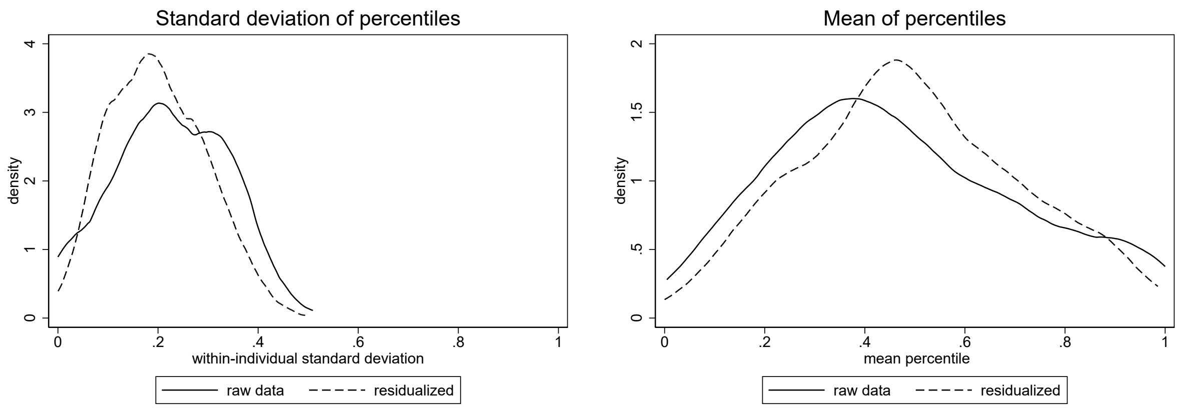


Figure I.4: Subjective error distributions and effort. The solid vertical red line denotes the average subjective error, while the dashed vertical red line plots the 95% confidence interval of the mean. Low effort/high effort is defined as the effort that is put into forming the probability distributions, whereby the 'low effort' group includes all respondents who provide a perfectly triangular distribution.

we show both results.

To get a sense of how consistently respondents answer the second question of the Subjective Error Method, we calculate the within-individual standard deviation of the percentiles of their subjective error distributions, that is, the location of the subjective error, across all questions. The larger the standard deviation, the more inconsistently individuals answer. We compute this measure only for individuals who answered at least four distribution questions. In addition to this raw measure, we also construct residualized percentiles corrected for the difficulty of the answer and the amount of effort provided by the respondent.[55] Figure I.5a plots the density of both measures. The figure shows that controlling for the individual effort per question, as well as the proxy for knowledge, reduces the within-individual dispersion of the location. The median standard deviation of

[55]For each question, we regress the observed percentile on the realized error, the time spent on the distribution, a dummy that equals one if the respondent provided a perfectly triangular distribution, and individual specific control variables. We then subtract the estimated marginal effects of knowledge and effort from the observed percentiles to control for their effects. This part of the analysis was not preregistered.



(a) Standard deviation **(b)** Mean location

Figure I.5: Densities of the within-individual standard deviation of the location of the subjective error and the individual-specific mean percentile. 'Raw data' denotes using the raw data while 'residualized' denotes the analysis using data of which the effect of knowledge and effort have been isolated.

the residualized within-individual standard deviation is 0.19 with relatively little dispersion around the median. Thus, respondents demonstrate relatively consistent answering behavior for the second question after controlling for the question characteristics.

Additionally, we compute the individual-specific mean of the (up to) six percentiles to obtain an aggregate measure for each respondent and to control for potential within-individual noise. The resulting aggregate measure allows us to examine what respondents, on average, have in mind when answering the second question in the Subjective Error Method. As an exploratory extension aimed at accounting for potential question characteristics, we compute a residualized variable following the same approach as above. Figure I.5b shows the resulting density distributions of both variables. Similar to before, controlling for question characteristics decreases the dispersion. The aggregated percentiles are centered around 0.48, with approximately 50% of respondents providing a location of the subjective error corresponding to a percentile between 33% and 64%.

Finally, inspired by a referee, we conduct a validation exercise to demonstrate that the Subjective Error Method provides a good approximation of respondents' intuitive perception of uncertainty. To do so, we calculate the variance in the distribution for each question (including the "Google control"). By regressing the subjective error on the variance, we obtain a direct quantitative benchmark against which to assess the subjective error measure. We perform this regression for both the original units and their logarithmic transformation. In all specifications, we control for individual characteristics or individual fixed effects, as well as for question fixed effects. Columns (1)-(4) of Table I.1 report the results for both the subjective error and the variance in levels, while Columns (5)-(8) present the results for both variables in logs.[56] Across all specifications, there is a strong and positive correlation between the subjective error and the variance estimate. These results provide compelling evidence that respondents' subjective errors reflect their underlying intuitive uncertainty, validating the Subjective Error Method as an effective method for measuring individuals' confidence miscalibration.

[56]Columns (5)-(8) contain fewer observations because logarithmic transformations exclude cases with a variance or subjective error of zero.

	Levels				Logarithmic			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Subjective Error							
variance	0.082***	0.030***	0.126***	0.130***	0.179***	0.098***	0.307***	0.301***
	(0.012)	(0.008)	(0.014)	(0.015)	(0.016)	(0.015)	(0.017)	(0.017)
age		-0.065***	-0.012	-0.014		-0.009***	-0.001	-0.001
		(0.017)	(0.012)	(0.012)		(0.002)	(0.001)	(0.001)
female		-0.320	-0.049	0.105		-0.011	0.042	0.019
		(0.548)	(0.379)	(0.358)		(0.051)	(0.029)	(0.029)
years education		0.057	-0.122	-0.119		-0.005	-0.019**	-0.018**
		(0.174)	(0.104)	(0.101)		(0.018)	(0.008)	(0.008)
answered		0.372	-0.905***	-0.940**		0.074	-0.057*	-0.074**
		(0.523)	(0.324)	(0.364)		(0.057)	(0.032)	(0.033)
gross income		-0.179	-0.091	-0.080		-0.012	-0.003	-0.004
		(0.109)	(0.070)	(0.068)		(0.011)	(0.006)	(0.006)
income missing		-1.910***	-0.419	-0.341		-0.172**	-0.022	-0.038
		(0.556)	(0.441)	(0.458)		(0.080)	(0.055)	(0.053)
expertise		-0.034***	-0.002	-0.004		-0.005***	-0.002**	-0.001**
		(0.011)	(0.009)	(0.008)		(0.001)	(0.001)	(0.001)
math. literacy		0.230	-1.467**	-1.414**		0.100	-0.081*	-0.092**
		(0.753)	(0.670)	(0.645)		(0.074)	(0.043)	(0.042)
mean(percentile)			19.894***	20.196***			2.329***	2.354***
			(2.057)	(2.138)			(0.110)	(0.114)
sd(percentile)				-4.346				0.617***
				(2.676)				(0.161)
N	3442	3437	3437	3433	2670	2679	2679	2677
adj. R^2	0.586	0.174	0.404	0.408	0.595	0.317	0.553	0.557
Question Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Fixed Effects	Yes	No	No	No	Yes	No	No	No

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table I.1: Relationship between the subjective error and the variance estimates. Columns (1)-(4) show the results for both the subjective error and the variance in levels, and Columns (5)-(8) for both variables in logs. Question fixed effects are included in all specifications. Columns (1) and (5) include individual fixed effects. The other columns use individual characteristics instead. Standard errors are clustered at the individual level.